



VIBRATING MACHINERY FOUNDATIONS ON SOIL

PURPOSE

This document establishes guidelines, recommended procedures, and sample calculations for the design of soil supported foundations for large reciprocating compressors, centrifugal compressors, and other similar vibrating equipment. The vibration analysis is based on frequency dependent soil stiffness and damping in a procedure originally described by Gazetas and Novak.

SCOPE

This document includes information about the following major topics:

- Definitions of several terms related to dynamic design.
- A detailed list of required design data.
- A description of the formulae and sequence to perform a vibration analysis.
- A listing of acceptable design results.
- Additional design conditions including situations not normally encountered.

A sample design is included as an aid in producing actual designs. A foundation for a reciprocating compressor is analyzed and designed using a computer program.

Attachment 10 is included for detailed information on the vibration analysis procedure. Basic theory, dynamic equations, and a sequence of analysis are covered. Tables are included to aid in evaluating complex soil stiffness and damping criteria. A detailed reference list identifies sources of information for further examination of guidelines, practices, and procedures. A hand calculation is provided to illustrate how the vibration analysis formulae are applied.

APPLICATION

This document applies to reciprocating machines greater than 200 horsepower and centrifugal machines greater than 500 horsepower supported on block type foundations. The design of foundations for small pumps is described in Structural Engineering Guideline 000.215.1227: Pump Foundations.

Frequency dependent stiffness and damping is applicable to a wide variety of soil conditions and driving frequencies. For uniform soil conditions, frequency dependent and independent methods should yield very similar results. For non-uniform soil conditions, and for higher excitation frequencies (above $A_{\#} = 2.0$, where $A_{\#}$ represents the four values defined in Section D4 of Attachment 10), frequency dependent criteria is more applicable.

DEFINITIONS

This section presents a discussion of a few topics relating to dynamics that need elaboration.



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Reciprocating Machine Forces

Two types of forces act within a reciprocating compressor: gas forces and inertia forces.

Gas forces result from the action of piston motion and valve action which generate time varying head and crank end pressures. The magnitude of gas forces depend on the differential between suction and discharge pressure, the area of the cylinder bore, rod area, pulsations, and external resistance. The near uniform pressure within each compressed volume make the force on the cylinder a direct reaction to the force on the piston. Thus, gas forces act on the crankshaft with an equal and opposite reaction on the cylinder.

Inertia forces result from varying accelerations of rotating and reciprocating machine parts. The magnitude of inertia forces depend on speed, rotating inertia, and on reciprocating inertias of crosshead, piston rod, and piston. Equations for the forces resulting from a single cylinder are indicated in Attachment 01. Inertia forces act at the crankshaft bearings without any opposing reaction on the frame.

For the dynamic analysis of a mat, gas forces tend to cancel each other within the machine frame and within the foundation pier. Thus only inertia forces need be considered. However, for machine anchorage and pier design purposes, the magnitude of gas forces transmitted into the pier is very important and is dependent on machine rigidity. To obtain reasonably accurate anchorage design forces, the rigidity of the machine must be realistically considered. For detailed pier and anchorage criteria, refer to the appropriate sections in this document.

Centrifugal Machine Forces

These inertia forces are caused by imperfect balancing of a rotor. Centrifugal machine forces can be determined by the supplier; however, this information is usually not voluntarily furnished. Either of the equations shown in Attachment 02 may be used in calculating centrifugal machine forces with the following criteria for eccentricity:

- A value of $Q_g = 0.615$ inch/second produces the commonly used equation from Arya:

$$\text{force, kips} = \frac{(\text{rotor weight, kips})(\text{rotor speed, rpm})}{6,000} \quad (\text{Equation 1})$$

- For large steam turbine compressors, ISO 1940, mentions a value of $Q_g = 0.2$ inch/second for use without supplier data, or

$$\text{force, kips} = \frac{(\text{rotor weight, kips})(\text{rotor speed, rpm})}{18,450} \quad (\text{Equation 2})$$

- For machines built in accordance with API 617, Arya recommends using twice the initial test eccentricity. This results in the following equation:



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$$\text{force, kips} = \frac{(\text{rotor weight, kips})(\text{rotor speed, rpm})^{1.5}}{321,673} \quad (\text{Equation 3})$$

If the above criteria is not adequate to produce a reasonable design, vibration sensors may be examined to give an indication about an upper limit of shaft misalignment. If such sensors are used, then an eccentricity may be computed based on the shaft vibration level that will cause the machine to be shut down. Judgment must be applied on an individual case basis with the assistance of the Mechanical Engineer.

Whichever formula is selected in lieu of supplier information, the design criteria should be reviewed with the machine supplier for acceptance.

Shear Modulus

The shear modulus is the ratio of shear stress to shear strain and is the most significant factor in computing soil impedance. This value should be obtained by a soil consultant and is usually supplied as a range of values or as a value for each soil layer. Because soil properties can easily vary, a range of shear modulus values (plus or minus 20 percent) should be checked in the foundation design. If a range of values is supplied, that range may be used instead.

The reliability and accuracy of obtaining the shear modulus must be carefully evaluated. Laboratory triaxial compression tests or field plate bearing tests should not be used. There are at least 3 ways a soil consultant may use to determine the shear modulus.

- A crosshole test is the preferred method for determining the shear modulus. This is basically a field test which can be performed at various depths to obtain a shear modulus profile.
- The Resonant Column Test, ASTM D4015, is a laboratory test performed on undisturbed samples of soil. The shear modulus is determined indirectly through simple elastic equations.
- Published correlations relate shear modulus to other more easily measured soil properties. Though simple to use, correlations are better used to examine the results of other methods.

For further detailed information on soil testing methods, refer to R.D. Woods, Measurement of Dynamic Soil Properties.

Soil Impedance

Basically, this is the soil's reaction to dynamic loads and is a combination of stiffness and damping. One set of impedance values will be determined at the mat base, and another set derived for mat embedment. Various methods are available to evaluate soil impedance.

Current state of the art procedures take into account the fact that soil impedance varies



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with frequency. Numerous published articles use various methods to determine soil impedance under a variety of conditions including elastic half space, viscoelastic half space (soil damping), multiple soil layers, elastic layer over a rigid layer, rectangular foundations, and flexible foundations. This document does not preclude the use of any recognized derivation; however, alternate methods should be carefully studied. The results in some reports may be limited in scope, consist of small hard to read graphs, or applicable to a narrow frequency range.

The soil impedance criteria presented in Attachment 10 are from sources that identify relatively simple equations valid for an elastic half space over a wide range of frequencies. Base stiffness and damping values are from Veletsos. Embedment stiffness and damping are from Novak. For different soil conditions, other criteria will be needed. Details and references are provided in Attachment 10.

INFORMATION NECESSARY FOR DESIGN

This section identifies all data required for analysis and design. The paragraphs are grouped by the source of the data.

Machine data should be requested from the machine supplier. Refer to Structural Engineering Specification 000.215.00920: Structural Data For Mechanical Equipment.

Soil information should be requested from the soil consultant. Refer to Civil Engineering Specification 000.210.02010: Geotechnical Investigation

Client and project data will be in the form of project specifications, client specifications, or meeting notes.

Basic Machine Data

Operating Speed: This is the frequency or range of frequencies of the dynamic machine forces. Individual rotor speeds should be provided if different from overall machine speed.

Outline Drawing: A layout of the machine provides dimensions to the cylinders, shaft, and other components.

Anchor Bolts and Layout: Bolt materials and sizes should be provided in order to verify bolt forces and pier dimensions. Whenever possible, anchor bolts for large compressors should be 12 inches minimum from bolt centerline to face of concrete.

Pier Layout: This layout should provide all dimensions and elevations for machine support. If not provided, pier dimensions may be determined using a minimum of 4 inches from the compressor base to the face of the pier.

Jacking Post Locations and Details: This provides the installation methods to properly align machine components.



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Machine Forces

Primary and Secondary Dynamic Forces: These loads are normally provided for reciprocating machines with a brief explanation of their application.

Rotor Weight and Eccentricity: This determines loads for centrifugal machines. If eccentricities are not provided, the supplier should be informed of the criteria to be used.

Location of Dynamic Forces: This should indicate the location at which reciprocating forces or rotor eccentricities are applied. Normally, this is somewhere along the shaft.

Machine Component Data

Component Weights: Usually, only the motor and compressor are broken out as separate parts.

Component CG: A center of gravity for each machine component should be indicated, usually on the machine outline drawing. Sometimes, one center of gravity is provided for the entire machine.

Component Mass Moments of Inertia: If provided, mass moments about each axis should be used in the vibration analysis. If not provided, mass moments are normally judged to be minor enough to be neglected.

Additional Machine Criteria

Grout Requirements: The machine supplier is the most important source for grout requirements. Supplier criteria must be carefully examined versus client or construction preferences.

Bolt Post Tensioning Criteria: Inappropriate bolt post tensioning can easily cause excessive movement between the machine and base.

Allowable Amplitudes: If provided, use supplier criteria instead of the criteria provided in this document. If not provided, the supplier should be informed of the criteria to be used.

Measurement Point Locations: Amplitude measurement locations should be requested if allowable amplitudes are specified.

Geotechnical Conditions

Soil Properties: Unit Weight, Shear Modulus, and Poisson's Ratio values should be provided for each layer. Separate values may be needed for backfill if embedment is to be included. Comparison values are as follows:



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Description:	Unit Weight: (lb/ft ³)	Shear Modulus: (K/in ²)	Poisson's Ratio:
Granite	150 to 160	4000 to 6000	0.15 to 0.2
Limestone	145 to 155	2000 to 5000	0.16 to 0.22
Sandstone	145 to 155	1000 to 4000	0.17 to 0.24
Dense Sand	115 to 140	10 to 19	0.28 to 0.34
Medium Sand	110 to 130	8 to 15	0.30 to 0.36
Loose Sand	95 to 125	5 to 11	0.32 to 0.38
Hard Clay	125 to 145	11 to 15	0.38 to 0.41
Medium Clay	115 to 135	7 to 11	0.41 to 0.44
Soft Clay	100 to 125	3 to 7	0.44 to 0.47

Allowable Net Soil Bearing: This is a standard component of every soil investigation.

Material Damping: Also called internal or hysteretic damping, this is the energy loss within the soil due to interparticle friction.

Construction Recommendations: Also a standard component; recommendations concerning excavation, backfill compaction, and vibration isolation from other foundations may be provided.

Embedment Recommendations: These should be provided to confirm characteristics of backfill materials.

Optional Geotechnical Assistance

Soil Layer Evaluation: The evaluation of soil layers to provide soil stiffness and damping could be computed by the soil consultant. This would be recommended especially if unusual soil conditions are encountered.

Foundation Vibration Analysis: Some soil consultants have the expertise to provide a partial or complete amplitude calculation. This service could become effective if highly unusual soil conditions need to be evaluated or if computer software is not readily available.

Client Specifications

Allowable Amplitudes: If provided, these requirements should be used if more conservative than supplier requirements.

Measurement Point Locations: Location of amplitude measurement points may be specified if allowable amplitudes are provided.

Anchor Bolt Materials and Details: Supplier criteria should still be carefully considered.



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Project Requirements

Concrete Strength and Reinforcing Grade: Standard material strengths for foundations are normally used (grade 60 reinforcing and 3,000 or 4,000 psi concrete). A higher strength of concrete may be advisable since tensile strength is important in preventing cracking in the pier.

Equipment Plot Location: This determines how much area the foundation mat can cover without interfering with other structures and equipment. Reciprocating machines with an odd number of cylinders require larger mats because the combined cylinder unbalanced forces are larger. Separation of compressor foundations from vibration sensitive equipment or structures should be considered. Problems should be worked out with Piping as early as possible.

VIBRATION ANALYSIS PROCEDURE

This section gives a description of how a vibration analysis is used to design a foundation mat.

Mat Plan Dimensions

General rules are used for an initial mat size. This criteria need not be literally applied; it is provided only for trial sizing information. The analysis will determine the acceptability of the mat size.

The mat width perpendicular to the shaft should be 1.5 times the height from the shaft to the bottom of the mat (0.75 times for centrifugal machines). Second, the mat length parallel to the shaft should be 2 feet longer than the length of the pier. The location of the mat relative to the pier may have to be adjusted in order to reduce foundation eccentricities. Refer to acceptance criteria for alignment offsets.

Mat Thickness

The mat thickness must be checked for rigidity because the impedance determination and analysis procedure described in Attachment 10 uses the common assumption that the foundation and machine are rigid relative to the soil. If the mat is not rigid, then the impedance should be determined by another, applicable, method and a dynamic finite element analysis should be used. The mat thickness must be verified by hand because the vibration analysis results will not confirm this assumption.

The mat thickness should not be less than 2 feet., in addition one of the following alternative equations should be applied:

1. The following formula is derived from beam on elastic foundation theory using a flexural deflection pattern that approximates essentially rigid mat behavior:

$$L_z \geq 0.02 \left[\frac{G L^4}{b} \right]^{1/3} \quad (\text{Equation 4})$$



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where,

- L_z = thickness of foundation mat (ft)
- G = soil shear modulus (psi)
- L = mat cantilever dimension in rocking direction (ft)
- b = width of mat perpendicular to rocking direction (ft)

2. The following equation is commonly specified in many client specifications:

$$L_z \geq 2 + \frac{L}{30} \quad (\text{Equation 5})$$

where,

L = longest mat dimension (ft)

3. The following formula from Gazetas is based on a published study of the subject:

$$\left(\frac{E_c}{E_s} \right) \left(\frac{L_z}{d} \right)^3 \geq 1.0 \quad (\text{Equation 6})$$

where

- E_c = Modulus of elasticity for concrete (K/in²)
- E_s = Modulus of elasticity for soil (K/in²)
- d = Mat cantilever beyond face of pier, in either direction (ft)

Mat Embedment

Increased stiffness and damping result from soil along the vertical sides of a mat embedded into the ground. Embedment must be carefully designed, not just selected as an option in a computer program. Compacted cohesionless soils (sand) should be used to ensure full embedment. Cohesive soils (clay) can easily shrink away from the sides of a foundation mat. If isolation details are not provided, foundations in cohesive soils should be analyzed for the extremes of full embedment and no embedment. Attachment 03 shows typical details used to embed a mat, including the use of thickened edges to increase embedment depth.

A conservative value of the embedded depth should be used in the analysis. Normally, 2/3 of the total embedment should be used in the vibration analysis.

Amplitude Locations

Locations should be established separately for machine tolerance and for human tolerance. If supplier or client criteria is not provided, locations will be determined from the following:



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- Machine Tolerance - The shaft bearing locations at each end of the shaft can be selected. If bearing locations are not known, points at each end of the shaft should be selected.
- Human Tolerance - Locations should be selected at grade elevation on top of the mat corners.

Analysis Calculations

After all input data has been obtained, a vibration analysis is performed to compute the dynamic response using the selected foundation dimensions. A computer program is normally used to obtain the following results:

MR	= Mass ratio
SB _{net}	= Net soil bearing (ksf)
OS _x , OS _y	= Alignment offset, in each horizontal direction (percent)
NF	= Natural frequency, for each mode of vibration (rpm)
FR	= Frequency ratio, for each frequency and operating speed
DA _x , DA _y , DA _z	= Double, or peak to peak, amplitudes, in each axis (mils)

The basic steps of the vibration analysis procedure are as follows:

1. Calculate the machine foundation soil center of gravity.
2. Calculate the mass and mass moments of inertia.
3. Compute mat stiffness and damping.
4. Compute embedment stiffness and damping.
5. Resolve all forces at the center of gravity.
6. Perform a 1 dof analysis for vertical translation and for torsional rotation.
7. Perform a 2 dof analysis for horizontal and rocking vibration in each direction.
8. Determine resulting double amplitudes at the desired location.

Attachment 10 provides the technical basis, detailed equations, and a list of references used in a standard calculation procedure. The procedures employed by specific computer programs may vary somewhat. Program documentation should be examined for significant calculation procedure variations.

Coastdown Effects

Centrifugal machines with high operating speeds are usually started and shut down slowly enough to be subject to temporary periods of resonance. If any of the 6 vibration modes have natural frequencies less than the machine speed, coastdown amplitudes should be investigated. Judgment may be used to skip modes which contribute little to total amplitudes. Computed resonant frequencies are used as coastdown machine speeds and reduced dynamic forces are calculated using the following formula:

$$\text{coastdown force, lb} = (\text{force at operating speed, lb})(CS / MS)^2 \quad (\text{Equation 7})$$

where

$$CS = \text{coastdown speed (rad/sec)}$$



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MS = machine speed (rad/sec)

The vibration analysis is repeated for each applicable coastdown frequency. Higher allowable amplitudes are provided for temporary coastdown resonance conditions as indicated in Attachment 04.

Combined Foundations

For multiple machines on a common mat, the vibration amplitude calculations are to be based upon the simultaneous operation of the maximum number of machines representing the design condition. Spare and standby machines are assumed stopped. For those machines operating, all unbalanced forces and moments are to be assumed acting in whichever combination will produce the highest amplitudes.

For relatively small machines spaced close together, the mat may be thick enough to be considered rigid. This evaluation will need to be carefully studied and confirmed. For most cases of multiple machine foundations, the mat is flexible and the rigid mat basis of this practice does not apply. Recommended analysis procedures involve a computer finite element analysis using appropriate soil stiffness and damping computed in observance of a flexible mat.

Because a finite element analysis is cumbersome, a simpler alternate is sometimes used. The following procedure is approximate at best and has a limited technical basis compared to the finite element solution. One can approximate effects of mat flexibility in the following way:

1. Consider a single machine supported on its tributary area of mat as an isolated foundation. That is, disregard the other machines and the remainder of the mat. Evaluate the vibration response of this system and calculate the ratio, R1, of total coupled translation rocking displacement at the compressor shaft to translation displacement at the mat. This represents the upper bound of rocking contribution.
2. Evaluate the vibration response of the common mat system using the procedures outlined herein. Calculate the ratio, R2, of total coupled translation rocking displacement at the compressor shaft to translation displacement at the mat. This represents the lower bound of rocking contribution because the half space solutions are based upon a rigid mat assumption.
3. An estimate of coupled translation rocking displacement at the compressor shaft for a flexible mat is calculated as the translation displacement at the common mat times the average value of R1 and R2 computed above.
4. The displacement from the torsional mode of vibration must be added to the above translation plus rocking displacement to obtain the total vibration amplitude.

Multiple Rotors

Multiple rotors can be analyzed separately with the resulting amplitudes added together. If the rotor speeds match, unbalanced forces can be combined prior to the vibration



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analysis. Combinations should include in-phase and out-of-phase rotors as necessary to compute conservative amplitude values.

VIBRATION ACCEPTANCE CRITERIA

This section describes the criteria used to accept the analysis results. Analysis by hand or by computer will be evaluated in the same manner.

Mass Ratio

Mass ratios are used periodically as a traditional gage of how much foundation mass is being provided relative to the machine. The following guideline may be used as an indicator of foundation performance:

$$\begin{aligned} &\text{for reciprocating machines, } MR \geq 5 \\ &\text{for centrifugal machines, } MR \geq 3 \end{aligned} \quad (\text{Equation 8})$$

where,

MR = Weight of machine & foundation divided by weight of machine

Foundations with a low mass ratio should not necessarily be resized. Instead, the dynamic analysis input and results should be carefully examined and used (refer to API 686).

Soil Pressure

Static soil pressure should be kept low to ensure elastic soil behavior. The following is generally considered acceptable:

$$SB_{\text{net}} \leq 0.5 (SB_{\text{allow}}) \quad (\text{Equation 9})$$

where,

$$\begin{aligned} SB_{\text{allow}} &= \text{net static allowable soil pressure} \\ SB_{\text{net}} &= \text{maximum net soil contact pressure} \end{aligned}$$

If this criteria is not met, try a larger mat.

Alignment Offsets

These should be kept to a minimum to prevent unintended coupled modes and to reduce differential settlement.

$$\begin{aligned} OS_x &\leq 5 \text{ percent} \\ OS_y &\leq 5 \text{ percent} \end{aligned} \quad (\text{Equation 10})$$

where,



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OS_x, OS_y = distance between CG and center of mat divided by mat length

If these criteria are not met, relocate the mat as appropriate.

Frequency Ratios

Foundations should be outside the resonant range to avoid high amplitudes. The following criteria for frequency ratios apply when soil properties are checked for plus or minus 20 percent variation.

$$FR \leq 0.8 \text{ or } FR \geq 1.2 \quad (\text{Equation 11})$$

where,

FR = machine speed divided by natural frequency

Note that client or manufacturers criteria can vary. Some specifications may use different range values (such as 0.7 and 1.4). Other specifications use a resonant frequency ratio (machine speed divided by resonant frequency). For low to moderate amounts of damping, the difference should be negligible. If needed, resonant frequency can be computed from the following:

$$RF = NF / \sqrt{1 - 2(DR)^2} \quad (\text{Equation 12})$$

where,

DR = damping ratio

NF = natural frequency

RF = resonant frequency

Sometimes, the frequency ratio criteria will be difficult to achieve. For reciprocating machines, 6 primary and 6 secondary frequency ratios will need to fall outside the resonant range. If one or more modes are within the resonant range, check the corresponding unbalanced forces and damping. The offending modes may be ignored if there are no applied forces or if the mode is overdamped, (i.e. damping is greater than twice the square root of stiffness times mass). If dynamic forces are small, an amplitude calculation using half the computed damping may be necessary. Otherwise, try a different mat size or embedment characteristics to alter the stiffness and damping.

Trying to live within the resonant range carries with it a certain level of risk. On one hand, computed damping levels are usually high enough to preclude significant resonance. On the other hand, machines have failed before due to resonance, especially under unusual soil conditions. Centrifugal machines are especially vulnerable because the magnitude of applied forces is only an educated guess.

Amplitudes

Vibration amplitudes must be kept low in order to prevent machine damage. Computed



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double amplitudes shall be less than supplier criteria, client requirements, or the limiting values from Attachment 04. Sometimes supplier or client requirements will include velocity limits. If needed, velocity can be computed from the following:

$$\text{velocity, in/sec} = (\text{MS})(\text{DA}) / 2,000 \quad (\text{Equation 13})$$

where,

MS = machine speed (rad/sec)

DA = double amplitude (mils)

If amplitude criteria is not met, the mat size and embedment criteria must be revised. The vibration analysis is performed again until amplitude criteria is acceptable. In unusual cases, acceptable amplitudes may not be achievable with a reasonable foundation size. Amplitude criteria may then need to be evaluated again with the client and supplier until a mutually agreeable solution is obtained.

ADDITIONAL DESIGN CONSIDERATIONS

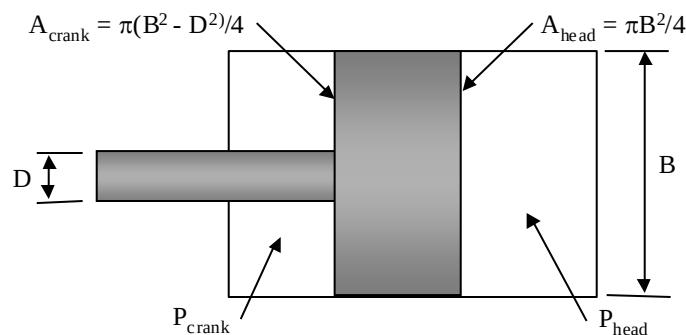
The vibration analysis is usually of primary concern in designing foundations for vibrating machines. However, several other considerations need to be addressed in completing the design.

Reinforcing Design

Reinforcing is used to eliminate or reduce concrete cracking. As a minimum, mats should be provided with temperature reinforcing (top and bottom bars > 0.0018 times gross area), and piers should be provided with a cage of #5 at 12 inches each way. Refer to Attachment 05 for typical reinforcing configurations.

Piers for Reciprocating Machines

Gas pressure forces must be considered in the design of reciprocating machine piers. The magnitude of these forces depend on the rigidity of the machine frame; the more rigid the frame, the lower the force transmitted to the pier. Frames are rarely rigid enough to eliminate these forces. In lieu of a detailed finite element analysis of the machine frame, The following equation from Smalley shall be used to compute pier forces:





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$$F_{fdn} = [(P_{head})(A_{head}) - (P_{crank})(A_{crank})] F_{cr} / F_{red} \quad \text{(Equation 14)}$$



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where,

- A_{head} = Area of piston head (in²)
- A_{crank} = Area of piston head on crank side (in²)
- B = Cylinder bore diameter (in)
- D = Rod diameter (in)
- P_{head} = Instantaneous head pressure (ksi)
- P_{crank} = Instantaneous crank pressure (ksi)
- F_{cr} = Correction factor (1.15 to 1.2)
- F_{fdn} = Lateral force on foundation, tributary to cylinder (kips)
- F_{red} = Reduction factor, use 2.0 unless better data is available

The instantaneous head and crank pressures can be taken from the maximum and minimum suction and discharge pressures obtained from the mechanical engineer.

The anchor bolts tributary to the cylinder shall be used to resist F_{fdn} .

Grouting Methods

Grout requirements are determined in accordance with the project grout specifications, the compressor supplier's requirements, and the grout supplier's criteria. Conflicts with supplier, client, construction, or design requirements will need to be resolved and a mutual agreement obtained. Normally, epoxy grout is used for compressor foundations. Steel chocks or epoxy chocks may also be used to thermally isolate the machine from the supporting foundation. Anchor bolt sleeves should normally not be filled with grout; machine supplier drawings should be consulted on this matter.

Anchor Bolt Design

Bolt requirements are provided by the machine supplier. This always includes the number and diameter of anchor bolts. Bolt material and post-tensioning requirements may or may not be provided. If not, then these requirements are selected based on the applied forces. In addition, because slippage cannot be tolerated, lateral resistance must be provided through a sufficient clamping force between machine and foundation.

$$T_{\text{min}} = F / \mu - W_r \quad (\text{Equation 15})$$

where,

- T_{min} = minimum tensile clamping force per bolt
- F = applied lateral force per bolt
- μ = friction coefficient (use 0.15 based on oily steel on cast iron)
- W_r = applicable machine weight (could conservatively use zero)

For centrifugal machines, the applied force per bolt can be determined from the overall unbalanced dynamic force. For reciprocating machines, the applied force per bolt shall be determined from the gas pressure forces on each cylinder.

If post tensioning is required, bolt tightening instructions to the field should be provided



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with due consideration for tension measurement procedures and for retensioning at a later time to minimize relaxation.

Concrete embedment should be provided in accordance with Structural Engineering Guideline 000.215.1207: Anchor Bolt Design Criteria. Because the level of actual post tensioning in the field is difficult to control, and because cracking the pier due to overtightened bolts is to be avoided, bolts should be provided with a ductile embedment. The ductile criteria for embedment should be followed for tension and lateral bursting embedment without the use of additional reinforcing. The need for additional embedment reinforcing should be avoided because it requires cracking to become effective. Shear embedment criteria should be skipped because lateral forces are resisted through post tensioning of bolts and friction between machine base and foundation. If not dictated by the machine supplier, anchor bolts for large compressors should be 12 inches minimum from bolt centerline to face of concrete.

Water Table

In a vibration analysis, mass and not weight is what is important. A high water table may affect the apparent foundation weight, but the mass remains unaffected. The water mass is not rigidly connected to any foundations elements and should not be included.

The effect of a high water table on soil impedance should be verified with the soil consultant.

Elevated Pipe Anchors

Dynamic forces from connected piping are rarely significant enough to be used for the mat design.

Truly rigid anchor points to resist mechanical or pulsation loads are nearly impossible to provide by tall, slender piers without the addition of congested, unsightly bracing. Thus, it is inadvisable to provide elevated pipe anchors. However, if required, such pipe supports attached to the foundation pier or mat should be designed so that their natural frequencies of horizontal vibration are either less than 0.5 times the compressor primary frequency or greater than 1.5 times the compressor secondary frequency.

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Structural Engineering
Guideline 000.215.1207: Anchor Bolt Design Criteria

Structural Engineering
Guideline 000.215.1227 Pump Foundations

Civil Engineering
Specification 000.210.02010: Geotechnical Investigation

Structural Engineering
Specification 000.215.00920: Structural Data For Mechanical Equipment

ATTACHMENTS

Attachment 01:
Reciprocating Machine Forces

Attachment 02:
Centrifugal Machine Forces

Attachment 03:
Embedment Details

Attachment 04:
Vibration Limits

Attachment 05:
Reinforcing Details



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Attachment 06:

Sample Design: Reciprocating Machine Foundation

Attachment 07:

SVAP Analysis, Run #1 (Soil Properties As Given)

Attachment 08:

SVAP Analysis, Run #2 (Soil 20 Percent Stronger)

Attachment 09:

SVAP Analysis, Run #3 (Soil 20 Percent Weaker)

Attachment 10:

Analysis Procedure

Attachment 11:

Directional Nomenclature

Attachment 12:

Mass Moments Of Inertia

Attachment 13:

Base Impedance Coefficients

Attachment 14:

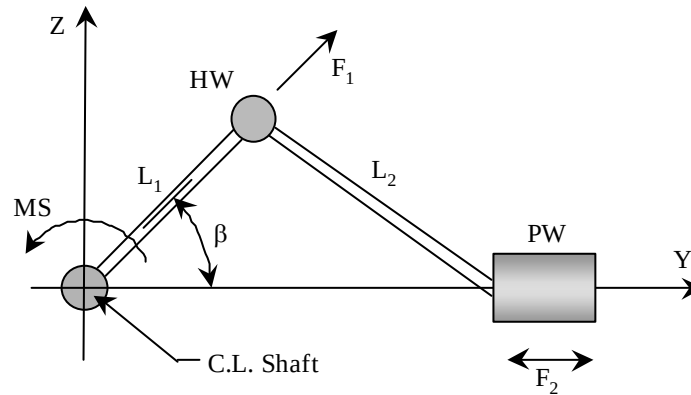
Embedment Impedance Coefficients

Attachment 15:

Vibration Calculation

VIBRATING MACHINERY FOUNDATIONS ON SOIL

Reciprocating Machine Forces



For one cylinder:

$$F_x(t) = 0$$

$$F_y(t) = \{(HW + PW)(L_1)(MS)^2 \cos [(MS)(t) + \beta] + (PW / L_2)(L_1 MS)^2 \cos 2 [(MS)(t) + \beta]\} / \text{gravity}$$

$$F_z(t) = (HW)(L_1)(MS)^2 \sin [(MS)(t) + \beta] / \text{gravity}$$

Where:

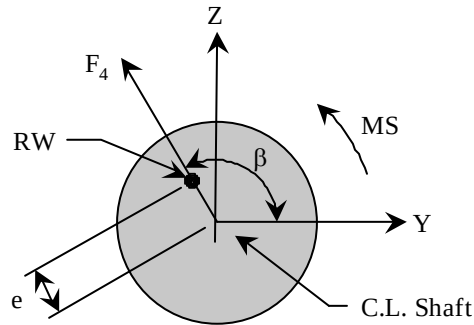
- HW = weight of hinge (kips)
- PW = weight of piston (kips)
- L_1 = length from shaft to hinge (ft)
- L_2 = length from hinge to piston (ft)
- gravity = 32.2 (ft/sec²)
- MS = machine speed (rad/sec)
- t = time (sec)
- F_x, F_y, F_z = forces in each coordinate direction (kips)
- β = crank angle (rad)

For multiple cylinders, the individual piston forces are combined using the appropriate distances between cylinders and the appropriate crank angles. These combined forces are usually provided midway between the cylinders by the machine supplier.



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Centrifugal Machine Forces



For one rotor:

$$F_4 = (RW)(e)(MS)^2 / 12(\text{gravity})$$

or

$$F_4 = (RW)(Q_g)(MS) / 12$$

$$F_x(t) = 0$$

$$F_y(t) = F_4 \cos [(MS)(t) + \beta]$$

$$F_z(t) = F_4 \sin [(MS)(t) + \beta]$$

Where:

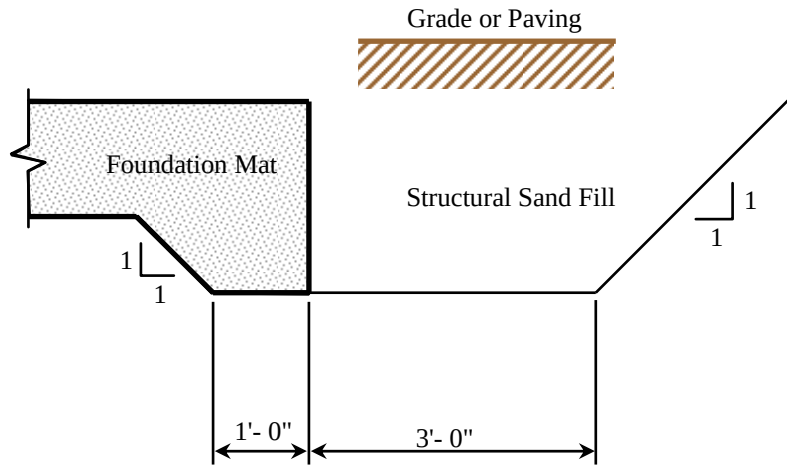
F_4	= inertia force of unbalanced rotor (kips)
RW	= rotor weight (kips)
e	= eccentricity of rotor (in)
Q_g	= measure of quality grade of rotor (in/sec)
gravity	= 32.2 (ft/sec ²)
MS	= machine speed (rad/sec)
t	= time (sec)
F_x, F_y, F_z	= forces in each coordinate direction (kips)
β	= crank angle (rad)

Multiple rotors should be handled independently. Each rotor can be analyzed separately and the results added together. Alternately, forces from each rotor can be combined prior to analysis using crank angles in worst case orientations. For this method, the speeds of each rotor must match.

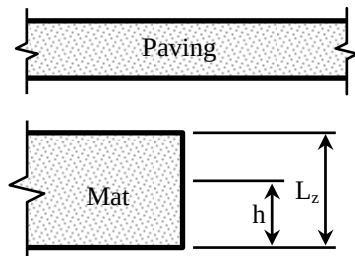


VIBRATING MACHINERY FOUNDATIONS ON SOIL

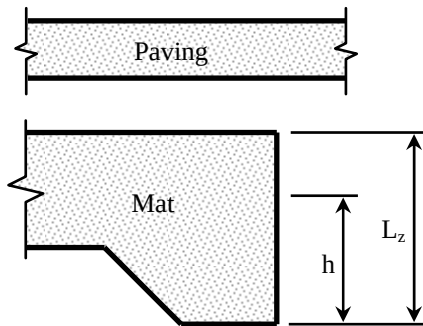
Embedment Details



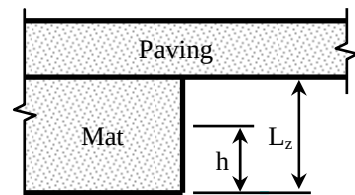
(a)



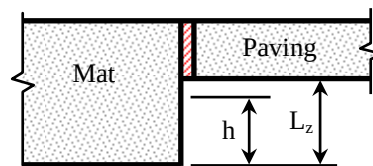
(b)



(c)



(d)

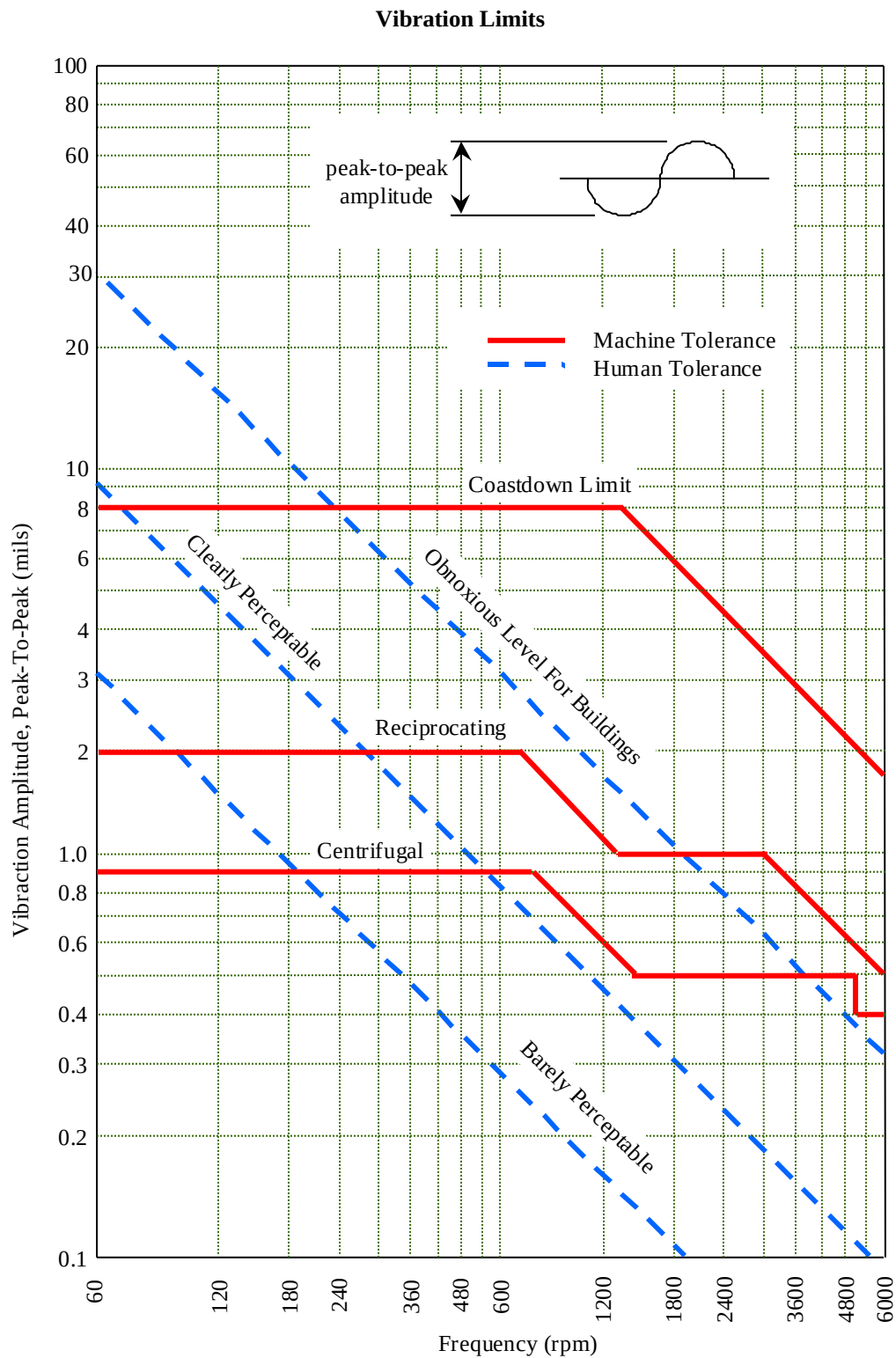


(e)

Effective Embedded Depth, $h = (2/3)(L_z)$



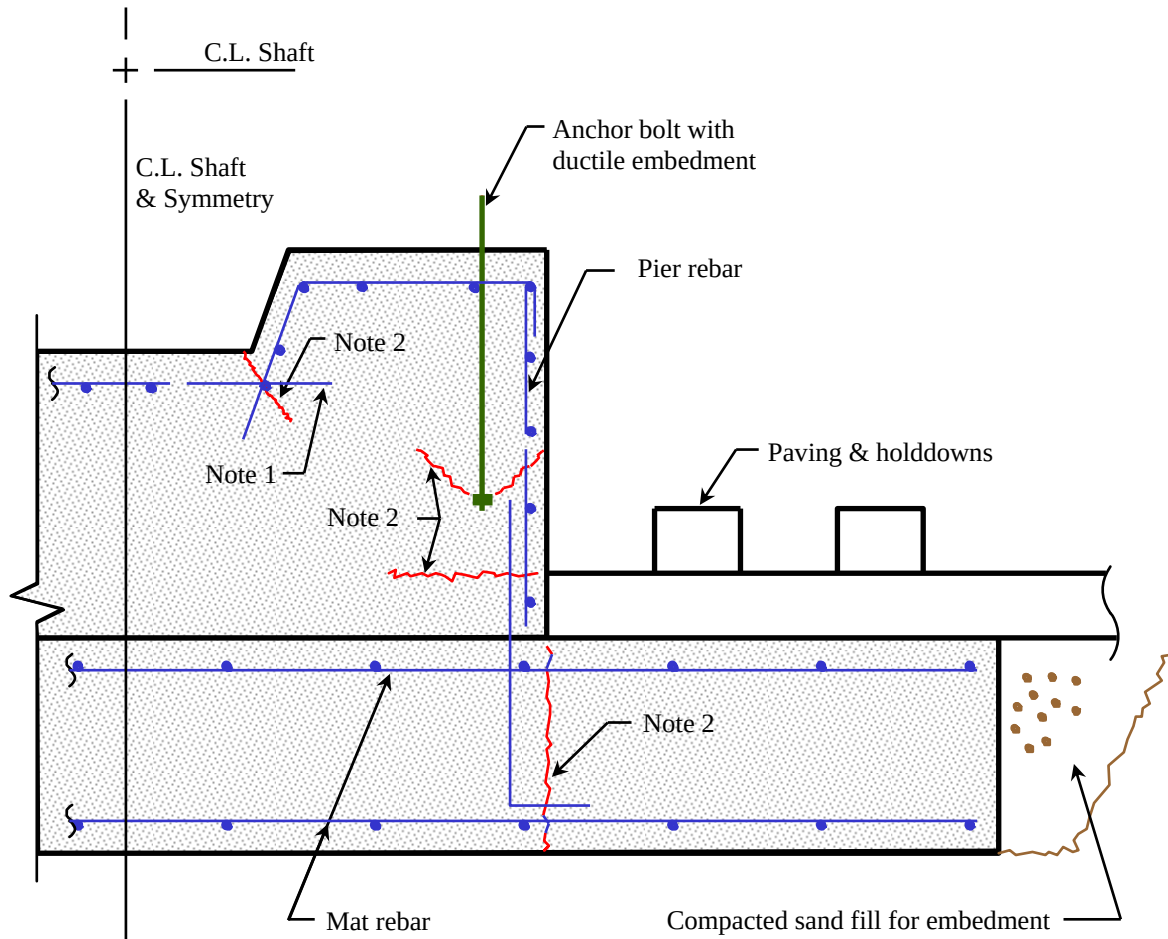
VIBRATING MACHINERY FOUNDATIONS ON SOIL





VIBRATING MACHINERY FOUNDATIONS ON SOIL

Reinforcing Details



Notes:

1. Develop all reinforcing, especially at re-entrant corners where pier cracking can originate.
2. Potential cracking to be resisted by concrete and reinforcing.



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation

Machine: Reciprocating, 300 rpm

motor = 15000 lb (C.G. at point #1)

compressor = 20,000 lb (C.G. at point #2)

Dynamic Loads: applied at point #3
 (provided by the machine supplier)

	(primary)	(secondary)
F_x	1,000 lb	0
F_y	1,500 lb	800 lb
F_z	875 lb	0
M_x	2,100 ft-lb	900 ft-lb
M_y	200 ft-lb	0
M_z	3,100 ft-lb	1,500 ft-lb

Soil: clay, net bearing = 3000 psf

existing soil:

unit weight = 110 pcf

shear modulus = 5,000 psi

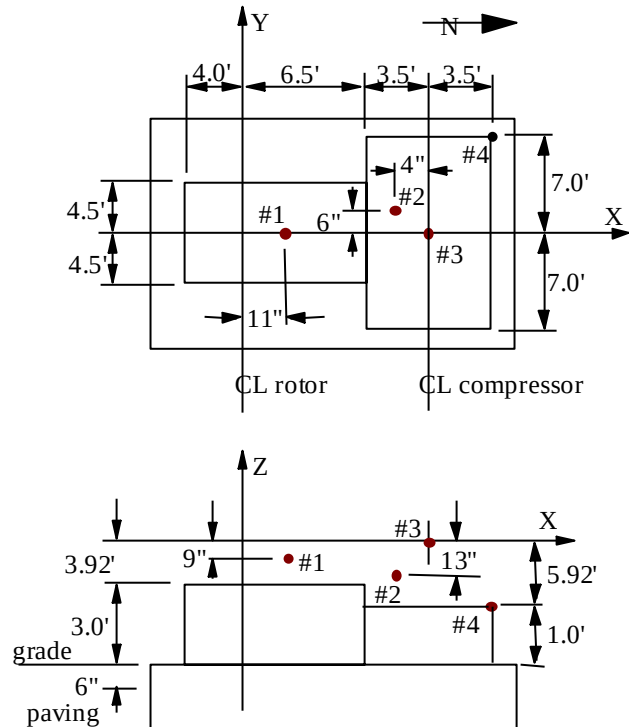
poisson's ratio = 0.44

purchased sand backfill:

unit weight = 120 pcf

shear modulus = 10,000 psi

poisson's ratio = 0.35



Cylinder Data: 4 - 1.5 in diameter bolts per cylinder

cylinder diameter = 3 inches

rod diameter = 0.625 inches

maximum gas pressure = 6,200 psi

minimum gas pressure = 500 psi

Trial Mat Size: assume 2' 0" thick

$$\begin{aligned}
 L_x &= \text{pier dimension plus 2 ft} \\
 &= [(4.0 \text{ ft}) + (6.5 \text{ ft}) + (3.5 \text{ ft}) + (3.5 \text{ ft})] + 2.0 \text{ ft} \\
 &= 19.5 \text{ ft, from previous trials, this is too small}
 \end{aligned}$$

USE $L_x = 21' 0''$

$$\begin{aligned}
 L_y &= 1.5 [\text{distance from shaft to bottom of mat}] \\
 &= 1.5 [(2.0 \text{ ft mat}) + (0.5 \text{ ft paving}) + (3.0 \text{ ft pier}) + (3.92 \text{ ft to shaft})] \\
 &= 14.88 \text{ ft, from previous trials, this is too small}
 \end{aligned}$$

USE $L_y = 26' 0''$

verify thickness, $L_z = 2' 0''$

$$\text{pier cantilever, } d = [(26.0 \text{ ft mat}) - (9 \text{ ft motor pier})] / 2 = 8.5 \text{ ft}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation

elastic modulus of concrete, $E_c = 57,000 = 57,000 = 3,122,019$ psi

elastic modulus of soil, $E_s = 2$ (shear modulus)(1 + poisson's ratio) = 2 (5,000 psi)(1 + 0.44) = 14,400 psi

$$\begin{aligned} [E_c / E_s] [L_z / d]^3 &\geq 1 && \text{(Equation 6)} \\ [(3,122,019 \text{ psi}) / (14,400 \text{ psi})] [(2.0 \text{ ft}) / (8.5 \text{ ft})]^3 & \\ 2.8 > 1 &\text{ OK, rigid} && \text{USE } L_z = 2' 0'' \end{aligned}$$

Note: A thickness of 2.5 feet will be used in the analysis in order to account for the 6 inches of concrete paving.
Pier heights will be reduced by 6 inches.

Select Amplitude Measurement Locations:

Use Point #4 for human tolerance:

$$X = 13.5 \text{ ft} \quad Y = 7.0 \text{ ft} \quad Z = -5.92 \text{ ft}$$

Select two points along shaft for machine tolerance: (point #3, and origin)

(point #3)	(origin)
$X = 10.0 \text{ ft}$	$X = 0.0 \text{ ft}$
$Y = 0.0 \text{ ft}$	$Y = 0.0 \text{ ft}$
$Z = 0.0 \text{ ft}$	$Z = 0.0 \text{ ft}$

Analysis Options:

For embedment use 2/3 of embedded mat thickness, $h = 2/3 (2.0 \text{ ft}) = 1.33 \text{ ft}$

Let computer relocate mat to machine/pier center of gravity

First Run: (soil properties as given)

1. Mass ratio = 7.49 > 5, OK
2. Soil pressure = 544 psf gross < 1500 psf, half net allowable, OK
3. Alignment offsets = automatically centered, OK

The initial southwest corner of the mat was entered as the origin ($X = 0.0$ & $Y = 0.0$). The program responded by relocating the mat 6' 1" south and 13' 1" west. These results provide the dimensions used in the design sketch.

This location, the mat size, and pier dimensions were checked and no interferences were found.



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation

4. Frequency ratios:

mode	natural frequency	primary	secondary
1 (vertical)	835 rpm	overdamped	overdamped
2 (torsion)	1121 rpm	0.26 OK	0.51, OK
3 (transverse)	704 rpm	0.24, OK	0.49, OK
4 (transverse)	1065 rpm	0.23, OK	0.46, OK
5 (longitudinal)	695 rpm	0.25, OK	0.50, OK
6 (longitudinal)	1047 rpm	0.23, OK	0.47, OK

5. Amplitudes:

Point #4: double amplitude
X 1.02 mils, barely perceptible, OK
Y 1.75 mils, barely perceptible, OK
Z 1.14 mils, barely perceptible, OK

Point #3: double amplitude
X 0.94 mils < 2, OK
Y 1.92 mils < 2, OK
Z 0.55 mils < 2, OK

Origin: double amplitude
X 0.94 mils < 2, OK
Y 1.86 mils < 2, OK
Z 0.49 mils < 2, OK

Second Run: (soil 20% stronger)

unit weight = 110 pcf
shear modulus = 5,000 psi * 1.2 = 6,000 psi
poisson's ratio = 0.44 / 1.2 = 0.37

1. Mass ratio, same as run #1, OK
2. Soil pressure, same as run #1, OK
3. Alignment offsets, same as run #1, OK



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation

4. Frequency ratios:

mode	natural frequency	primary	secondary
1 (vertical)	862 rpm	overdamped	overdamped
2 (torsion)	1228 rpm	0.23, OK	0.47, OK
3 (transverse)	756 rpm	0.23, OK	0.46, OK
4 (transverse)	1116 rpm	0.22, OK	0.44, OK
5 (longitudinal)	745 rpm	0.24, OK	0.48, OK
6 (longitudinal)	1099 rpm	0.22, OK	0.44, OK

5. Amplitudes:

Point #4: double amplitude
X 0.87 mils, barely perceptible, OK
Y 1.51 mils, barely perceptible, OK
Z 1.07 mils, barely perceptible, OK

Point #3: double amplitude
X 0.84 mils < 2, OK
Y 1.69 mils < 2, OK
Z 0.51 mils < 2, OK

Origin: double amplitude
X 0.84 mils < 2, OK
Y 1.64 mils < 2, OK
Z 0.46 mils < 2, OK

Third Run: (soil 20% weaker)

unit weight = 110 pcf
shear modulus = $5,000 / 1.2 = 4,167$ psi
poisson's ratio = $0.44 * 1.2 = 0.53$

Note: A poisson's ratio of 0.53 is unrealistic for an elastic material. Use a maximum value of 0.47 for a very weak clay.

1. Mass ratio, same as run #1, OK
2. Soil pressure, same as run #1, OK
3. Center of gravity offsets = automatically centered, OK



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation

4. Frequency ratios:

mode	natural frequency	primary	secondary
1 (vertical)	783 rpm	overdamped	overdamped
2 (torsion)	1023 rpm	0.28, OK	0.56, OK
3 (transverse)	649 rpm	0.26, OK	0.53, OK
4 (transverse)	993 rpm	0.25, OK	0.50, OK
5 (longitudinal)	641 rpm	0.27, OK	0.54, OK
6 (longitudinal)	975 rpm	0.25, OK	0.50, OK

5. Amplitudes:

Point #4: total
X 1.23 mils, barely perceptible, OK
Y 2.06 mils, clearly perceptible, OK
Z 1.29 mils, barely perceptible, OK

Point #3: total
X 1.09 mils < 2, OK
Y 2.23 mils > 2
Z 0.62 mils < 2, OK

Origin: total
X 1.09 mils < 2, OK
Y 2.16 mils > 2
Z 0.55 mils < 2, OK

Note: The Y axis double amplitude is slightly high for soil 20% weaker than measured. The design may need to be adjusted accordingly.

Anchor Bolt Check:

use maximum and minimum gas pressures for head and crank pressures.

cylinder head area,

$$A_{\text{head}} = \pi (3.0 \text{ in})^2 / 4 = 7.07 \text{ in}^2$$

crank area,

$$A_{\text{crank}} = A_{\text{head}} - \pi (3.0 \text{ in})^2 / 4 = (7.07 \text{ in}^2) - \pi (0.625 \text{ in})^2 / 4 = 6.76 \text{ in}^2$$

cylinder force:

$$F_{\text{fdn}} = [(P_{\text{head}})(A_{\text{head}}) - (P_{\text{crank}})(A_{\text{crank}})] F_1 / F_{\text{red}} \quad (\text{Equation 14})$$

$$= [(6,200 \text{ psi})(7.07 \text{ in}^2) - (500 \text{ psi})(6.76 \text{ in}^2)] 1.15 / 2.0$$

$$= 23,261 \text{ lb}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation

bolt force,

$$F_{\text{bolt}} = (F_{\text{fdn}}) / N = (23,262 \text{ lb}) / (4 \text{ bolts}) = 5,815 \text{ lb}$$

bolt pre-tensioning,

(Equation 15)

$$T_{\text{min}} = F_{\text{bolt}} / \mu - W_r = (5,815 \text{ lb}) / 0.15 - (\text{say } 0 \text{ lb}) = 64,513 \text{ lb}$$

bolt tensile stress area, galvanized, $A_b = 1.41 \text{ in}^2$

(###.215.1207, Attachment 01)

limit pre-tensioning to 80% of yield for A36 bolt material,

$$T_{\text{allow}} = 0.8 (F_y)(A_b) = 0.8 (36,000 \text{ psi})(1.41 \text{ in}^2) = 40,608 \text{ lb} > T_{\text{min}}, \text{ OK}$$

Note: For brevity, the anchor bolt embedment check is not shown here. See Guideline 000.215.1207, Anchor Bolt Design Criteria, for details.

Mat Reinforcing Design:

$$\text{mat cantilever} = [(26 \text{ ft mat}) - (9 \text{ ft motor pier})] / 2 = 8.5 \text{ ft}$$

By Inspection, an 8.5 ft cantilever and a 2 ft thick mat will not require more than nominal reinforcing.

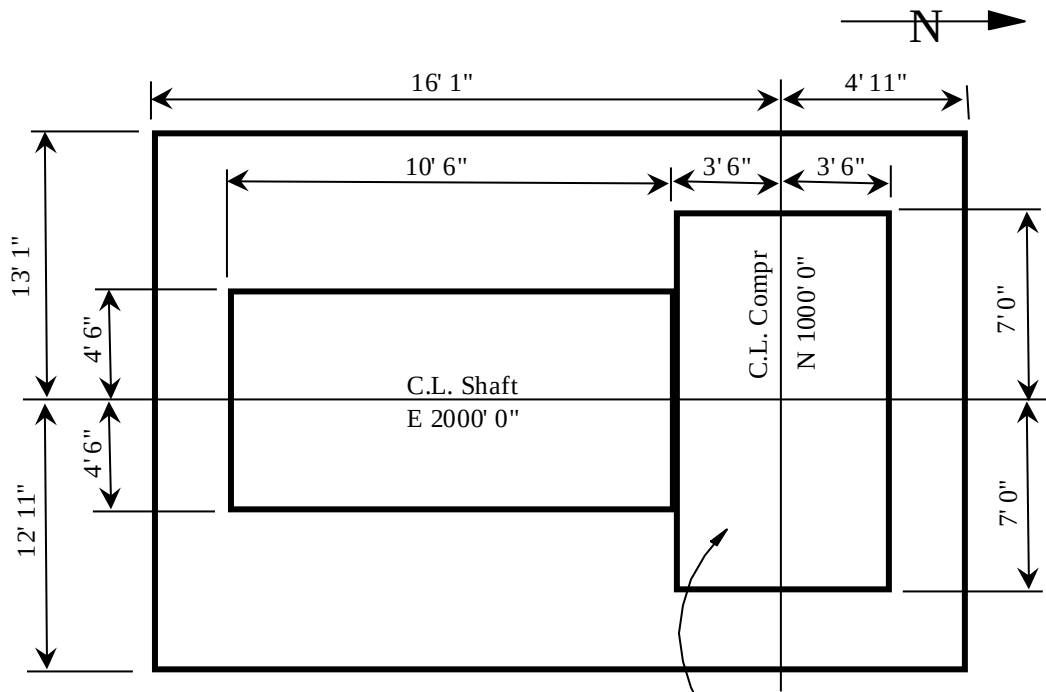
$$A_s = (0.0018)(24 \text{ in depth})(12 \text{ in unit width}) = 0.52 \text{ in}^2$$

$$\text{USE \#6 at 12 inches each way, top and bottom} \quad A_s (\text{provided}) = 2 (0.44 \text{ in}^2) = 0.88 \text{ in}^2$$



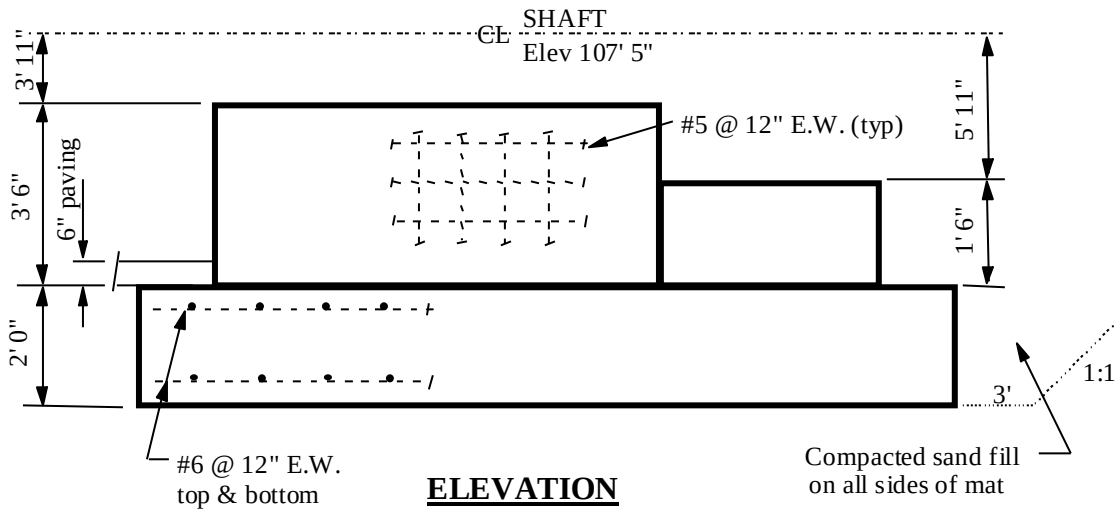
VIBRATING MACHINERY FOUNDATIONS ON SOIL

Sample Design: Reciprocating Machine Foundation



PLAN

Designer to locate anchor bolts and pier details per vendor drawing dated 4-1-90



ELEVATION



VIBRATING MACHINERY FOUNDATIONS ON SOIL

SVAP Analysis, Run #1 (Soil Properties as Given)

The current version of SVAP uses a slightly different method in the computation of base and embedment impedance from that described in Attachment 10. Results are not exact.

FLUOR	Date: 1/4/00
SVAP (rev: 2.1h, 1/6/99)	Contract: 123456
Technical Guideline, Sample Design	By: Bounds
Soil Properties as Given	Page: 1

Machine Forces:

Machine Data-	Location of Forces-
Reciprocating machine.	N/S coordinate = N 10.000 ft
Speed = 300 rpm	E/W coordinate = E 0.000 ft
	Elevation = 107.420 ft
Direction-	Primary- Secondary-
N/S force	1.00 K 0.00 K
E/W force	1.50 K 0.80 K
Vertical force	0.88 K 0.00 K
Moment about N/S axis	2.10 K-ft 0.90 K-ft
Moment about E/W axis	0.20 K-ft 0.00 K-ft
Torsional moment	3.10 K-ft 1.50 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec2-ft)
motor	N 0.920	E 0.000	106.670	15.000	0 0 0
compressor	N 9.667	W 0.500	106.330	20.000	0 0 0

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
point #4	N 13.500	E 7.000	101.500
point #3	N 10.000	E 0.000	107.420
origin	N 0.000	E 0.000	107.420

Foundation Data:

- * Foundation geometry will not be checked by the analysis.
- * Group stiffness and damping will be computed from soil layers.
- * The mat will be located at the foundation center of gravity.
- * Embedment effects will be computed.

Grade elevation	=	100.50 ft
Effective embedment height	=	1.33 ft
Backfill density	=	120.00 lb/ft3
Poisson's ratio	=	0.35
Stiffness variation	=	1.00
Damping variation	=	1.00
Internal damping	=	5.00 %



VIBRATING MACHINERY FOUNDATIONS ON SOIL

SVAP Analysis, Run #1 (Soil Properties as Given)

FLUOR

Date: 1/4/00

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Contract: 123456

By: Bounds

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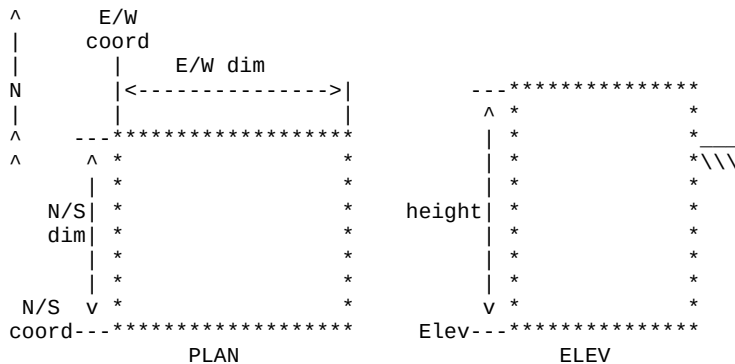
Technical Guideline, Sample Design

Soil Properties as Given

Soil Layer Data

Top elev: (ft)	Thickness: (ft)	Density: (lb/ft3)	Shear modulus: (ksf)	Poisson's ratio:
100.500	100.000	110.000	720.000	0.440

Typical Pier & Mat dimensioning:



Mat Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Mat	21.000	26.000	2.500	N 0.000	E 0.000	98.000

Pier Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
Motor Pier	10.500	9.000	3.000	S 4.000	W 4.500	100.500
Comp. Pier	7.000	14.000	1.000	N 6.500	W 7.000	100.500

Results of mass computation:

Center of gravity:	Weight & Mass:
N/S coordinate = N 4.416 ft	Total weight = 296.98 K
E/W coordinate = W 0.108 ft	Machine weight = 35.00 K
elevation = 100.582 ft	Mass = 9.22 K-sec2/ft
	Mass ratio = 7.49 to 1
Footing data:	Mass moments @ CG:
Max gross soil brg = 0.544 ksf	about N/S axis = 877 K-sec2-ft
Min gross soil brg = 0.544 ksf	about E/W axis = 681 K-sec2-ft
	about Vert axis = 673 K-sec2-ft
Mat relocated south = 6.084 ft	
Mat relocated west = 13.108 ft	



VIBRATING MACHINERY FOUNDATIONS ON SOIL

SVAP Analysis, Run #1 (Soil Properties as Given)

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Soil Properties as Given

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Base Stiffness Results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	48878 K/ft	60 %
E/W translation	48878 K/ft	60 %
vertical	67799 K/ft	105 %
about N/S axis	9544746 K-ft/rad	44 %
about E/W axis	6928405 K-ft/rad	37 %
torsional	9263812 K-ft/rad	26 %

Embedment stiffness results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	4475 K/ft	16 %
E/W translation	4475 K/ft	16 %
vertical	2668 K/ft	11 %
about N/S axis	706534 K-ft/rad	0 %
about E/W axis	571174 K-ft/rad	2 %
torsional	0 K-ft/rad	0 %

Total Stiffness Results:

Center of stiffness:	Rocking Correction Factors:
N/S coordinate = N 4.416 ft	Nr (about N/S) = 2.036
E/W coordinate = W 0.108 ft	Nr (about E/W) = 1.923
Elevation = 98.000 ft	

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	53353 K/ft	60 %
E/W translation	53353 K/ft	60 %
vertical	70467 K/ft	95 %
about N/S axis	10251279 K-ft/rad	40 %
about E/W axis	7499579 K-ft/rad	40 %
torsional	9263812 K-ft/rad	20 %

Results of modal analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	695	704	835	1047	1065	1121
Resonant freq (rpm)	1191	1234	O.D.	1282	1300	1168
Frequency ratio (p)	0.25	0.24	O.D.	0.23	0.23	0.26
(s)	0.50	0.49	O.D.	0.47	0.46	0.51
Dynamic Magnification (p)	1.05	1.05	0.90	1.06	1.05	1.07
(s)	0.98	0.97	0.69	1.22	1.22	1.34
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	1.00	-0.01	0.00
E/W trans (ft)	0.00	1.00	0.00	0.00	1.00	0.00
Vertical trans (ft)	0.00	0.00	1.00	0.00	0.00	0.00
about N/S axis (rad)	0.00	0.02	0.00	-0.00	-0.45	0.00
about E/W axis (rad)	-0.03	0.00	0.00	0.42	-0.00	0.00
Torsional axis (rad)	0.00	0.00	0.00	0.00	0.00	1.00



VIBRATING MACHINERY FOUNDATIONS ON SOIL

SVAP Analysis, Run #1 (Soil Properties as Given)

FLUOR

Date: 1/4/00

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By: Bounds

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Technical Guideline, Sample Design

Soil Properties as Given

Double Amplitudes at point #4

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.60	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.81	0.00
	(s)	0.00	0.40	0.00
Vert translation	(p)	0.00	0.00	0.27
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.04	0.28
	(s)	0.00	0.02	0.14
Rocking about E/W axis	(p)	0.04	0.00	0.44
	(s)	0.00	0.00	0.00
Torsional	(p)	0.23	0.29	0.00
	(s)	0.15	0.19	0.00
Total @ 300 rpm		1.02	1.75	1.14

Double Amplitudes at point #3

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.60	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.81	0.00
	(s)	0.00	0.40	0.00
Vert translation	(p)	0.00	0.00	0.27
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.27	0.00
	(s)	0.00	0.14	0.00
Rocking about E/W axis	(p)	0.33	0.00	0.27
	(s)	0.00	0.00	0.00
Torsional	(p)	0.00	0.18	0.00
	(s)	0.00	0.12	0.00
Total @ 300 rpm		0.94	1.92	0.55

Double Amplitudes at origin

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.60	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.81	0.00
	(s)	0.00	0.40	0.00
Vert translation	(p)	0.00	0.00	0.27
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.27	0.00
	(s)	0.00	0.14	0.00
Rocking about E/W axis	(p)	0.33	0.00	0.21
	(s)	0.00	0.00	0.00
Torsional	(p)	0.00	0.14	0.00
	(s)	0.00	0.09	0.00
Total @ 300 rpm		0.94	1.86	0.49



VIBRATING MACHINERY FOUNDATIONS ON SOIL

SVAP Analysis, Run #2 (Soil Properties 20% Stronger)

The current version of SVAP uses a slightly different method in the computation of base and embedment impedance from that described in Attachment 10. Results are not exact.

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Machine Forces:

Machine Data-	Location of Forces-	
Reciprocating machine.	N/S coordinate =	N 10.000 ft
Speed = 300 rpm	E/W coordinate =	E 0.000 ft
	Elevation =	107.420 ft
Direction-	Primary-	Secondary-
N/S force	1.00 K	0.00 K
E/W force	1.50 K	0.80 K
Vertical force	0.88 K	0.00 K
Moment about N/S axis	2.10 K-ft	0.90 K-ft
Moment about E/W axis	0.20 K-ft	0.00 K-ft
Torsional moment	3.10 K-ft	1.50 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec2-ft)			
motor	N 0.920	E 0.000	106.670	15.000	0	0	0	
compressor	N 9.667	W 0.500	106.330	20.000	0	0	0	

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
point #4	N 13.500	E 7.000	101.500
point #3	N 10.000	E 0.000	107.420
origin	N 0.000	E 0.000	107.420

Foundation Data:

- * Foundation geometry will not be checked by the analysis.
- * Group stiffness and damping will be computed from soil layers.
- * The mat will be located at the foundation center of gravity.
- * Embedment effects will be computed.

Grade elevation	=	100.50 ft
Effective embedment height	=	1.33 ft
Backfill density	=	120.00 lb/ft3
Poisson's ratio	=	0.35
Stiffness variation	=	1.00
Damping variation	=	1.00
Internal damping	=	5.00 %



VIBRATING MACHINERY FOUNDATIONS ON SOIL

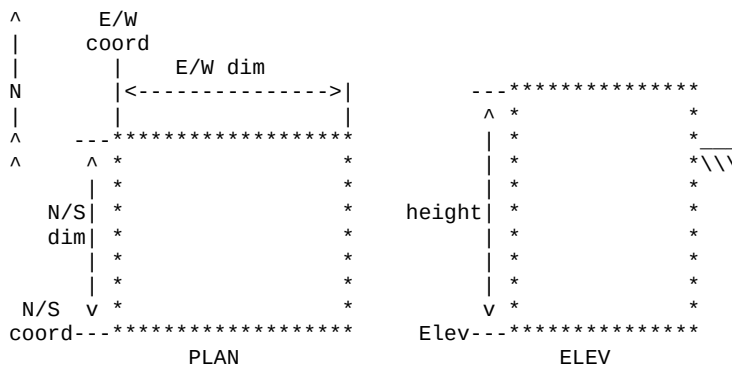
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Soil Layer Data

Top elev:	Thickness:	Density:	Shear modulus:	Poisson's ratio:
(ft)	(ft)	(lb/ft ³)	(ksf)	
100.500	100.000	110.000	864.000	0.370

Typical Pier & Mat dimensioning:



Mat Data:

Name:	N/S dim:	E/W dim:	Height:	N/S coord:	E/W coord:	Elev:
	(ft)	(ft)	(ft)	(ft)	(ft)	(ft)
The Mat	21.000	26.000	2.500	N 0.000	E 0.000	98.000

Pier Data:

Name:	N/S dim:	E/W dim:	Height:	N/S coord:	E/W coord:	Elev:
	(ft)	(ft)	(ft)	(ft)	(ft)	(ft)
Motor Pier	10.500	9.000	3.000	S 4.000	W 4.500	100.500
Comp. Pier	7.000	14.000	1.000	N 6.500	W 7.000	100.500

Results of mass computation:

Center of gravity:	Weight & Mass:
N/S coordinate = N 4.416 ft	Total weight = 296.98 K
E/W coordinate = W 0.108 ft	Machine weight = 35.00 K
elevation = 100.582 ft	Mass = 9.22 K-sec ² /ft
	Mass ratio = 7.49 to 1
Footing data:	Mass moments @ CG:
Max gross soil brg = 0.544 ksf	about N/S axis = 860 K-sec ² -ft
Min gross soil brg = 0.544 ksf	about E/W axis = 668 K-sec ² -ft
	about Vert axis = 673 K-sec ² -ft
Mat relocated south = 6.084 ft	
Mat relocated west = 13.108 ft	



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SVAP Analysis, Run #2 (Soil Properties 20% Stronger)

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Technical Guideline, Sample Design

Soil Properties 20% Stronger

Base Stiffness Results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	56839 K/ft	59 %
E/W translation	56839 K/ft	59 %
vertical	72319 K/ft	99 %
about N/S axis	10181062 K-ft/rad	41 %
about E/W axis	7390299 K-ft/rad	35 %
torsional	11116575 K-ft/rad	26 %

Embedment stiffness results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	5204 K/ft	15 %
E/W translation	5204 K/ft	15 %
vertical	2845 K/ft	10 %
about N/S axis	753636 K-ft/rad	0 %
about E/W axis	609252 K-ft/rad	2 %
torsional	0 K-ft/rad	0 %

Total Stiffness Results:

Center of stiffness:	Rocking Correction Factors:
N/S coordinate = N 4.416 ft	Nr (about N/S) = 1.996
E/W coordinate = W 0.108 ft	Nr (about E/W) = 1.885
Elevation = 98.000 ft	

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	62043 K/ft	60 %
E/W translation	62043 K/ft	60 %
vertical	75165 K/ft	95 %
about N/S axis	10934698 K-ft/rad	40 %
about E/W axis	7999551 K-ft/rad	40 %
torsional	11116575 K-ft/rad	20 %

Results of modal analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	745	756	862	1099	1116	1228
Resonant freq (rpm)	1260	1309	O.D.	1349	1365	1280
Frequency ratio (p)	0.24	0.23	O.D.	0.22	0.22	0.23
(s)	0.48	0.46	O.D.	0.44	0.44	0.47
Dynamic Magnification (p)	1.05	1.04	0.91	1.05	1.05	1.06
(s)	1.02	1.01	0.70	1.20	1.20	1.27
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	1.00	-0.01	0.00
E/W trans (ft)	0.00	1.00	0.00	0.01	1.00	0.00
Vertical trans (ft)	0.00	0.00	1.00	0.00	0.00	0.00
about N/S axis (rad)	0.00	0.03	0.00	-0.00	-0.40	0.00
about E/W axis (rad)	-0.04	0.00	0.00	0.38	-0.00	0.00
Torsional axis (rad)	0.00	0.00	0.00	0.00	0.00	1.00



VIBRATING MACHINERY FOUNDATIONS ON SOIL

SVAP Analysis, Run #2 (Soil Properties 20% Stronger)

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Soil Properties 20% Stronger

Double Amplitudes at point #4

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.53	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.70	0.00
	(s)	0.00	0.36	0.00
Vert translation	(p)	0.00	0.00	0.25
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.03	0.26
	(s)	0.00	0.02	0.14
Rocking about E/W axis	(p)	0.04	0.00	0.41
	(s)	0.00	0.00	0.00
Torsional	(p)	0.19	0.24	0.00
	(s)	0.12	0.15	0.00
Total @ 300 rpm		0.87	1.51	1.07

Double Amplitudes at point #3

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.53	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.70	0.00
	(s)	0.00	0.36	0.00
Vert translation	(p)	0.00	0.00	0.25
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.25	0.00
	(s)	0.00	0.13	0.00
Rocking about E/W axis	(p)	0.31	0.00	0.25
	(s)	0.00	0.00	0.00
Torsional	(p)	0.00	0.15	0.00
	(s)	0.00	0.09	0.00
Total @ 300 rpm		0.84	1.69	0.51

Double Amplitudes at origin

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.53	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.70	0.00
	(s)	0.00	0.36	0.00
Vert translation	(p)	0.00	0.00	0.25
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.25	0.00
	(s)	0.00	0.13	0.00
Rocking about E/W axis	(p)	0.31	0.00	0.20
	(s)	0.00	0.00	0.00
Torsional	(p)	0.00	0.12	0.00
	(s)	0.00	0.07	0.00
Total @ 300 rpm		0.84	1.64	0.46



Vibrating Machinery Foundations on Soil

SVAP Analysis, Run #3 (Soil Properties 20% Weaker)

The current version of SVAP uses a slightly different method in the computation of base and embedment impedance from that described in Attachment 10. Results are not exact.

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Machine Forces:

Machine Data-	Location of Forces-	
Reciprocating machine.	N/S coordinate =	N 10.000 ft
Speed = 300 rpm	E/W coordinate =	E 0.000 ft
	Elevation =	107.420 ft
Direction-	Primary-	Secondary-
N/S force	1.00 K	0.00 K
E/W force	1.50 K	0.80 K
Vertical force	0.88 K	0.00 K
Moment about N/S axis	2.10 K-ft	0.90 K-ft
Moment about E/W axis	0.20 K-ft	0.00 K-ft
Torsional moment	3.10 K-ft	1.50 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec2-ft)			
motor	N 0.920	E 0.000	106.670	15.000	0	0	0	
compressor	N 9.667	W 0.500	106.330	20.000	0	0	0	

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
point #4	N 13.500	E 7.000	101.500
point #3	N 10.000	E 0.000	107.420
origin	N 0.000	E 0.000	107.420

Foundation Data:

- * Foundation geometry will not be checked by the analysis.
- * Group stiffness and damping will be computed from soil layers.
- * The mat will be located at the foundation center of gravity.
- * Embedment effects will be computed.

Grade elevation	=	100.50 ft
Effective embedment height	=	1.33 ft
Backfill density	=	120.00 lb/ft3
Poisson's ratio	=	0.35
Stiffness variation	=	1.00
Damping variation	=	1.00
Internal damping	=	5.00 %



Vibrating Machinery Foundations on Soil

SVAP Analysis, Run #3 (Soil Properties 20% Weaker)

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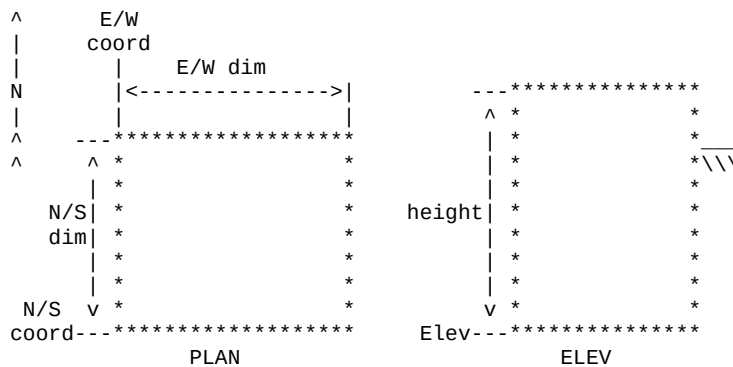
Technical Guideline, Sample Design

Soil Properties 20% Weaker

Soil Layer Data

Top elev: (ft)	Thickness: (ft)	Density: (lb/ft ³)	Shear modulus: (ksf)	Poisson's ratio:
100.500	100.000	110.000	600.000	0.470

Typical Pier & Mat dimensioning:



Mat Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
The Mat	21.000	26.000	2.500	N 0.000	E 0.000	98.000

Pier Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
Motor Pier	10.500	9.000	3.000	S 4.000	W 4.500	100.500
Comp. Pier	7.000	14.000	1.000	N 6.500	W 7.000	100.500

Results of mass computation:

Center of gravity:	Weight & Mass:
N/S coordinate = N 4.416 ft	Total weight = 296.98 K
E/W coordinate = W 0.108 ft	Machine weight = 35.00 K
elevation = 100.582 ft	Mass = 9.22 K-sec ² /ft
	Mass ratio = 7.49 to 1
Footing data:	Mass moments @ CG:
Max gross soil brg = 0.544 ksf	about N/S axis = 885 K-sec ² -ft
Min gross soil brg = 0.544 ksf	about E/W axis = 687 K-sec ² -ft
	about Vert axis = 673 K-sec ² -ft
Mat relocated south 6.084 ft	
Mat relocated west 13.108 ft	



Vibrating Machinery Foundations on Soil

SVAP Analysis, Run #3 (Soil Properties 20% Weaker)

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Base Stiffness Results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	41405 K/ft	61 %
E/W translation	41405 K/ft	61 %
vertical	59698 K/ft	108 %
about N/S axis	8404179 K-ft/rad	45 %
about E/W axis	6100483 K-ft/rad	38 %
torsional	7719844 K-ft/rad	26 %

Embedment stiffness results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	3791 K/ft	16 %
E/W translation	3791 K/ft	16 %
vertical	2349 K/ft	11 %
about N/S axis	622105 K-ft/rad	0 %
about E/W axis	502920 K-ft/rad	2 %
torsional	0 K-ft/rad	0 %

Total Stiffness Results:

Center of stiffness:	Rocking Correction Factors:
N/S coordinate = N 4.416 ft	Nr (about N/S) = 2.055
E/W coordinate = W 0.108 ft	Nr (about E/W) = 1.941
Elevation = 98.000 ft	

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	45196 K/ft	60 %
E/W translation	45196 K/ft	60 %
vertical	62046 K/ft	95 %
about N/S axis	9026284 K-ft/rad	40 %
about E/W axis	6603403 K-ft/rad	40 %
torsional	7719844 K-ft/rad	20 %

Results of modal analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	641	649	783	975	993	1023
Resonant freq (rpm)	1105	1143	0.D.	1193	1211	1067
Frequency ratio (p)	0.27	0.26	0.D.	0.25	0.25	0.28
(s)	0.54	0.53	0.D.	0.50	0.50	0.56
Dynamic Magnification (p)	1.05	1.05	0.89	1.06	1.06	1.09
(s)	0.92	0.92	0.66	1.25	1.25	1.44
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	1.00	-0.01	0.00
E/W trans (ft)	0.00	1.00	0.00	0.00	1.00	0.00
Vertical trans (ft)	0.00	0.00	1.00	0.00	0.00	0.00
about N/S axis (rad)	0.00	0.02	0.00	-0.00	-0.47	0.00
about E/W axis (rad)	-0.03	0.00	0.00	0.44	-0.00	0.00
Torsional axis (rad)	0.00	0.00	0.00	0.00	0.00	1.00



Vibrating Machinery Foundations on Soil

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Soil Properties 20% Weaker

Double Amplitudes at point #4

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.71	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.96	0.00
	(s)	0.00	0.45	0.00
Vert translation	(p)	0.00	0.00	0.30
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.04	0.32
	(s)	0.00	0.02	0.16
Rocking about E/W axis	(p)	0.05	0.00	0.50
	(s)	0.00	0.00	0.00
Torsional	(p)	0.28	0.35	0.00
	(s)	0.19	0.24	0.00
		-----	-----	-----
Total @	300 rpm	1.23	2.06	1.29

Double Amplitudes at point #3

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.71	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.96	0.00
	(s)	0.00	0.45	0.00
Vert translation	(p)	0.00	0.00	0.30
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.31	0.00
	(s)	0.00	0.15	0.00
Rocking about E/W axis	(p)	0.38	0.00	0.31
	(s)	0.00	0.00	0.00
Torsional	(p)	0.00	0.22	0.00
	(s)	0.00	0.15	0.00
		-----	-----	-----
Total @	300 rpm	1.09	2.23	0.62

Double Amplitudes at origin

Case:		N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	(p)	0.71	0.00	0.00
	(s)	0.00	0.00	0.00
E/W translation	(p)	0.00	0.96	0.00
	(s)	0.00	0.45	0.00
Vert translation	(p)	0.00	0.00	0.30
	(s)	0.00	0.00	0.00
Rocking about N/S axis	(p)	0.00	0.31	0.00
	(s)	0.00	0.15	0.00
Rocking about E/W axis	(p)	0.38	0.00	0.24
	(s)	0.00	0.00	0.00
Torsional	(p)	0.00	0.17	0.00
	(s)	0.00	0.12	0.00
		-----	-----	-----
Total @	300 rpm	1.09	2.16	0.55



VIBRATING MACHINERY FOUNDATIONS ON SOIL

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VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

A General Considerations

A1 The Equation of Motion

- A1a The vibration analysis of compressor foundations involves solving the following standard dynamic equation: (Several basic dynamics references are listed in Section K1.)

$$M y'' + C_t y' + K_t y = F(t) \quad \text{for translation} \quad (\text{Equation 10-1a})$$

$$I \phi'' + C_r \phi' + K_r \phi = M(t) \quad \text{for rocking} \quad (\text{Equation 10-1b})$$

where,

- M = mass (K-sec²/ft)
- I = mass moment of inertia (K-sec²-ft)
- C_t = damping constant (K-sec/ft)
- C_r = rotational damping constant (ft-K-sec/rad)
- K_t = stiffness (K/ft)
- K_r = rotational stiffness (ft-K/rad)
- F(t) = dynamic force (kips)
- M(t) = dynamic moment (ft-K)
- y'' = acceleration (ft/sec²)
- y' = velocity (ft/sec)
- y = displacement (feet)
- φ'' = rotational acceleration (rad/sec²)
- φ' = rotational velocity (rad/sec)
- φ = rotational displacement (radians)

- A1b Three of each of the above equations are needed to describe the 3 translation and 3 rocking modes of vibration. Forces are determined at the foundation-machine-soil center of gravity.
- A1c The first term, mass times acceleration, is the inertia force. Details of mass computations are provided in Section B.
- A1d The next two terms, damping times velocity and stiffness times displacement, represent forces from the soil and are also termed the soil impedance. Details of the impedance calculation are provided in Sections D and E.
- A1e The dynamic force is normally a combination of sinusoidal functions. This force is preferably provided by the machine supplier. Alternately, standard criteria can be used as described in the main guideline.

A2 Equation Solutions

- A2a Natural Frequencies - The use of frequency dependent impedance implies that natural frequencies cannot be directly determined. The method indicated in Sections G5 and H5 use an iteration where a trial frequency is assumed, the impedance is determined, and a resulting frequency is computed from undamped conditions. The correct natural frequency is found when the trial and resulting frequencies match.
- A2b Forced response - The equations for acceleration, velocity, and displacement provided in Sections G and H



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

are derived using the Complex Method (References 2d and 4c). This method is suitable for either frequency dependent or independent soil impedance. It is also a direct solution in that natural frequencies and mode shapes are not a part of the solution sequence. Details of this solution method are provided in Sections G and H.

A3 Solution Notes

- A3a The solution method provided in this attachment takes advantage of independent vibration modes. Instead of having to solve 6 coupled equations, a decoupling takes place when the system center of gravity is directly above the mat centerlines. With this decoupling, vertical translation and torsional rocking are solved as single degree of freedom modes. Translation and rocking in each horizontal direction are solved as coupled modes with 2 degrees of freedom. The alignment of the center of gravity over the mat centerline is computed in Section B and checked according to criteria provided in the main guideline.
- A3b A rigid foundation is assumed. The 6 vibration modes described above and the soil impedance criteria presented in this attachment depend on a rigid foundation. Otherwise, a finite element analysis modeling the machine, pier, mat, and possibly the soil may be required for an accurate analysis. Criteria for checking the mat rigidity is presented in the main guideline.
- A3c Phase angles are neglected in this procedure. The direction of reciprocating machine forces are usually known, but may be subject to change at a later date. The direction of centrifugal machine forces cannot be predicted, normally worst case orientations are used. Amplitudes are conservatively computed in this procedure by adding peak values from each mode.
- A3d A reference coordinate system is required. Any reasonably located origin will do. An origin at the centerline and base of the foundation mat is often used. An alternate would be to use an origin at the bottom of baseplate in order to minimize location changes when mat or pier sizes are revised. In this technical attachment, an X axis parallel to the shaft, a Y axis perpendicular to the shaft, and a vertical Z axis will be utilized as shown in Attachment 11.

A4 Software

Computer programs are available which are capable of performing most or all of the calculations described in this attachment. Though some time is required to input data and to print results, the analysis usually requires only a matter of seconds. Since performing these calculations by hand is quite tedious and time consuming, use of a computer is strongly recommended. Several computer programs are listed in the reference section. It should be noted that exact solution methods will vary for particular programs.

B Mass Calculations

- B1 Mass properties can be calculated using standard equations. In the following procedure, the machine weight, total weight, mass moments, and center of gravity are computed simultaneously from the reference origin described in Section A10. The vibration calculation (refer to Attachment 15) uses a slight variation in order to locate the mat under the center of gravity.

TW	= Foundation, machine, and soil weight (kips)
MW	= Machine weight (kips)
I_x, I_y, I_z	= Mass moments of inertia, about each axis ($K\text{-sec}^2\text{-ft}$)



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Analysis Procedure

CG_x, CG_y, CG_z = Center of gravity location with respect to the reference origin (ft)

It is assumed that the machine and foundation are symmetric enough to ignore cross mass moments of inertia. Connected pipe could contribute to the mass if it vibrates rigidly with the machine. However, connected pipe is usually insignificant compared to the total machine foundation mass.

- B2 The concrete foundation, soil, and the machine are divided into a series of components using the following definitions. Attachment 11 provides a pictorial definition of the dimensional terms.

CW = Weight of component (kips)
 $\partial_x, \partial_y, \partial_z$ = Distance from the origin to the component's center in each axis (ft)
 L_x, L_y, L_z = Dimensions of the component in each axis (ft)
 E_x, E_y, E_z = Component weight times distance from origin (ft-K)
 Q_x, Q_y, Q_z = Component mass moment of inertia, about component CG (K-sec²-ft)
 J_x, J_y, J_z = Component's translated mass moment of inertia (K-sec²-ft)
 W_c = Density of concrete (K/ft³)
 W_f = Density of compacted fill material (K/ft³)
 gravity = Acceleration of gravity (32.2 ft/sec²)

- B3 The equations given in the following sections for component properties are standard formulae for rectangular blocks. Nonrectangular blocks can be used if the proper equations for weight (CW) and mass moment (Q_x, Q_y, Q_z) are substituted. Attachment 12 provides some criteria for nonrectangular blocks.

- B4 The concrete foundation and soil above the mat are divided into a series of components with the following properties and calculated values:

$CW = (L_x)(L_y)(L_z)(W_c)$	Concrete components	(Equation 10-2a)
$CW = (L_x)(L_y)(L_z)(W_f)$	Soil components	(Equation 10-2b)

$E_x = CW (\partial_x)$	(Equation 10-3a)
-------------------------	------------------

$E_y = CW (\partial_y)$	(Equation 10-3b)
-------------------------	------------------

$E_z = CW (\partial_z)$	(Equation 10-3c)
-------------------------	------------------

$J_x = CW [\partial_y^2 + \partial_z^2]$	(Equation 10-4a)
--	------------------

$J_y = CW [\partial_x^2 + \partial_z^2]$	(Equation 10-4b)
--	------------------

$J_z = CW [\partial_x^2 + \partial_y^2]$	(Equation 10-4c)
--	------------------

$Q_x = CW [L_y^2 + L_z^2] / 12$	(Equation 10-5a)
---------------------------------	------------------

$Q_y = CW [L_x^2 + L_z^2] / 12$	(Equation 10-5b)
---------------------------------	------------------

$Q_z = CW [L_x^2 + L_y^2] / 12$	(Equation 10-5c)
---------------------------------	------------------

- B5 Machine components are now included with the following properties:

CW, Q_x, Q_y, Q_z = provided by the machine supplier

Equations 10-3a- c and 10-4a-c are used to compute needed properties about the origin.



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

B6 The center of gravity can now be computed by summing individual computed values:

$$\begin{aligned} MW &= \sum (CW) && \text{Machine components only} \\ TW &= \sum (CW) && \text{Concrete, soil, and machine components} \end{aligned}$$

$$CG_x = \sum (E_x) / TW \quad (\text{Equation 10-6a})$$

$$CG_y = \sum (E_y) / TW \quad (\text{Equation 10-6b})$$

$$CG_z = \sum (E_z) / TW \quad (\text{Equation 10-6c})$$

B7 The mass moments are computed by summing individual computed values, adjusting for the actual center of gravity, and dividing by gravity.

$$I_x = [\sum (Q_x) + \sum (J_x) - TW (CG_y^2 + CG_z^2)] / \text{gravity} \quad (\text{Equation 10-7a})$$

$$I_y = [\sum (Q_y) + \sum (J_y) - TW (CG_x^2 + CG_z^2)] / \text{gravity} \quad (\text{Equation 10-7b})$$

$$I_z = [\sum (Q_z) + \sum (J_z) - TW (CG_x^2 + CG_y^2)] / \text{gravity} \quad (\text{Equation 10-7c})$$

B8 The mass and mass ratio can now be computed. The mass ratio is compared using the acceptance criteria given in the main guideline.

$$\begin{aligned} \text{mass} &= \text{Translational mass (K-sec}^2/\text{ft)} \\ \text{MR} &= \text{Mass ratio} \end{aligned}$$

$$\text{mass} = TW / \text{gravity} \quad (\text{Equation 10-8})$$

$$\text{MR} = (TW - MW) / MW \quad (\text{Equation 10-9})$$

B9 Alignment offsets are now computed to confirm the assumptions of Section A6. Comparisons are made using the acceptance criteria provided in the main guideline.

$$OS_x, OS_y = \text{Center of gravity offset in each direction (percent)}$$

$$\partial_x, \partial_y = \text{Distance from origin to center of mat (ft)}$$

$$L_x, L_y = \text{Mat dimensions (ft)}$$

$$OS_x = \text{abs} \{ 100 [CG_x - \partial_x] / L \} \quad (\text{Equation 10-10a})$$

$$OS_y = \text{abs} \{ 100 [CG_y - \partial_y] / L \} \quad (\text{Equation 10-10b})$$

C Mat and Soil Properties

C1 The net soil bearing must be kept low to assure an elastic and predictable soil response. Net soil bearing will be computed and compared using the acceptance criteria provided in the main guideline (refer to Attachment 11).

$$SB_{\text{net}} = \text{Maximum net static soil bearing (K/ft}^2\text{)}$$

$$\text{GradeHt} = \text{Height from origin to grade (ft)}$$

$$\text{MatHt} = \text{Height from origin to bottom of mat (ft)}$$

$$W_s = \text{Density of soil (K/ft}^3\text{)}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$$SB_{net} = [TW / (L_x)(L_y)] - WS (MatHt - GradeHt) \quad (\text{Equation 10-11})$$

- C2 Equivalent mat radii will be used to compute the base and embedment impedances defined in Sections D and E. If soil impedance is to be computed using alternate criteria, these radii may not be needed. These formulae provide accurate results for length to width ratios from 0.25 to 4.

- R_t = Equivalent mat radius for translation (ft)
 R_x, R_y, R_z = Equivalent mat radii for rotation, about each axis (ft)
 L_x = Mat dimension along X axis (ft)
 L_y = Mat dimension along Y axis (ft)

$$R_t = \sqrt{(L_x)(L_y)} / \pi \quad (\text{Equation 10-12a})$$

$$R_x = \sqrt[4]{(L_x)(L_y)^3 / 3 \pi} \quad (\text{Equation 10-12b})$$

$$R_y = \sqrt[4]{(L_x)^3 (L_y) / 3 \pi} \quad (\text{Equation 10-12c})$$

$$R_z = \sqrt[4]{(L_x)(L_y)[(L_x)^2 + (L_y)^2] / 6 \pi} \quad (\text{Equation 10-12d})$$

D Soil Impedance at Base of Mat

- D1 For the majority of cases, uniform soil conditions will be encountered. The impedance equations, from Veletsos, provided in this section are valid for uniform soil conditions. Only very modest variations in soil properties can be accommodated. Attachment 13 is provided for use in hand calculations as an aid to the listed equations.

Impedance results for nonuniform conditions vary widely. Because variations with frequency also change, effective soil properties should not be used to approximate layered soil. There are 2 options for obtaining accurate impedance values:

- From the soil consultant: - This is the preferred option if the consultant is experienced at the evaluation of dynamic soil impedance. The soil consultant should be more knowledgeable of site soil conditions and may have specialized software or evaluation methods.
- From alternate criteria: - Numerous reports have been published to study soil impedance under a variety of situations. Many contain a summary of results, simple equations, or tables for application to similar conditions. Section K3, Base Impedance, includes references for nonuniform soil conditions.

- D2 Apply Veletsos equations (References 3d and 3e) to compute stiffness and damping at the bottom of mat:

- $K_{tx}^c, K_{ty}^c, K_{tz}^c$ = Base translational stiffness (K/ft)
 $K_{rx}^c, K_{ry}^c, K_{rz}^c$ = Base rotational stiffness, about each axis (ft-K/rad)
 $C_{tx}^c, C_{ty}^c, C_{tz}^c$ = Base translational damping (K-sec/ft)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$C_{rx}^c, C_{ry}^c, C_{rz}^c$ = Base rotational damping, about each axis (ft-K-sec/rad)

D3 Static stiffnesses are computed from the following formulae.

$K_{tx}^s, K_{ty}^s, K_{tz}^s$ = Static translational stiffness (K/ft)

$K_{rx}^s, K_{ry}^s, K_{rz}^s$ = Static rotational stiffness, about each axis (ft-K/rad)

G_s = Shear modulus of soil (K/ft²)

μ_s = Poisson's ratio for soil

$$K_{tx}^s = K_{ty}^s = 8(G_s)(R_t)/(2 - \mu_s) \quad (\text{Equation 10-13a})$$

$$K_{tz}^s = 4(G_s)(R_t)/(1 - \mu_s) \quad (\text{Equation 10-13b})$$

$$K_{rx}^s = 8(G_s)(R_x)^3/3(1 - \mu_s) \quad (\text{Equation 10-14a})$$

$$K_{ry}^s = 8(G_s)(R_y)^3/3(1 - \mu_s) \quad (\text{Equation 10-14b})$$

$$K_{rz}^s = 16(G_s)(R_z)^3/3 \quad (\text{Equation 10-14c})$$

D4 Dimensionless frequencies are used to compute the base impedance coefficients defined in the next section. A second set of dimensionless frequencies is required for reciprocating machines with secondary forces using twice the machine speed.

A_t = Dimensionless frequency for translation

A_x, A_y, A_z = Dimensionless frequency for rotation

MS = Machine speed (rad/sec)

$$A_t = (MS)(R_t) \sqrt{W_s/(G_s)(\text{gravity})} \quad (\text{Equation 10-15a})$$

$$A_x = (MS)(R_x) \sqrt{W_s/(G_s)(\text{gravity})} \quad (\text{Equation 10-15b})$$

$$A_y = (MS)(R_y) \sqrt{W_s/(G_s)(\text{gravity})} \quad (\text{Equation 10-15c})$$

$$A_z = (MS)(R_z) \sqrt{W_s/(G_s)(\text{gravity})} \quad (\text{Equation 10-15d})$$

D5 Stiffness and damping coefficients are determined next. Coefficients may be computed from the following criteria or may be taken directly from the precomputed coefficients provided in Attachment 13. A second set of coefficients and resulting impedance must be determined for reciprocating machines with secondary forces.

$S_{tx}^c, S_{ty}^c, S_{tz}^c$ = Base translation stiffness coefficient



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$S_{rx}^c, S_{ry}^c, S_{rz}^c$ = Base rotational stiffness coefficient

$N_{tx}^c, N_{ty}^c, N_{tz}^c$ = Base translation damping coefficient

$N_{rx}^c, N_{ry}^c, N_{rz}^c$ = Base rotational damping coefficient

Determine variables b_1 through b_4 for each mode of vibration from the following; interpolation will probably be required:

Poisson's Ratio = 0	Vertical	Horizontal	Rocking	Torsion
b_1	0.250	0.000	0.525	0.425
b_2	1.000	0.000	0.800	0.687
b_3	0.000	0.000	0.000	0.000
b_4	0.850	0.775	0.000	0.000
Poisson's Ratio = 0.33	Vertical	Horizontal	Rocking	Torsion
b_1	0.350	0.000	0.500	0.425
b_2	0.800	0.000	0.800	0.687
b_3	0.000	0.000	0.000	0.000
b_4	0.750	0.650	0.000	0.000
Poisson's Ratio = 0.5	Vertical	Horizontal	Rocking	Torsion
b_1	0.000	0.000	0.400	0.425
b_2	0.000	0.000	0.800	0.687
b_3	0.170	0.000	0.027	0.000
b_4	0.850	0.600	0.000	0.000

Compute the stiffness and damping coefficients from common formulae using the variables determined above and the applicable dimensionless frequency.

$$S_{tx}^c, S_{ty}^c, S_{tz}^c = 1 - \frac{b_1 [(b_2)(A_t)]^2}{1 + [(b_2)(A_t)]^2} - (b_3)(A_t)^2 \quad (\text{Equation 10-16a})$$

$$S_{rx}^c = 1 - \frac{b_1 [(b_2)(A_x)]^2}{1 + [(b_2)(A_x)]^2} - (b_3)(A_x)^2 \quad (\text{Equation 10-16b})$$

$$S_{ry}^c = 1 - \frac{b_1 [(b_2)(A_y)]^2}{1 + [(b_2)(A_y)]^2} - (b_3)(A_y)^2 \quad (\text{Equation 10-16c})$$

$$S_{rz}^c = 1 - \frac{b_1 [(b_2)(A_z)]^2}{1 + [(b_2)(A_z)]^2} - (b_3)(A_z)^2 \quad (\text{Equation 10-16d})$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$$N_{tx}^c, N_{ty}^c, N_{tz}^c = b_4 + \frac{(b_1)(b_2)[(b_2)(A_t)]^2}{1 + [(b_2)(A_t)]^2} \quad (\text{Equation 10-17a})$$

$$N_{rx}^c = b_4 + \frac{(b_1)(b_2)[(b_2)(A_x)]^2}{1 + [(b_2)(A_x)]^2} \quad (\text{Equation 10-17b})$$

$$N_{ry}^c = b_4 + \frac{(b_1)(b_2)[(b_2)(A_y)]^2}{1 + [(b_2)(A_y)]^2} \quad (\text{Equation 10-17c})$$

$$N_{rz}^c = b_4 + \frac{(b_1)(b_2)[(b_2)(A_z)]^2}{1 + [(b_2)(A_z)]^2} \quad (\text{Equation 10-17d})$$

D6 The stiffness and damping values that make up the base soil impedance can now be computed from the following formulae:

$$K_{tx}^c = (S_{tx}^c)(K_{tx}^s) \quad (\text{Equation 10-18a})$$

$$K_{ty}^c = (S_{ty}^c)(K_{ty}^s) \quad (\text{Equation 10-18b})$$

$$K_{tz}^c = (S_{tz}^c)(K_{tz}^s) \quad (\text{Equation 10-18c})$$

$$K_{rx}^c = (S_{rx}^c)(K_{rx}^s) \quad (\text{Equation 10-19a})$$

$$K_{ry}^c = (S_{ry}^c)(K_{ry}^s) \quad (\text{Equation 10-19b})$$

$$K_{rz}^c = (S_{rz}^c)(K_{rz}^s) \quad (\text{Equation 10-19c})$$

$$C_{tx}^c = (N_{tx}^c)(K_{tx}^s)(A_t) / MS \quad (\text{Equation 10-20a})$$

$$C_{ty}^c = (N_{ty}^c)(K_{ty}^s)(A_t) / MS \quad (\text{Equation 10-20b})$$

$$C_{tz}^c = (N_{ty}^c)(K_{ty}^s)(A_t) / MS \quad (\text{Equation 10-20c})$$

$$C_{rx}^c = (N_{rx}^c)(K_{rx}^s)(A_x) / MS \quad (\text{Equation 10-21a})$$

$$C_{ry}^c = (N_{ry}^c)(K_{ry}^s)(A_y) / MS \quad (\text{Equation 10-21b})$$

$$C_{rz}^c = (N_{rz}^c)(K_{rz}^s)(A_z) / MS \quad (\text{Equation 10-21c})$$

E Soil Impedance Due to Mat Embedment

E1 The majority of published design criteria for embedment impedance is by Novak. Other published reports are available; however, none are as comprehensive. In the following listing of Novak's equations, simplified versions are not used due to a limited frequency range. Attachment 14 is provided for use in hand calculations as an aid to the listed equations.

E2 Apply Novak's equations (References 4a, 4b, and 4c) to compute stiffness and damping along the sides of the mat:

$$\begin{aligned} K_{tx}^e, K_{ty}^e, K_{tz}^e &= \text{Embedment translational stiffnesses (K/ft)} \\ K_{rx}^e, K_{ry}^e, K_{rz}^e &= \text{Embedment rotational stiffnesses, about each axis (ft-K/rad)} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$C_{tx}^e, C_{ty}^e, C_{tz}^e$ = Embedment translational damping (K-sec/ft)
 $C_{rx}^e, C_{ry}^e, C_{rz}^e$ = Embedment rotational damping, about each axis (ft-K-sec/rad)

- E3 An effective mat embedment depth should be used in order to compute a conservative value of stiffness and damping (refer to Attachment 03).

h = Effective mat embedment (ft)

$h = (2/3)(\text{underground thickness of mat})$ (Equation 10-22)

- E4 Dimensionless frequencies are used to compute the impedance coefficients defined in the next section. A second set of dimensionless frequencies is required for reciprocating machines with secondary forces using twice the machine speed.

G_f = Shear modulus of compacted fill material (K/ft²)

$A_t = (MS)(R_t)\sqrt{W_f / (G_f)(\text{gravity})}$ (Equation 10-23a)

$A_x = (MS)(R_x)\sqrt{W_f / (G_f)(\text{gravity})}$ (Equation 10-23b)

$A_y = (MS)(R_y)\sqrt{W_f / (G_f)(\text{gravity})}$ (Equation 10-23c)

$A_z = (MS)(R_z)\sqrt{W_f / (G_f)(\text{gravity})}$ (Equation 10-23d)

- E5 Stiffness and damping coefficients are determined next. Coefficients may be computed from the following criteria or may be taken directly from the precomputed coefficients provided in Attachment 14. Since Bessel functions and complex arithmetic are involved, direct calculations are not recommended. A second set of coefficients and resulting impedance must be determined for reciprocating machines with secondary forces

$S_{tx}^e, S_{ty}^e, S_{tz}^e$ = Embedment translation stiffness coefficient
 $S_{rx}^e, S_{ry}^e, S_{rz}^e$ = Embedment rocking stiffness coefficient, about each axis

$N_{tx}^e, N_{ty}^e, N_{tz}^e$ = Embedment translation damping coefficient
 $N_{rx}^e, N_{ry}^e, N_{rz}^e$ = Embedment rocking damping coefficient, about each axis

X_0 = Dimensionless frequency

f = Dimensionless variable

g = Complex variable

μ_f = Poisson's ratio for compacted fill material

$H_0(x)$ = Bessel function of the third kind, or order 0, evaluated at x (complex)

$H_1(x)$ = Bessel function of the third kind, or order 1, evaluated at x (complex)

$H_2(x)$ = Bessel function of the third kind, or order 2, evaluated at x (complex)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$$f = \sqrt{2(1 - \mu_f) / 1 - 2(\mu_f)} \quad (\text{Equation 10-24})$$

$$X_o = A_t / f \quad (\text{Equation 10-25})$$

$$g = \frac{f (H_2(A_t))(H_1(X_o)) + (H_2(X_o))(A_t)}{(H_0(A_t))(H_2(X_o)) + (H_0(X_o))(H_2(A_t))} \quad (\text{Equation 10-26})$$

$$S_{tx}^e + iN_{tx}^e = S_{ty}^e + iN_{ty}^e = 2 \pi (A_t)(g) \quad (\text{Equation 10-27a})$$

$$S_{tz}^e + iN_{tz}^e = 2 \pi (A_t) (H_1(A_t)) / (H_0(A_t)) \quad (\text{Equation 10-27b})$$

$$S_{rx}^e + iN_{rx}^e = \pi - \pi (A_x) (H_0(A_x)) / (H_1(A_x)) \quad (\text{Equation 10-27c})$$

$$S_{ry}^e + iN_{ry}^e = \pi - \pi (A_y) (H_0(A_y)) / (H_1(A_y)) \quad (\text{Equation 10-27d})$$

$$S_{rz}^e + iN_{rz}^e = 4 \pi - 2 \pi (A_z) (H_0(A_z)) / (H_1(A_z)) \quad (\text{Equation 10-27e})$$

E6 The stiffness and damping values that make up embedment impedance can now be computed from the following formulae:

$$K_{tx}^e = (S_{tx}^e)(G_f)(h) \quad (\text{Equation 10-28a})$$

$$K_{ty}^e = (S_{ty}^e)(G_f)(h) \quad (\text{Equation 10-28b})$$

$$K_{tz}^e = (S_{tz}^e)(G_f)(h) \quad (\text{Equation 10-28c})$$

$$K_{rx}^e = \left[S_{rx}^e + \left(\frac{h}{R_x} \right)^2 \left(\frac{S_{ty}^e}{12} \right) \right] (G_f)(h)(R_x)^2 \quad (\text{Equation 10-28d})$$

$$K_{ry}^e = \left[S_{ry}^e + \left(\frac{h}{R_y} \right)^2 \left(\frac{S_{tx}^e}{12} \right) \right] (G_f)(h)(R_y)^2 \quad (\text{Equation 10-28e})$$

$$K_{rz}^e = (S_{rz}^e)(G_f)(h)(R_z)^2 \quad (\text{Equation 10-28f})$$

$$C_{tx}^e = (N_{tx}^e)(G_f)(h) / MS \quad (\text{Equation 10-29a})$$

$$C_{ty}^e = (N_{ty}^e)(G_f)(h) / MS \quad (\text{Equation 10-29b})$$

$$C_{tz}^e = (N_{tz}^e)(G_f)(h) / MS \quad (\text{Equation 10-29c})$$

$$C_{rx}^e = \left[N_{rx}^e + \left(\frac{h}{R_x} \right)^2 \left(\frac{N_{ty}^e}{12} \right) \right] \frac{(G_f)(h)(R_x)^2}{MS} \quad (\text{Equation 10-29d})$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$$C_{ry}^e = \left[N_{ry}^e + \left(\frac{h}{R_y} \right)^2 \left(\frac{N_{tx}^e}{12} \right) \right] \frac{(G_f)(h)(R_y)^2}{MS} \quad (\text{Equation 10-29e})$$

$$C_{rz}^e = (N_{rz}^e)(G_f)(h)(R_z)^2 / MS \quad (\text{Equation 10-29f})$$

F Forces at the Center of Gravity

F1 All dynamic forces must be determined at the center of gravity in order to apply the basic dynamic equation described in Section A1. Mass and mass moments of inertia have already been computed at the CG in Section B7. Impedance and machine forces will be resolved at the CG in the following sections.

F2 Compute total stiffness and damping at the center of gravity. A second set of stiffness and damping values must be computed for reciprocating machines with secondary forces.

$$\begin{aligned} K_{tx}, K_{ty}, K_{tz} &= \text{Translational stiffness at CG (K/ft)} \\ K_{rx}, K_{ry}, K_{rz} &= \text{Rotational stiffness at CG, about each axis (ft-K/rad)} \\ K_{cx}, K_{cy} &= \text{Cross stiffness at CG (K/rad)} \\ \\ C_{tx}, C_{ty}, C_{tz} &= \text{Translational damping at CG (K-sec/ft)} \\ C_{rx}, C_{ry}, C_{rz} &= \text{Rotational damping at CG, about each axis (ft-K-sec/rad)} \\ C_{cx}, C_{cy} &= \text{Cross damping at CG (K-sec/rad)} \end{aligned}$$

F3 Two dimensions need to be determined first.

$$\begin{aligned} \partial_c &= \text{Height from CG to bottom of mat (ft)} \\ \partial_e &= \text{Height from CG to center of embedment (ft)} \end{aligned}$$

$$\partial_c = \text{MatHt} - \text{CG}_z \quad (\text{Equation 10-30})$$

$$\partial_e = \partial_c - 0.5 (h) \quad (\text{Equation 10-31})$$

F4 Total stiffness values are computed from the following:

$$K_{tx} = K_{tx}^c + K_{tx}^e \quad (\text{Equation 10-32a})$$

$$K_{ty} = K_{ty}^c + K_{ty}^e \quad (\text{Equation 10-32b})$$

$$K_{tz} = K_{tz}^c + K_{tz}^e \quad (\text{Equation 10-32c})$$

$$K_{rx} = K_{rx}^c + K_{rx}^e + (K_{ty}^c)(\partial_c)^2 + (K_{ty}^e)(\partial_e)^2 \quad (\text{Equation 10-32d})$$

$$K_{ry} = K_{ry}^c + K_{ry}^e + (K_{tx}^c)(\partial_c)^2 + (K_{tx}^e)(\partial_e)^2 \quad (\text{Equation 10-32e})$$

$$K_{rz} = K_{rz}^c + K_{rz}^e \quad (\text{Equation 10-32f})$$

$$K_{cx} = - (K_{ty}^c)(\partial_c) - (K_{ty}^e)(\partial_e) \quad (\text{Equation 10-32g})$$

$$K_{cy} = - (K_{tx}^c)(\partial_c) - (K_{tx}^e)(\partial_e) \quad (\text{Equation 10-32h})$$

F5 Total damping values are computed from the following:



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$$C_{tx} = C_{tx}^c + C_{tx}^e \quad (\text{Equation 10-33a})$$

$$C_{ty} = C_{ty}^c + C_{ty}^e \quad (\text{Equation 10-33b})$$

$$C_{tz} = C_{tz}^c + C_{tz}^e \quad (\text{Equation 10-33c})$$

$$C_{rx} = C_{rx}^c + C_{rx}^e + (C_{ty}^c)(\partial_c)^2 + (C_{ty}^e)(\partial_e)^2 \quad (\text{Equation 10-33d})$$

$$C_{ry} = C_{ry}^c + C_{ry}^e + (C_{tx}^c)(\partial_c)^2 + (C_{tx}^e)(\partial_e)^2 \quad (\text{Equation 10-33e})$$

$$C_{rz} = C_{rz}^c + C_{rz}^e \quad (\text{Equation 10-33f})$$

$$C_{cx} = - (C_{ty}^c)(\partial_c) - (C_{ty}^e)(\partial_e) \quad (\text{Equation 10-33g})$$

$$C_{cy} = - (C_{tx}^c)(\partial_c) - (C_{tx}^e)(\partial_e) \quad (\text{Equation 10-33h})$$

- F6 Normally, unbalance forces are provided along the shaft, usually at a rotor or between cylinders. The unbalanced forces and moments must then be translated to the center of gravity. For reciprocating machines, this calculation must be performed for primary loads, and again for secondary loads.

$$\begin{aligned} F_x, F_y, F_z &= \text{Unbalanced force at CG (kips)} \\ M_x, M_y, M_z &= \text{Unbalanced moment at CG (ft-K)} \end{aligned}$$

$$\begin{aligned} F_x^v, F_y^v, F_z^v &= \text{Force provided by supplier (kips)} \\ M_x^v, M_y^v, M_z^v &= \text{Moment provided by supplier (ft-kips)} \end{aligned}$$

$$\partial_x, \partial_y, \partial_z = \text{Distance from origin to location of supplier forces in each axis (ft)}$$

$$F_x = F_x^v \quad (\text{Equation 10-34a})$$

$$F_y = F_y^v \quad (\text{Equation 10-34b})$$

$$F_z = F_z^v \quad (\text{Equation 10-34c})$$

$$M_x = M_x^v + F_y^v [\text{abs}(\partial_z - \text{CG}_z)] + F_z^v [\text{abs}(\partial_y - \text{CG}_y)] \quad (\text{Equation 10-34d})$$

$$M_y = M_y^v + F_x^v [\text{abs}(\partial_z - \text{CG}_z)] + F_z^v [\text{abs}(\partial_x - \text{CG}_x)] \quad (\text{Equation 10-34e})$$

$$M_z = M_z^v + F_x^v [\text{abs}(\partial_y - \text{CG}_y)] + F_y^v [\text{abs}(\partial_x - \text{CG}_x)] \quad (\text{Equation 10-34f})$$

G Single DOF Analysis

- G1 A single degree of freedom analysis is used for vertical translation and for torsional rocking. (Reference 2d)

- G2 The analysis for vertical translation, designated mode #1, computes the following variables:

$$\begin{aligned} \text{Mode \#1:} & \quad (\text{Z axis translation}) \\ NF_1 &= \text{Vertical natural frequency (rad/sec)} \\ FR_1 &= \text{Vertical frequency ratio} \\ \Delta_z &= \text{Vertical translation at CG (ft)} \end{aligned}$$

- G3 Compute 2 quantities:

$$a_1, a_2 = \text{Impedance variables (K/ft)}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$$a_1 = (K_{tz}) - (\text{mass})(MS)^2 \quad (\text{Equation 10-35})$$

$$a_2 = (C_{tz})(MS) \quad (\text{Equation 10-36})$$

G4 Compute amplitude at the center of gravity:

$$\Delta_z = F_z / \sqrt{(a_1)^2 + (a_2)^2} \quad (\text{Equation 10-37})$$

G5 Iterate to find the natural frequency:

1. Select a trial frequency, NF_1 .
2. Determine S_{tz}^c and K_{tz}^c from the criteria given in Section D.
3. If embedded, determine S_{tz}^e and K_{tz}^e from the criteria given in Section E.
4. Compute K_{tz} from the equations given in Section F4.
5. Compute NF_1 :

$$NF_1 = \sqrt{K_{tz} / \text{mass}} \quad (\text{Equation 10-38})$$

6. Repeat the above steps until the trial and computed frequencies match.

G6 Compute the frequency ratio and compare using the acceptance criteria provided in the main guideline.

$$FR_1 = MS / NF_1 \quad (\text{Equation 10-39})$$

G7 The analysis for Z axis rocking or torsion, designated mode #2, computes the following variables:

Mode #2: (Z axis rocking)
 NF_2 = Torsional natural frequency (rad/sec)
 FR_2 = Torsional frequency ratio
 ϕ_z = Torsional rotation at CG (ft)

G8 The computations described in Sections G3 through G7 are repeated for mode #2 with the following substitutions:

I_z	for mass	(mass)
K_{tz}	for K_{tz}	(stiffness)
C_{tz}	for C_{tz}	(damping)
M_z	for F_z	(unbalanced load)
ϕ_z	for Δ_z	(amplitude)
NF_2	for NF_1	(natural frequency)
FR_2	for FR_1	(frequency ratio)

G9 Secondary amplitudes must be computed for reciprocating machines with a secondary force. The above steps are repeated with twice the machine speed and with secondary values replacing all primary values.



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Analysis Procedure

H Two DOF Analysis

H1 A two degree of freedom analysis is used in each horizontal direction for coupled translation and rocking. (Reference 4c)

H2 The analysis for lateral translation and rocking, designated Modes #3 and #4, computes the following variables:

Modes #3 and #4: (Y axis translation & X axis rocking)
 NF_3, NF_4 = Natural frequencies (rad/sec)
 FR_3, FR_4 = Frequency ratios
 Δ_y = Translation at CG (ft)
 ϕ_x = Rotation at CG (rad)

H3 Compute 6 quantities:

a_1, a_2 = Impedance variables (ft-K²)
 b_1, b_2, c_1, c_2 = Impedance variables (kips²)

$$a_1 = [(K_{rx}) - (I_x)(MS)^2](F_y) - (K_{cx})(M_x) \quad \text{(Equation 10-40)}$$

$$a_2 = (C_{rx})(MS)(F_y) - (C_{cx})(MS)(M_x) \quad \text{(Equation 10-41)}$$

$$b_1 = [(K_{ty}) - (\text{mass})(MS)^2](M_x) - (K_{cx})(F_y) \quad \text{(Equation 10-42)}$$

$$b_2 = (C_{ty})(MS)(M_x) - (C_{cx})(MS)(F_y) \quad \text{(Equation 10-43)}$$

$$c_1 = (\text{mass})(I_x)(MS)^4 + (K_{ty})(K_{rx}) - (K_{cx})^2 \\ - [(\text{mass})(K_{rx}) + (I_x)(K_{ty}) + (C_{ty})(C_{rx}) - (C_{cx})^2](MS)^2 \quad \text{(Equation 10-44)}$$

$$c_2 = [(C_{ty})(K_{rx}) + (C_{rx})(K_{ty}) - 2(C_{cx})(K_{cx})](MS) \\ - [(\text{mass})(C_{rx}) + (I_x)(C_{ty})](MS)^3 \quad \text{(Equation 10-45)}$$

H4 Compute amplitudes:

$$\Delta_y = \sqrt{(a_1)^2 + (a_2)^2 / (c_1)^2 + (c_2)^2} \quad \text{(Equation 10-46)}$$

$$\phi_x = \sqrt{(b_1)^2 + (b_2)^2 / (c_1)^2 + (c_2)^2} \quad \text{(Equation 10-47)}$$

H5 Iterate to find the natural frequency:

1. Select a trial frequency, NF.
2. Determine S_{ty}^c , S_{rx}^c , K_{ty}^c , and K_{rx}^c from the criteria given in Section D.
3. If embedded, determine S_{ty}^e , S_{rx}^e , K_{ty}^e , and K_{rx}^e from the criteria given in Section E.



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Analysis Procedure

4. Compute K_{ty} , K_{rx} , and K_{cx} from the equations given in Section F4.
5. Compute NF_3 and NF_4 :

$$\begin{aligned} b &= \text{Frequency variable (1/sec}^2\text{)} \\ c &= \text{Frequency variable (1/sec}^4\text{)} \end{aligned}$$

$$b = (K_{ty} / \text{mass}) + (K_{rx} / I_x) \quad (\text{Equation 10-48})$$

$$c = (K_{ty})(K_{rx}) - (K_{cx})^2 / (\text{mass})(I_x) \quad (\text{Equation 10-49})$$

$$NF_3 = 0.5(b) - 0.5\sqrt{(b)^2 - 4(c)} \quad (\text{Equation 10-50})$$

$$NF_4 = 0.5(b) + 0.5\sqrt{(b)^2 - 4(c)} \quad (\text{Equation 10-51})$$

6. Repeat the above steps until the trial and either of both computed frequencies match.

H6 Compute the frequency ratios and compare using the acceptance criteria provided in the main guideline.

$$FR_3 = MS / NF_3 \quad (\text{Equation 10-52a})$$

$$FR_4 = MS / NF_4 \quad (\text{Equation 10-52b})$$

H7 The analysis for lateral translation and rocking, designated modes #5 and #6, computes the following variables:

$$\begin{aligned} \text{Modes \#5 and \#6:} & \quad (\text{X axis translation and Y axis rocking}) \\ NF_5, NF_6 &= \text{Natural frequencies (rad/sec)} \\ FR_5, FR_6 &= \text{Frequency ratios} \\ \Delta_x &= \text{Translation at CG (ft)} \\ \phi_y &= \text{Rotation at CG (rad)} \end{aligned}$$

H8 The computations described in Sections H3 through H7 are repeated for Modes #5 and #6 with the following substitutions:

I_y	for I_x	(mass moment of inertia)
K_{tx}	for K_{ty}	(translation stiffness)
K_{ry}	for K_{rx}	(rocking stiffness)
C_{tx}	for C_{ty}	(translation damping)
C_{ry}	for C_{rx}	(rocking damping)
F_x	for F_y	(unbalanced force)
M_y	for M_x	(unbalanced moment)
Δ_x	for Δ_y	(translation)
ϕ_y	for ϕ_x	(rotation)
NF_5, NF_6	for NF_3, NF_4	(natural frequency)
FR_5, FR_6	for FR_3, FR_4	(frequency ratio)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

- H9 Secondary amplitudes must be computed for reciprocating machines with secondary loads. The above steps are repeated with twice the machine speed and with secondary values replacing all primary values.

I Double Amplitudes at Selected Locations

- I1 Double amplitudes (peak to peak) are computed at a selected point on the foundation. A comparison is then made using the acceptance criteria provided in the main guideline.

DA_x, DA_y, DA_z = Double, or peak to peak, amplitude, at specified location (mils)
 PT_x, PT_y, PT_z = Distance from origin to selected point (ft)
 D_x, D_y, D_z = Distance from CG to selected point, along each axis (ft)

$$D_x = \text{abs} [PT_x - CG_x] \quad (\text{Equation 10-53a})$$

$$D_y = \text{abs} [PT_y - CG_y] \quad (\text{Equation 10-53b})$$

$$D_z = \text{abs} [PT_z - CG_z] \quad (\text{Equation 10-53c})$$

$$DA_x = \{\Delta_x + D_y (\phi_z) + D_z (\phi_y)\} (24000) \quad (\text{Equation 10-54a})$$

$$DA_y = \{\Delta_y + D_x (\phi_z) + D_z (\phi_x)\} (24000) \quad (\text{Equation 10-54b})$$

$$DA_z = \{\Delta_z + D_x (\phi_y) + D_y (\phi_x)\} (24000) \quad (\text{Equation 10-54c})$$

- I2 For reciprocating machines, amplitudes must be computed for secondary forces and added to those obtained for primary forces.

J NOMENCLATURE (including variables used elsewhere in the Document)

J1 Superscripts and Subscripts

1 = Vertical vibration mode
 2 = Torsional vibration mode
 3, 4 = Coupled Y axis translation and X axis rocking vibration modes
 5, 6 = Coupled X axis translation and Y axis rocking vibration modes
 c = Mat centroid quantity
 e = Embedment quantity
 r = Rotation
 s = Static quantity
 t = Translation
 v = Supplier provided load
 x = Coordinate axis parallel to shaft
 y = Coordinate axis perpendicular to shaft
 z = Vertical coordinate axis

J2 Variables

a_1, a_2 = Impedance variables (K/ft)
 A_t = Dimensionless frequency for translation
 A_x, A_y, A_z = Dimensionless frequencies for rotation about each axis
 A_{crank} = Area of piston head (in²)
 A_{head} = Area of piston head on crank side (in²)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

b	= Frequency variable (1/sec ²)
b ₁ , b ₂	= Impedance variables (K/ft)
B	= Cylinder bore diameter (in)
c	= Frequency variable (1/sec ⁴)
c ₁ , c ₂	= Impedance variables (K/ft)
C _{tx} ..C _{tz}	= Translation damping at CG (K-sec/ft)
C _{rx} ..C _{rz}	= Rotational damping at CG (ft-K-sec/rad)
C _{cx} , C _{cy}	= Cross damping at CG (K-sec/rad)
C _{tx} ^c ..C _{tz} ^c	= Base translational damping (K-sec/ft)
C _{rx} ^c ..C _{rz} ^c	= Base rotational damping (ft-K-sec/rad)
C _{tx} ^e ..C _{tz} ^e	= Embedment translational damping (K-sec/ft)
C _{rx} ^e ..C _{rz} ^e	= Embedment rotational damping (ft-K-sec/rad)
CG _x , CG _y , CG _z	= Center of gravity location with respect to the referenced origin (ft)
CS	= Coastdown speed (rad/sec)
CW	= Weight of component (kips)
d	= Mat cantilever beyond face of pier, in either direction (ft)
D	= Rod diameter (in)
D _x , D _y , D _z	= Distance from CG to selected point, along each axis (ft)
DA _x , DA _y , DA _z	= Double, or peak to peak, amplitude at specified location (mils)
e	= Eccentricity of rotor (ft)
E _x , E _y , E _z	= Component weight times distance from origin (ft-K)
E _c	= Modulus of elasticity for concrete (K/in ²)
E _s	= Modulus of elasticity for soil (K/in ²)
f	= Embedment variable
F _x , F _y , F _z	= Unbalanced force at CG (kips)
F _x ^v , F _y ^v , F _z ^v	= Force provided by supplier (kips)
F ₁	= Inertia force of hinge (kips)
F ₂	= Primary inertia force of piston (kips)
F ₃	= Secondary inertia force of piston (kips)
F ₄	= Inertia force of unbalanced rotor (kips)
F _{cr}	= Correction factor
F _{fdn}	= Lateral force on foundation, tributary to cylinder (kips)
F _{red}	= Reduction factor, use 2.0 unless better data is available
FR ₁ ..FR ₆	= Frequency ratio
g	= Embedment variable (complex)
G _f	= Shear modulus of compacted fill material (K/ft ²)
G _s	= Shear modulus of soil (K/ft ²)
GradeHt	= Height from origin to grade (ft)
gravity	= Acceleration of gravity (32.2 ft/sec ²)
h	= Effective mat embedment (ft)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

$H_0(x)$	= Bessel function of the third kind, or order 0, evaluated at x.
$H_1(x)$	= Bessel function of the third kind, or order 1, evaluated at x.
$H_2(x)$	= Bessel function of the third kind, or order 2, evaluated at x.
HW	= Weight of hinge (kips)
I_x, I_y, I_z	= Mass moment of inertia, about each axis (K-sec ² -ft)
J_x, J_y, J_z	= Component's translated mass moment of inertia (K-sec ² -ft)
$K_{tx}..K_{tz}$	= Translation stiffness at CG (K/ft)
$K_{rx}..K_{rz}$	= Rotational stiffness at CG (ft-K/rad)
K_{cx}, K_{cy}	= Cross stiffness at CG (K/rad)
$K_{tx}^c..K_{tz}^c$	= Base translational stiffness (K/ft)
$K_{rx}^c..K_{rz}^c$	= Base rotational stiffness (ft-K/rad)
$K_{tx}^e..K_{tz}^e$	= Embedment translational stiffness (K/ft)
$K_{rx}^e..K_{rz}^e$	= Embedment rotational stiffness (ft-K/rad)
$K_{tx}^s..K_{tz}^s$	= Static translational stiffness (K/ft)
$K_{rx}^s..K_{rz}^s$	= Static rotational stiffness (ft-K/rad)
L	= Length (ft)
L_1	= Length from shaft to hinge (ft)
L_2	= Length from hinge to piston (ft)
L_x, L_y, L_z	= Dimensions of foundation component (ft)
M_x, M_y, M_z	= Unbalanced moment at CG (ft-kips)
M_x^v, M_y^v, M_z^v	= Moment provided by supplier (ft-kips)
mass	= Translational mass (K-sec ² /ft)
MatHt	= Height from origin to bottom of mat (ft)
MR	= Mass ratio
MS	= Machine speed (rad/sec)
MW	= Machine weight (kips)
$N_{tx}^c..N_{tz}^c$	= Base translation damping coefficient
$N_{rx}^c..N_{rz}^c$	= Base rotational damping coefficient
$N_{tx}^e..N_{tz}^e$	= Embedment translation damping coefficient
$N_{rx}^e..N_{rz}^e$	= Embedment rotational damping coefficient
$NF_1..NF_6$	= Natural frequency (rad/sec)
OS_x, OS_y	= Center of gravity offset in each direction (percent)
P_{crank}	= Instantaneous crank pressure (ksi)
P_{head}	= Instantaneous head pressure (ksi)
PT_x, PT_y, PT_z	= Distance from origin to selected point (ft)
PW	= Weight of piston (kips)
Q_g	= Measure of rotor quality grade (in/sec)
Q_x, Q_y, Q_z	= Component mass moment of inertia about component CG (K-sec ² -ft)
r	= Radius (ft)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Analysis Procedure

R_t	= Equivalent mat radius for translation (ft)
R_x, R_y, R_z	= Equivalent mat radius for rotation, about each axis (ft)
RW	= Weight of rotor (kips)
R_1, R_2	= Combined foundation ratios
$S_{tx}^c \dots S_{tz}^c$	= Base translation stiffness coefficient
$S_{rx}^c \dots S_{rz}^c$	= Base rotational stiffness coefficient
$S_{tx}^e \dots S_{tz}^e$	= Embedment translation stiffness coefficient
$S_{rx}^e \dots S_{rz}^e$	= Embedment rotational stiffness coefficient
SB_{net}	= Maximum net static soil bearing (ksf)
t	= Time (sec)
TW	= Foundation, machine, and soil weight (kips)
W_c	= Density of concrete (K/ft ³)
W_f	= Density of compacted fill material (K/ft ³)
W_s	= Density of soil (K/ft ³)
X_0	= Dimensionless frequency
β	= Crank angle (rad)
∂_c	= Height from CG to bottom of mat (ft)
∂_e	= Height from CG to center of embedment (ft)
$\partial_x, \partial_y, \partial_z$	= Distance from origin to center of selected location (ft)
$\Delta_x, \Delta_y, \Delta_z$	= Translation at CG (ft)
ϕ_x, ϕ_y, ϕ_z	= Rotation at CG (rad)
μ_f	= Poisson's ratio for compacted fill material
μ_s	= Poisson's ratio for soil

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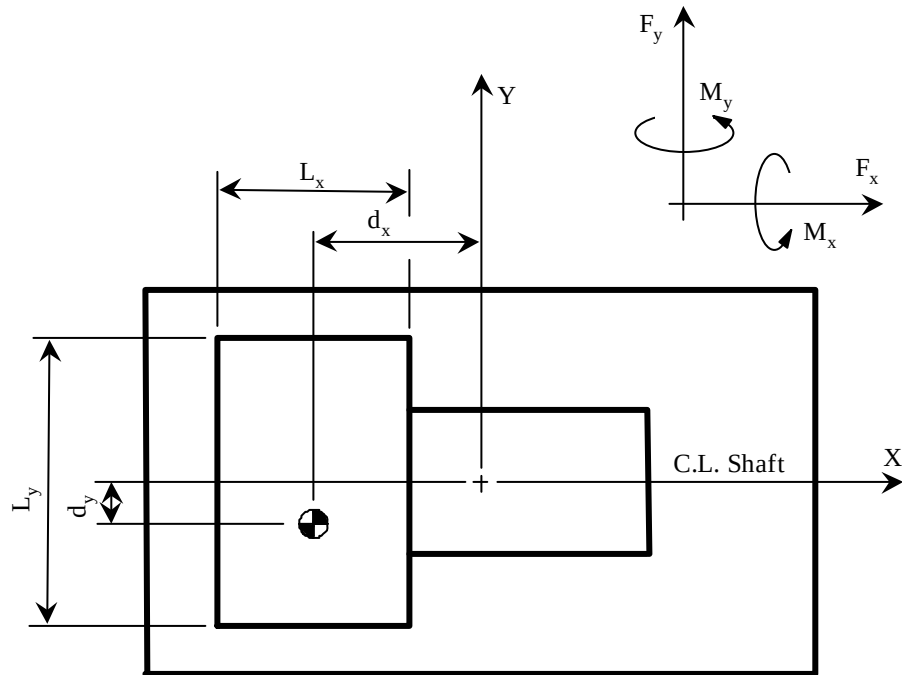
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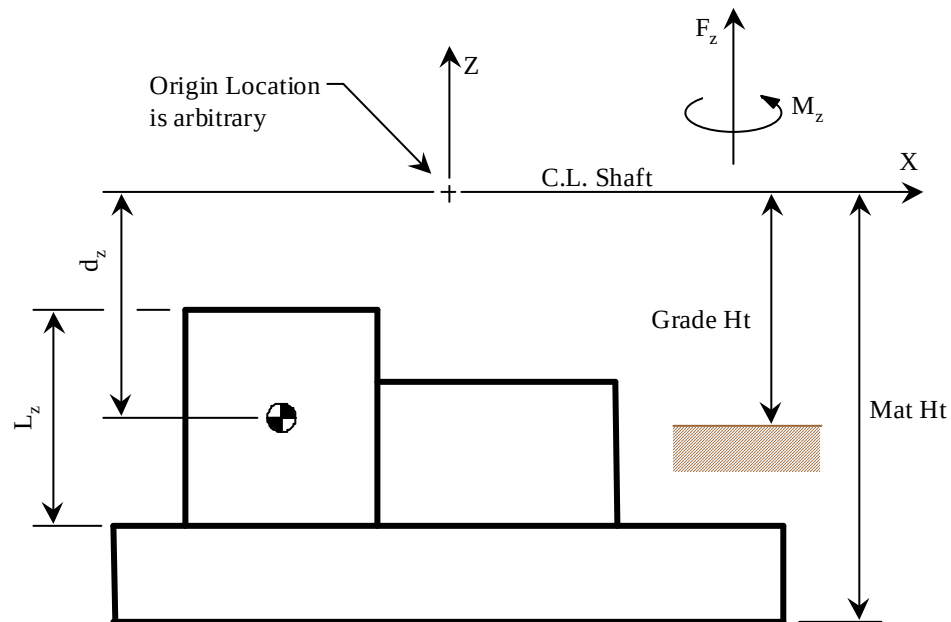


VIBRATING MACHINERY FOUNDATIONS ON SOIL

Directional Nomenclature



FOUNDATION PLAN

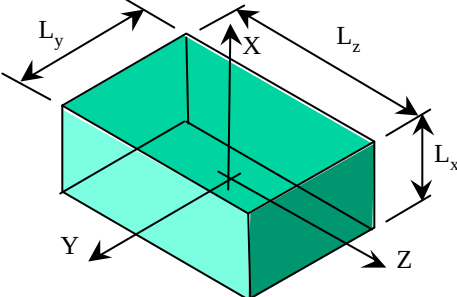
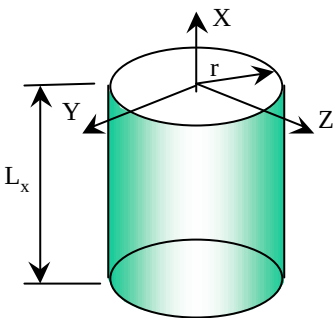
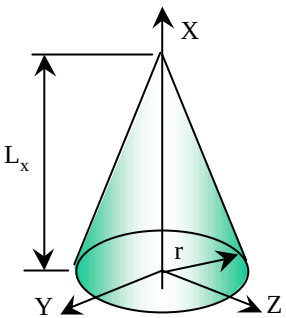
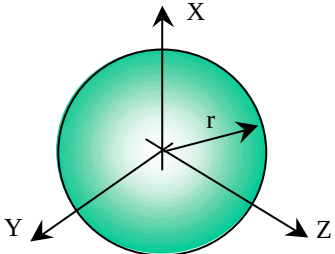


ELEVATION

VIBRATING MACHINERY FOUNDATIONS ON SOIL

Mass Moments of Intertia

Note that the results of these equations must be divided by "g" for the "mass" moment of inertia.

	Rectangular Prism $CW = (\text{Density})(L_x)(L_y)(L_z)$ $Q_x = (CW)[(L_y)^2 + (L_z)^2] / 12$ $Q_y = (CW)[(L_x)^2 + (L_z)^2] / 12$ $Q_z = (CW)[(L_x)^2 + (L_y)^2] / 12$
	Circular Cylinder $CW = (\text{Density})(\pi)(r)^2 (L_x)$ $Q_x = (CW)(r)^2 / 2$ $Q_y = (CW)[3(r)^2 + (L_x)^2] / 12$ $Q_z = Q_y$
	Circular Cone $CW = (\text{Density})(\pi)(r)^2 (L_x) / 3$ $Q_x = 3(CW)(r)^2 / 10$ $Q_y = 3(CW)[4(r)^2 + (L_x)^2] / 80$ $Q_z = Q_y$
	Sphere $CW = (\text{Density})(4\pi)r^3 / 3$ $Q_x = 2(CW)r^2 / 5$ $Q_y = Q_x$ $Q_z = Q_x$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Base Impedance Coefficients

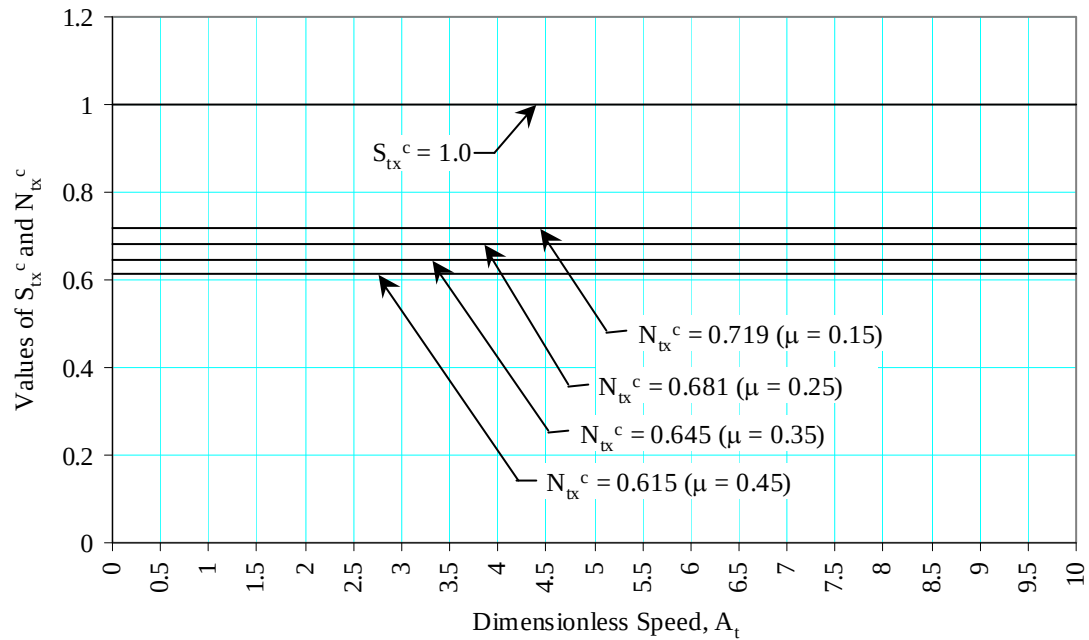


Figure 1: S_{tx}^c and N_{tx}^c (refer to Attachment 10, Section D5)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Base Impedance Coefficients

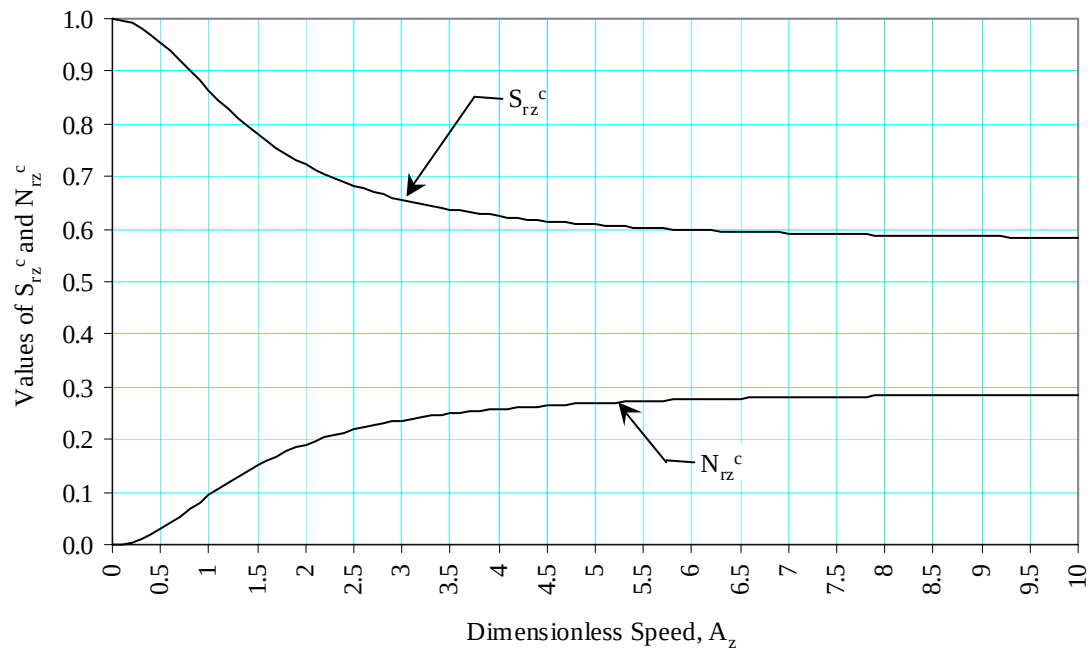


Figure 2: S_{rz}^c and N_{rz}^c (refer to Attachment 10, Section D5)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Base Impedance Coefficients

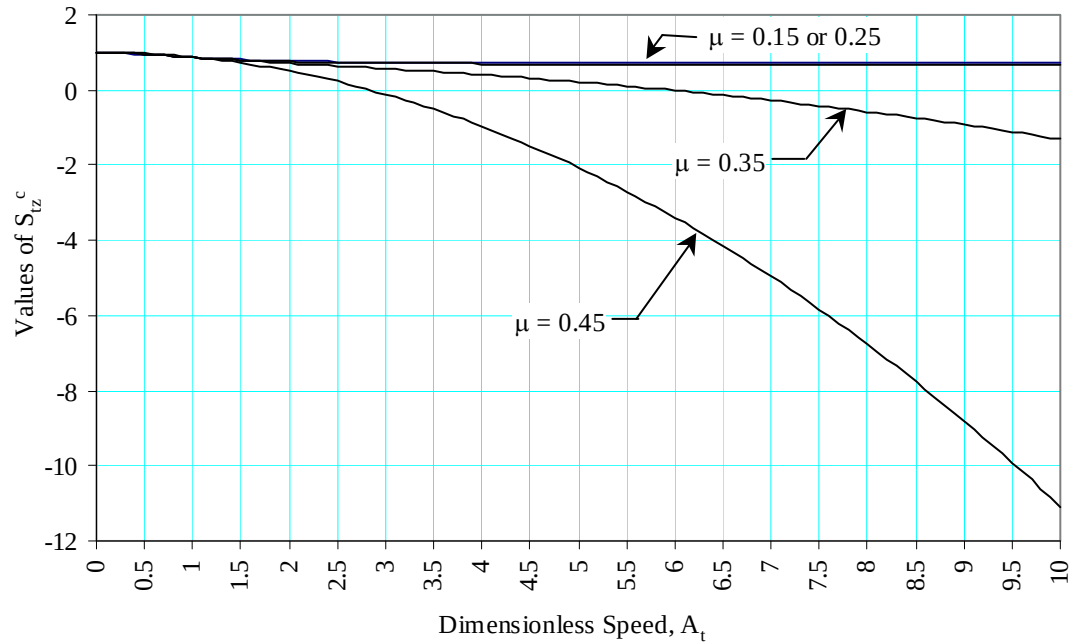


Figure 3: S_{tz}^c (refer to Attachment 10, Section D5)

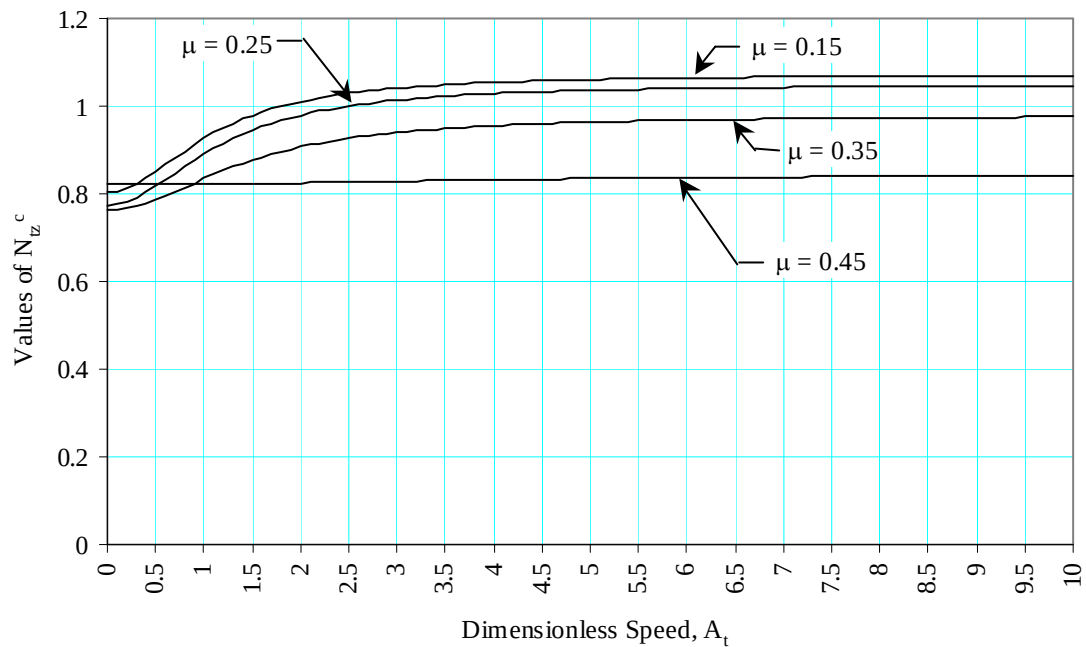


Figure 4: N_{tz}^c (refer to Attachment 10, Section D5)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Base Impedance Coefficients

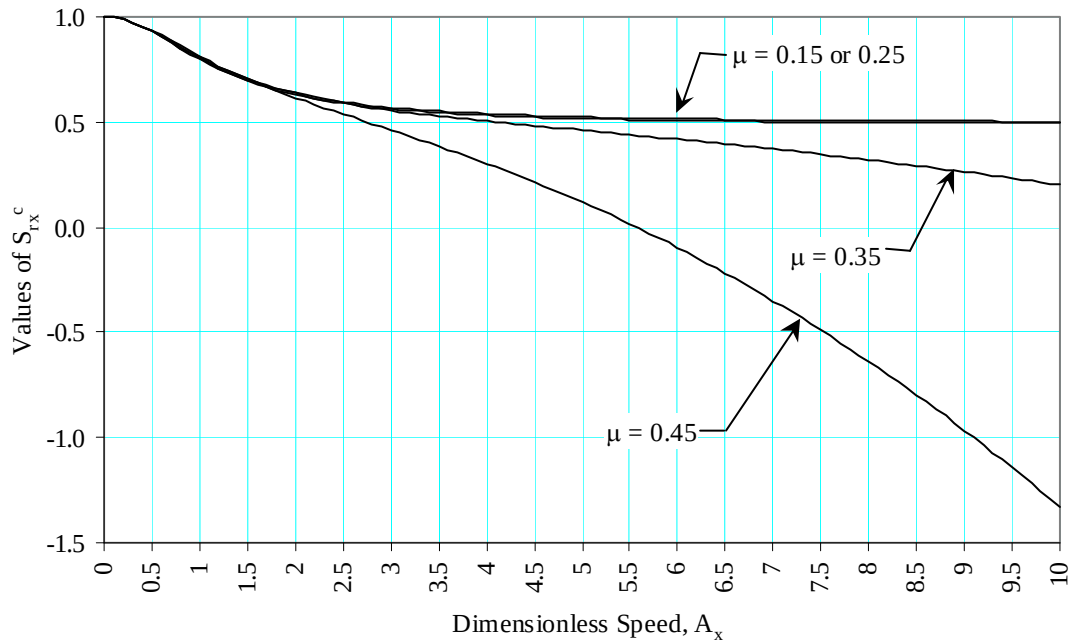


Figure 5: S_{rx}^c (refer to Attachment 10, Section D5)

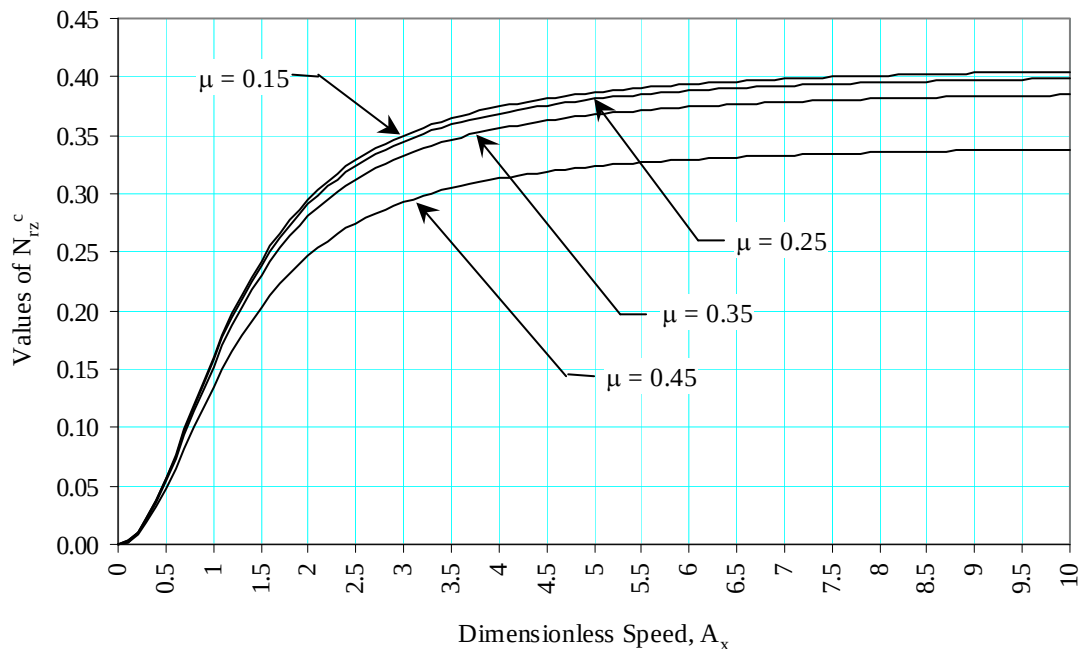


Figure 6: N_{rx}^c (refer to Attachment 10, Section D5)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Embedment Coefficient

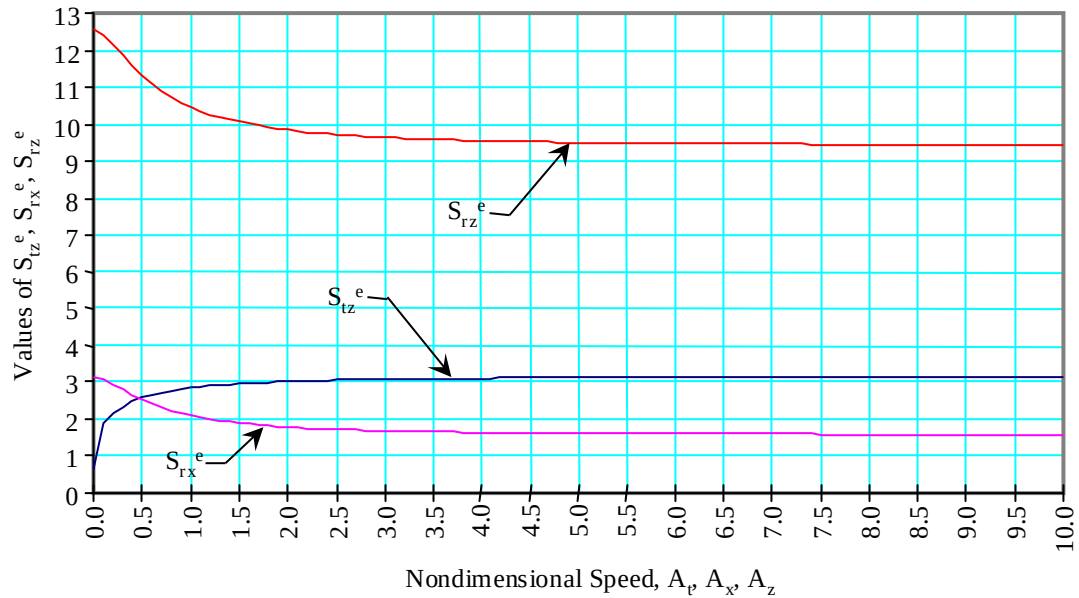


Figure 1: S_{tz}^e , S_{rx}^e , and S_{rz}^e (refer to Attachment 10, Section E5)

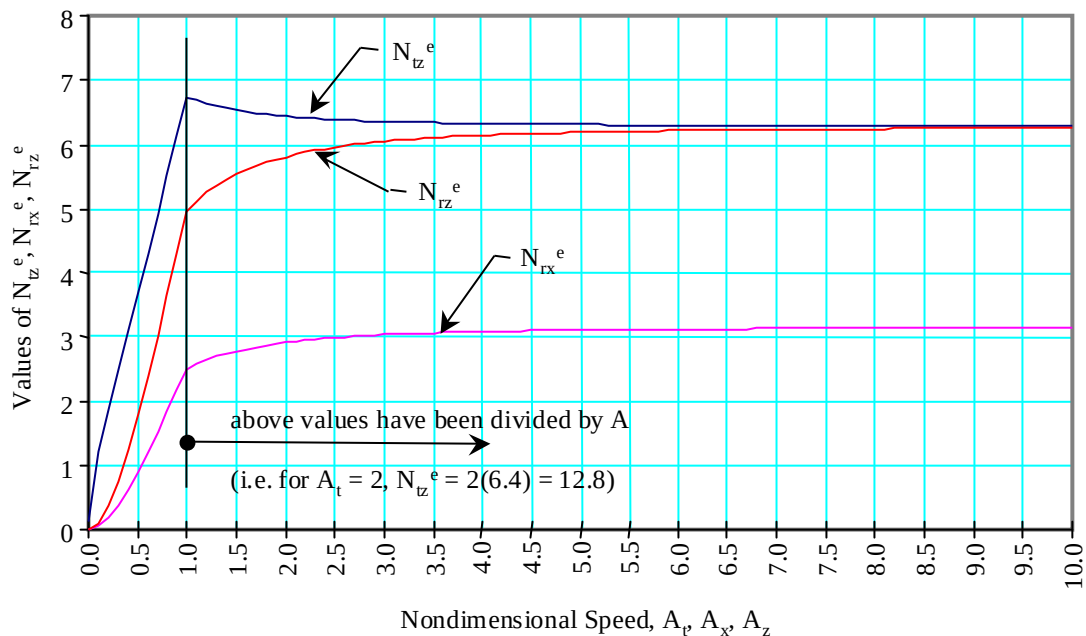


Figure 2: N_{tx}^e , N_{rx}^e , and N_{rz}^e (refer to Attachment 10, Section E5)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Embedment Coefficient

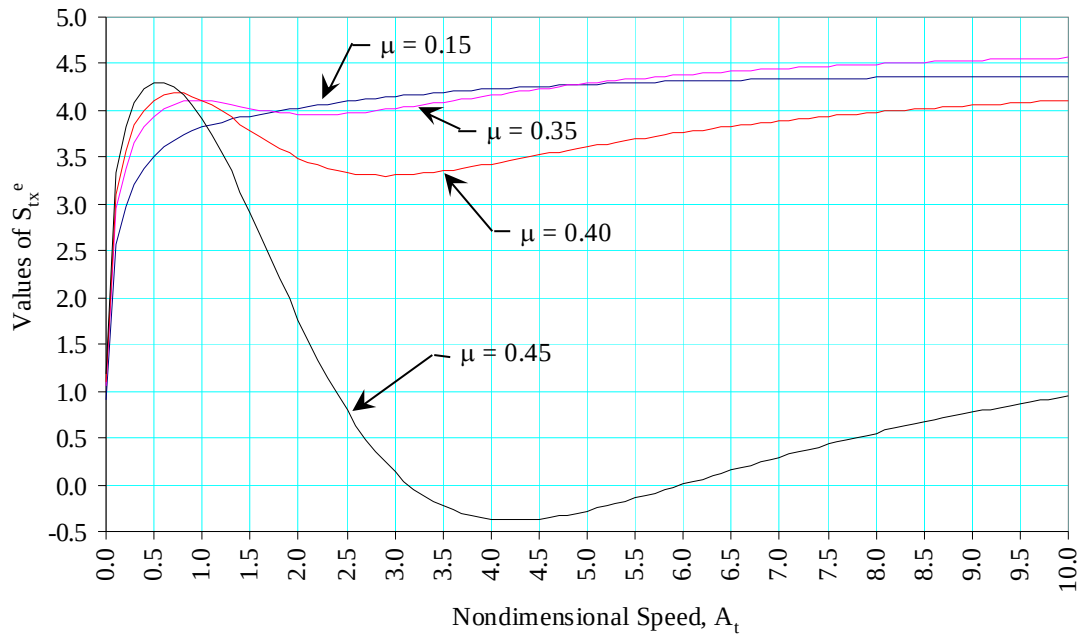


Figure 3: S_{tx}^e (refer to Attachment 10, Section E5)

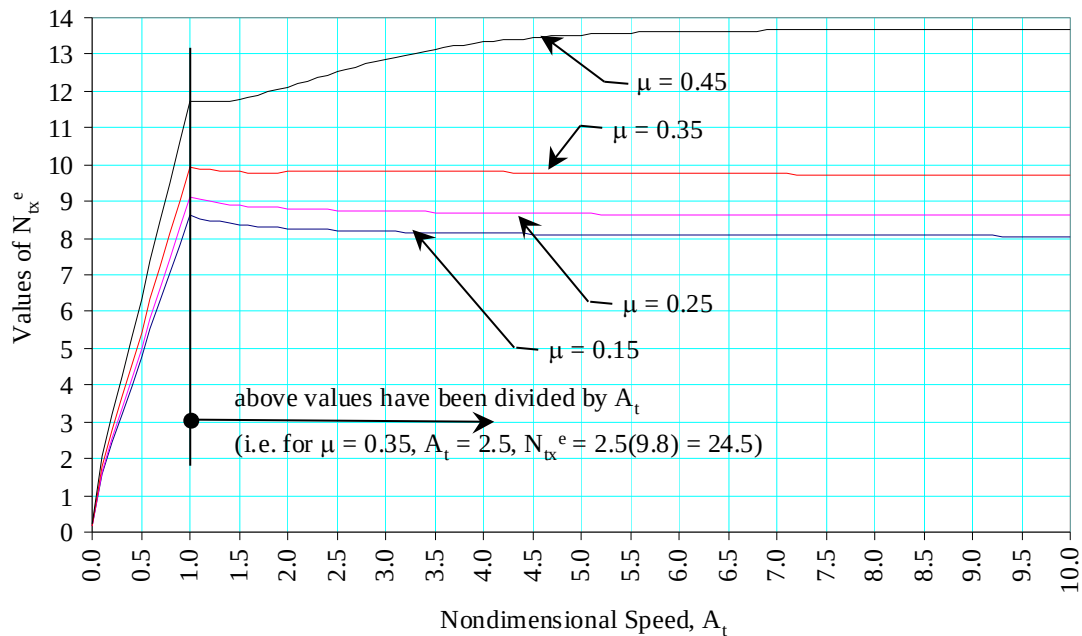


Figure 4: N_{tx}^e (refer to Attachment 10, Section E5)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Machine: reciprocating

speed = 300 rpm

motor = 15000 lb (C.G. at point #1)

compressor = 20,000 lb (C.G. at point #2)

Dynamic Loads: applied at point #3

	(primary)	(secondary)
F_x	1,000 lb	0
F_y	1,500 lb	800 lb
F_z	875 lb	0
M_x	2,100 ft-lb	900 ft-lb
M_y	200 ft-lb	0
M_z	3,100 ft-lb	1,500 ft-lb

Soil: clay, net bearing = 3000 psf

existing soil:

unit weight, $W_s = 110$ pcf

shear modulus, $G_s = 5,000$ psi

poisson's ratio, $\mu_s = 0.44$

purchased sand backfill:

unit weight, $W_f = 120$ pcf

shear modulus, $G_f = 10,000$ psi

poisson's ratio, $\mu_f = 0.35$

Miscellaneous:

compute amplitudes at point #4 (for human tolerance)

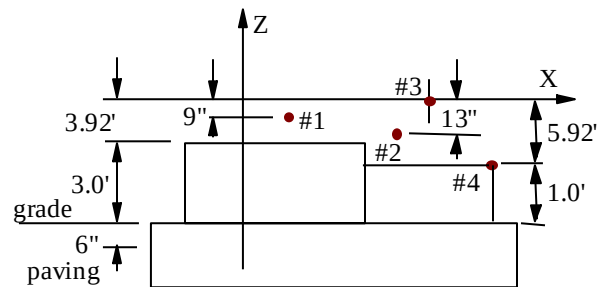
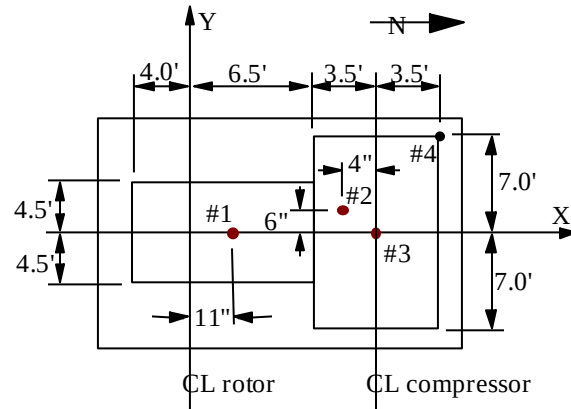
Compute C.G. of Machine & Pier to Set Center of Mat:

(Attachment 10, Sections B4, B5, B6)

Note: Pier heights are reduced by the thickness of paving. The mat thickness will be adjusted.

motor pier weight = $(150 \text{ pcf})(10.5 \text{ ft})(9.0 \text{ ft})(3.0 \text{ ft}) = 42,525 \text{ lb}$

compressor pier weight = $(150 \text{ pcf})(7.0 \text{ ft})(14.0 \text{ ft})(1.0 \text{ ft}) = 14,700 \text{ lb}$





VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

part:	CW (kips)	δ_x (ft)	E_x (ft-K)	δ_y (ft)	E_y (ft-K)	δ_z (ft)	E_z (ft-K)
motor	15	0.92	13.8	0	0	-0.75	-11.3
compressor	20	9.67	193.4	0.5	10	-1.08	-21.6
motor pier	42.5	1.25	53.1	0	0	-5.42	-230.4
comp. pier	14.7	10	147	0	0	-6.42	-94.4
Σ total	92.2		407.3		10		-357.7

$$CG_x = \Sigma (E_x) / TW = (407.3 \text{ ft-K}) / (92.2 \text{ kips}) = 4.42 \text{ ft} \quad \text{say } 4' 5''$$

$$CG_y = \Sigma (E_y) / TW = (10.0 \text{ ft-K}) / (92.2 \text{ kips}) = 0.11 \text{ ft} \quad \text{say } 1''$$

Trial Mat Size: assume 2' 0" minimum thickness

$$L_x = \text{pier dimension plus 2 ft} \\ = [(4.0 \text{ ft}) + (6.5 \text{ ft}) + (3.5 \text{ ft}) + (3.5 \text{ ft})] + 2.0 \text{ ft} = 19.5 \text{ ft} \quad \text{USE } L_x = 19' 6''$$

$$L_y = 1.5 [\text{distance from shaft to bottom of mat}] \\ = 1.5 [(2.0 \text{ ft mat}) + (0.5 \text{ ft paving}) + (3.0 \text{ ft pier}) + (3.92 \text{ ft to shaft})] \\ = 14.88 \text{ ft (checking pier dimensions, use pier width plus 2 ft)} \quad \text{USE } L_y = 16' 0''$$

mat cantilever,
 $d = [(16.0 \text{ ft mat}) - (9 \text{ ft motor pier})] / 2 = 3.5 \text{ ft}$

modulus of concrete,
 $E_c = 57,000 = 57,000 = 3,122,019 \text{ psi}$

modulus of soil,
 $E_s = 2 (G_s)(1 + \mu_s) = 2 (5,000 \text{ psi})(1 + 0.44) = 14,400 \text{ psi}$

mat rigidity, (Practice, Equation 6)

$$\frac{[E_c / E_s] [L_z / d]^3}{[(3,122,019 \text{ psi}) / (14,400 \text{ psi})] [(2.0 \text{ ft}) / (3.5 \text{ ft})]^3} \\ 40.5 > 1 \quad \text{OK, rigid} \quad \text{USE } L_z = 2' 0''$$

Note: A thickness of 2.5 feet will be used in the analysis in order to account for the 6" of concrete paving. Pier heights were previously reduced by 6".



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Compute Total Center of Gravity:

(Attachment 10, Sections B4, B6)

mat weight = (150 pcf)(19.5 ft)(16.0 ft)(2.5 ft) = 117,000 lb

part:	CW (kips)	δ_x (ft)	E_x (ft-K)	δ_y (ft)	E_y (ft-K)	δ_z (ft)	E_z (ft-K)
previous	92.2	-	407.3	-	10	-	-357.7
mat	117	4.42	517.1	0.08	9.4	-8.17	-955.9
Σ total	209.2		924.4		19.4		-1,313.6

total weight, TW = CW = 209.2 kips

$CG_x = \Sigma (E_x) / TW = (924.4 \text{ ft-K}) / (209.2 \text{ kips}) = 4.42 \text{ ft}$

$CG_y = \Sigma (E_y) / TW = (19.4 \text{ ft-K}) / (209.2 \text{ kips}) = 0.09 \text{ ft}$

$CG_z = \Sigma (E_z) / TW = (-1,313.6 \text{ ft-K}) / (209.2 \text{ kips}) = -6.28 \text{ ft}$

Mass Ratio:

(Attachment 10, Section B8)

Machine Weight, MW = 15.0 kips + 20.0 kips = 35.0 kips

mass = TW / gravity = (209.2 kips) / (32.2 ft/sec²) = 6.5 K-sec²/ft

MR = [TW - MW] / MW = [(209.2 kips) - (35.0 kips)] / (35.0 kips) = 4.98 $\approx$ 5, potential trouble

Mass Moment of Inertia about X axis:

(Attachment 10, Section B7)

part:	CW (kips)	L_y (ft)	L_z (ft)	Q_x (K-ft ²)	δ_y (ft)	δ_z (ft)	J_x (K-ft ²)
motor	15	-	-	0	0	-0.75	8.4
compressor	20	-	-	0	0.5	-1.08	28.3
motor pier	42.5	9	3	318.8	0	-5.42	1,248.5
comp pier	14.7	14	1	241.3	0	-6.42	605.9
mat	117	16	2.5	2,556.9	0.08	-8.17	7,810.4
Σ total	209.2			3,117.0			9,701.5

$$\begin{aligned}
 I_x &= \{ \Sigma (Q_x) + \Sigma (J_x) - \text{TotalWt} [(CG_y)^2 + (CG_z)^2] \} / \text{gravity} \\
 &= \{ (3,177.0 \text{ K-ft}^2) + (9,701.5 \text{ K-ft}^2) - (209.2 \text{ kips})[(0.09 \text{ ft})^2 + (6.28 \text{ ft})^2] \} / (32.2 \text{ ft/sec}^2) \\
 &= 141.8 \text{ ft-K-sec}^2
 \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Mass Moment of Inertia about Y axis:

(Attachment 10, Section B7)

part:	CW (kips)	L _x (ft)	L _z (ft)	Q _y (K-ft ²)	δ _x (ft)	δ _z (ft)	J _y (K-ft ²)
motor	15	-	-	0	0.92	-0.75	21.1
compressor	20	-	-	0	9.67	-1.08	1,893.5
motor pier	42.5	10.5	3	422.3	1.25	-5.42	1,314.9
comp pier	14.7	7	1	61.3	10	-6.42	2,075.9
mat	117	19.5	2.5	3,768.4	4.42	-8.17	10,095.4
Σ total	209.2			4,252.0			15,400.8

$$\begin{aligned}
 I_y &= \{ \Sigma (Q_y) + \Sigma (J_y) - \text{TotalWt} [(CG_x)^2 + (CG_z)^2] \} / \text{gravity} \\
 &= \{ (4,252.0 \text{ K-ft}^2) + (15,400.8 \text{ K-ft}^2) - (209.2 \text{ kips}) [(4.42 \text{ ft})^2 + (6.26 \text{ ft})^2] \} / (32.2 \text{ ft/sec}^2) \\
 &= 227.2 \text{ ft-K-sec}^2
 \end{aligned}$$

Mass Moment of Inertia about Z axis:

(Attachment 10, Section B7)

part:	CW (kips)	L _x (ft)	L _y (ft)	Q _z (K-ft ²)	δ _x (ft)	δ _y (ft)	J _z (K-ft ²)
motor	15	-	-	0	0.92	0	12.7
compressor	20	-	-	0	9.67	0.5	1,875.2
motor pier	42.5	10.5	9	677.3	1.25	0	66.4
comp pier	14.7	7	14	300.1	10	0	1,470.0
mat	117	19.5	16	6,203.4	4.42	0.08	2,286.5
Σ total	209.2			7,180.8			5,710.8

$$\begin{aligned}
 I_z &= \{ \Sigma (Q_z) + \Sigma (J_z) - \text{TotalWt} [(CG_x)^2 + (CG_y)^2] \} / \text{gravity} \\
 &= \{ (7,180.8 \text{ K-ft}^2) + (5,710.8 \text{ K-ft}^2) - (209.2 \text{ kips}) [(4.42 \text{ ft})^2 + (0.09 \text{ ft})^2] \} / (32.2 \text{ ft/sec}^2) \\
 &= 273.4 \text{ ft-K-sec}^2
 \end{aligned}$$

Compute Center of Gravity Offsets:

(Attachment 10, Section B9)

$$OS_x = \text{abs} \{ 100 [CG_x - \delta_x] / L_x \} = \text{abs} \{ (100) [(4.42 \text{ ft}) - (4.42 \text{ ft})] / (19.5 \text{ ft}) \} = 0\% \text{ as expected OK}$$

$$OS_y = \text{abs} \{ 100 [CG_y - \delta_y] / L_y \} = \text{abs} \{ (100) [(0.09 \text{ ft}) - (0.08 \text{ ft})] / (16.0 \text{ ft}) \} = 0.1\% < 5\% \text{ OK}$$

Soil Bearing:

(Attachment 10, Section C1)

$$\begin{aligned}
 SB_{\text{net}} &= TW / [(L_x)(L_y)] - [\text{MatHt} - \text{GradeHt}](W_s) \\
 &= (209.2 \text{ kips}) / [(16.0 \text{ ft})(19.5 \text{ ft})] - [(9.42 \text{ ft}) - (6.92 \text{ ft})](0.11 \text{ kcf}) \\
 &= 0.396 \text{ ksf} < 0.5 (3 \text{ ksf}) \text{ OK}
 \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Equivalent Radii:

(Attachment 10, Section C2)

$$R_t = \sqrt{(L_x)(L_y)/\pi} = \sqrt{(19.5 \text{ ft})(16.0 \text{ ft})/\pi} = 9.96 \text{ ft}$$

$$R_x = \sqrt[4]{(L_x)(L_y)^3/3\pi} = \sqrt[4]{(19.5 \text{ ft})(16.0 \text{ ft})^3/3\pi} = 9.59 \text{ ft}$$

$$R_y = \sqrt[4]{(L_y)(L_x)^3/3\pi} = \sqrt[4]{(16.0 \text{ ft})(19.5 \text{ ft})^3/3\pi} = 10.59 \text{ ft}$$

$$R_z = \sqrt[4]{(L_x)(L_y)[(L_x)^2 + (L_y)^2]/6\pi} = \sqrt[4]{(19.5 \text{ ft})(16.0 \text{ ft})[(19.5 \text{ ft})^2 + (16.0 \text{ ft})^2]/6\pi} = 10.13 \text{ ft}$$

Static stiffnesses:

(Attachment 10, Section D3)

Veletsos stiffness and damping equations will be used for the uniform soil conditions under this foundation.

$$K_{tx}^s = K_{ty}^s = 8 (G_s)(R_t) / (2 - \mu_s) = 8 (720 \text{ ksf})(9.97 \text{ ft}) / (2 - 0.44) = 36,812 \text{ K/ft}$$

$$K_{tz}^s = 4 (G_s)(R_t) / (1 - \mu_s) = 4 (720 \text{ ksf})(9.97 \text{ ft}) / (1 - 0.44) = 51,274 \text{ K/ft}$$

$$K_{rx}^s = 8 (G_s)(R_x)^3 / [3 (1 - \mu_s)] = 8 (720 \text{ ksf})(9.59 \text{ ft})^3 / [3 (1 - 0.44)] = 3,023,911 \text{ ft-K/rad}$$

$$K_{ry}^s = 8 (G_s)(R_y)^3 / [3 (1 - \mu_s)] = 8 (720 \text{ ksf})(10.59 \text{ ft})^3 / [3 (1 - 0.44)] = 4,071,937 \text{ ft-K/rad}$$

$$K_{rz}^s = 16 (G_s)(R_z)^3 / 3 = 16 (720 \text{ ksf})(10.13 \text{ ft})^3 / 3 = 3,991,715 \text{ ft-K/rad}$$

Dimensionless Frequencies for Base Impedance: (at primary machine speed)

(Attachment 10, Section D4)

$$MS = (300 \text{ rpm})(2\pi \text{ rad/rev}) / (60 \text{ sec/min}) = 31.42 \text{ rad/sec}$$

$$A_t = (MS)(R_t) \sqrt{W_s / [(G_s)(\text{gravity})]} = (31.42 \text{ rad/sec})(9.97 \text{ ft}) \sqrt{(0.11 \text{ kcf}) / [(720 \text{ K/ft}^2)(32.2 \text{ ft/sec}^2)]} = 0.68 \cong 0.70$$

$$A_x = 0.68 (9.59 \text{ ft}) / (9.97 \text{ ft}) = 0.66 \cong 0.65$$

$$A_y = 0.68 (10.59 \text{ ft}) / (9.97 \text{ ft}) = 0.72 \cong 0.70$$

$$A_z = 0.68 (10.13 \text{ ft}) / (9.97 \text{ ft}) = 0.69 \cong 0.70$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Stiffness & Damping of Underlying Soil: (at primary machine speed) (Attachment 10, Section D5, D6)

From Attachment 13 using rounded dimensionless frequencies and $\mu_s \cong 0.45$, select:

$$\begin{array}{ll} S_{tx}^c = 1.000 & N_{tx}^c = 0.615 \\ S_{ty}^c = 1.000 & N_{ty}^c = 0.615 \\ S_{tz}^c = 0.907 & N_{tz}^c = 0.855 \end{array}$$

$$\begin{array}{ll} S_{rx}^c = 0.900 & N_{rx}^c = 0.073 \\ S_{ry}^c = 0.888 & N_{ry}^c = 0.082 \\ S_{rz}^c = 0.920 & N_{rz}^c = 0.055 \end{array}$$

$$K_{tx}^c = K_{ty}^c = (S_{tx}^c)(K_{tx}^s) = (1.000)(36,812 \text{ K/ft}) = 36,812 \text{ K/ft}$$

$$K_{tz}^c = (S_{tz}^c)(K_{tz}^s) = (0.907)(51,274 \text{ K/ft}) = 46,506 \text{ K/ft}$$

$$K_{rx}^c = (S_{rx}^c)(K_{rx}^s) = (0.900)(3,023,911 \text{ ft-K/rad}) = 2,721,520 \text{ ft-K/rad}$$

$$K_{ry}^c = (S_{ry}^c)(K_{ry}^s) = (0.888)(4,071,937 \text{ ft-K/rad}) = 3,615,880 \text{ ft-K/rad}$$

$$K_{rz}^c = (S_{rz}^c)(K_{rz}^s) = (0.920)(3,991,715 \text{ ft-K/rad}) = 3,672,378 \text{ ft-K/rad}$$

$$C_{tx}^c = C_{ty}^c = (N_{tx}^c)(K_{tx}^s)(A_t) / MS = (0.615)(36,812 \text{ K/ft})(0.68) / (31.42 \text{ rad/sec}) = 490 \text{ K-sec/ft}$$

$$C_{tz}^c = (N_{tz}^c)(K_{tz}^s)(A_t) / MS = (0.855)(51,274 \text{ K/ft})(0.68) / (31.42 \text{ rad/sec}) = 949 \text{ K-sec/ft}$$

$$C_{rx}^c = (N_{rx}^c)(K_{rx}^s)(A_x) / MS = (0.073)(3,023,911 \text{ ft-K/rad})(0.66) / (31.42 \text{ rad/sec}) = 4,637 \text{ ft-K-sec/rad}$$

$$C_{ry}^c = (N_{ry}^c)(K_{ry}^s)(A_y) / MS = (0.082)(4,071,937 \text{ ft-K/rad})(0.72) / (31.42 \text{ rad/sec}) = 7,651 \text{ ft-K-sec/rad}$$

$$C_{rz}^c = (N_{rz}^c)(K_{rz}^s)(A_z) / MS = (0.055)(3,991,715 \text{ ft-K/rad})(0.69) / (31.42 \text{ rad/sec}) = 4,821 \text{ ft-K-sec/rad}$$

Dimensionless Frequencies for Embedment Impedance: (at primary machine speed) (Attachment 10, Section E4)

$$MS = (300 \text{ rpm})(2\pi \text{ rad/rev}) / (60 \text{ sec/min}) = 31.42 \text{ rad/sec}$$

$$A_t = (MS)(R_t) \sqrt{W_f / [(G_f)(\text{gravity})]} = (31.42 \text{ rad/sec})(9.97 \text{ ft}) \sqrt{(0.12 \text{ kcf}) / [(1,440 \text{ K/ft}^2)(32.2 \text{ ft/sec}^2)]} = 0.50$$

$$A_x = 0.50 (9.59 \text{ ft}) / (9.97 \text{ ft}) = 0.48 \cong 0.50$$

$$A_y = 0.50 (10.59 \text{ ft}) / (9.97 \text{ ft}) = 0.53 \cong 0.55$$

$$A_z = 0.50 (10.13 \text{ ft}) / (9.97 \text{ ft}) = 0.51 \cong 0.50$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Stiffness & Damping of Embedded Mat: (at primary machine speed) (Attachment 10, Sections E3, E5, E6)

effective mat embedment,

$$h = (2/3)(\text{depth of mat}) = (2/3)(2.0 \text{ ft}) = 1.33 \text{ ft}$$

from Attachment 14 using rounded dimensionless frequencies and $\mu_f = 0.35$, select:

$$\begin{array}{ll} S_{tx}^e = 3.944 & N_{tx}^e = 5.438 \\ S_{ty}^e = 3.944 & N_{ty}^e = 5.438 \\ S_{tz}^e = 2.568 & N_{tz}^e = 3.710 \end{array}$$

$$\begin{array}{ll} S_{rx}^e = 2.519 & N_{rx}^e = 0.899 \\ S_{ry}^e = 2.462 & N_{ry}^e = 1.046 \\ S_{rz}^e = 11.321 & N_{rz}^e = 1.799 \end{array}$$

$$K_{tx}^e = K_{ty}^e = (S_{tx}^e)(G_f)(h) = (3.944)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) = 7,554 \text{ K/ft}$$

$$K_{tze} = (S_{tz}^e)(G_f)(h) = (2.568)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) = 4,918 \text{ K/ft}$$

$$\begin{aligned} K_{rx}^e &= \{(S_{rx}^e) + [(h)/(R_x)]^2 (S_{ty}^e) / 12\} (G_f)(h)(R_x)^2 \\ &= \{(2.519) + [(1.33 \text{ ft})/(9.59 \text{ ft})]^2 (3.944) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(9.59 \text{ ft})^2 \\ &= 444,803 \text{ ft-K/rad} \end{aligned}$$

$$\begin{aligned} K_{ry}^e &= \{(S_{ry}^e) + [(h)/(R_y)]^2 (S_{tx}^e) / 12\} (G_f)(h)(R_y)^2 \\ &= \{(2.462) + [(1.33 \text{ ft})/(10.59 \text{ ft})]^2 (3.944) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.59 \text{ ft})^2 \\ &= 529,917 \text{ ft-K/rad} \end{aligned}$$

$$K_{rz}^e = (S_{rz}^e)(G_f)(h)(R_z)^2 = (11.321)(1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.13 \text{ ft})^2 = 2,224,937 \text{ ft-K/rad}$$

$$C_{tx}^e = C_{ty}^e = (N_{tx}^e)(G_f)(h) / MS = (5.438)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) / (31.42 \text{ rad/sec}) = 331 \text{ K-sec/ft}$$

$$C_{tze} = (N_{tz}^e)(G_f)(h) / MS = (3.710)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) / (31.42 \text{ rad/sec}) = 226 \text{ K-sec/ft}$$

$$\begin{aligned} C_{rx}^e &= \{(N_{rx}^e) + [(h)/(R_x)]^2 (N_{ty}^e) / 12\} (G_f)(h)(R_x)^2 / MS \\ &= \{(0.899) + [(1.33 \text{ ft})/(9.59 \text{ ft})]^2 (5.438) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(9.59 \text{ ft})^2 / (31.42 \text{ rad/sec}) \\ &= 5,089 \text{ ft-K-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{ry}^e &= \{(N_{ry}^e) + [(h)/(R_y)]^2 (N_{tx}^e) / 12\} (G_f)(h)(R_y)^2 / MS \\ &= \{(1.046) + [(1.33 \text{ ft})/(10.59 \text{ ft})]^2 (5.438) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.59 \text{ ft})^2 / (31.42 \text{ rad/sec}) \\ &= 7,199 \text{ ft-K-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{rz}^e &= (N_{rz}^e)(G_f)(h)(R_z)^2 / MS \\ &= (1.799)(1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.13 \text{ ft})^2 / (31.42 \text{ rad/sec}) \\ &= 11,253 \text{ ft-K-sec/rad} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Combined Stiffness & Damping: (at primary machine speed)

(Attachment 10, Sections F3, F4, F5)

$$\delta_c = (3.92 \text{ ft}) + (3.0 \text{ ft}) + (2.5 \text{ ft}) - (6.28 \text{ ft}) = 3.14 \text{ ft}$$

$$\delta_e = (3.14 \text{ ft}) - (1.33 \text{ ft}) / 2 = 2.48 \text{ ft}$$

$$K_{tx} = K_{ty} = K_{tx}^c + K_{tx}^e = (36,812 \text{ K/ft}) + (7,554 \text{ K/ft}) = 44,366 \text{ K/ft}$$

$$K_{tz} = K_{tz}^c + K_{tz}^e = (46,506 \text{ K/ft}) + (4,918 \text{ K/ft}) = 51,424 \text{ K/ft}$$

$$\begin{aligned} K_{rx} &= K_{rx}^c + K_{rx}^e + (K_{ty}^c)(\delta_c)^2 + (K_{ty}^e)(\delta_e)^2 \\ &= (2,721,520 \text{ ft-K/rad}) + (444,803 \text{ ft-K/rad}) + (36,812 \text{ K/ft})(3.14 \text{ ft})^2 + (7,554 \text{ K/ft})(2.48 \text{ ft})^2 \\ &= 3,575,735 \text{ ft-K/rad} \end{aligned}$$

$$\begin{aligned} K_{ry} &= K_{ry}^c + K_{ry}^e + (K_{tx}^c)(\delta_c)^2 + (K_{tx}^e)(\delta_e)^2 \\ &= (3,615,880 \text{ ft-K/rad}) + (529,917 \text{ ft-K/rad}) + (36,812 \text{ K/ft})(3.14 \text{ ft})^2 + (7,554 \text{ K/ft})(2.48 \text{ ft})^2 \\ &= 4,555,209 \text{ ft-K/rad} \end{aligned}$$

$$K_{rz} = K_{rz}^c + K_{rz}^e = (3,672,378 \text{ ft-K/rad}) + (2,224,937 \text{ ft-K/rad}) = 5,897,315 \text{ ft-K/rad}$$

$$K_{cx} = K_{cy} = - (K_{ty}^c)(\delta_c) - (K_{ty}^e)(\delta_e) = - (36,812 \text{ K/ft})(3.14 \text{ ft}) - (7,554 \text{ K/ft})(2.48 \text{ ft}) = -134,324 \text{ K/rad}$$

$$C_{tx} = C_{ty} = C_{tx}^c + C_{tx}^e = (490 \text{ K-sec/ft}) + (331 \text{ K-sec/ft}) = 821 \text{ K-sec/ft}$$

$$C_{tz} = C_{tz}^c + C_{tz}^e = (949 \text{ K-sec/ft}) + (226 \text{ K-sec/ft}) = 1,175 \text{ K-sec/ft}$$

$$\begin{aligned} C_{rx} &= C_{rx}^c + C_{rx}^e + (C_{ty}^c)(\delta_c)^2 + (C_{ty}^e)(\delta_e)^2 \\ &= (4,637 \text{ ft-K-sec/rad}) + (5,089 \text{ ft-K-sec/rad}) + (490 \text{ K-sec/ft})(3.14 \text{ ft})^2 + (331 \text{ K-sec/ft})(2.48 \text{ ft})^2 \\ &= 16,593 \text{ ft-K-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{ry} &= C_{ry}^c + C_{ry}^e + (C_{tx}^c)(\delta_c)^2 + (C_{tx}^e)(\delta_e)^2 \\ &= (7,651 \text{ ft-K-sec/rad}) + (7,199 \text{ ft-K-sec/rad}) + (490 \text{ K-sec/ft})(3.14 \text{ ft})^2 + (331 \text{ K-sec/ft})(2.48 \text{ ft})^2 \\ &= 21,717 \text{ ft-K-sec/rad} \end{aligned}$$

$$C_{rz} = C_{rz}^c + C_{rz}^e = (4,821 \text{ ft-K-sec/rad}) + (11,253 \text{ ft-K-sec/rad}) = 16,074 \text{ ft-K-sec/rad}$$

$$C_{cx} = C_{cy} = - (C_{ty}^c)(\delta_c) - (C_{ty}^e)(\delta_e) = - (490 \text{ K-sec/ft})(3.14 \text{ ft}) - (331 \text{ K-sec/ft})(2.48 \text{ ft}) = -2,359 \text{ K-sec/rad}$$

Determine Primary Loads at Center of Gravity:

(Attachment 10, Section F6)

$$F_x = F_x^v = 1.0 \text{ kips}$$

$$F_y = F_y^v = 1.5 \text{ kips}$$

$$F_z = F_z^v = 0.875 \text{ kips}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$\begin{aligned} M_x &= M_x^v + F_y^v [\text{abs}(\delta_z - CG_z)] + F_z^v [\text{abs}(\delta_y - CG_y)] \\ &= (2.1 \text{ ft-K}) + (1.5 \text{ kips}) [\text{abs}(0 - 6.28 \text{ ft})] + (0.875 \text{ kips}) [\text{abs}(0 - 0.09 \text{ ft})] \\ &= 11.599 \text{ ft-K} \end{aligned}$$

$$\begin{aligned} M_y &= M_y^v + F_x^v [\text{abs}(\delta_z - CG_z)] + F_z^v [\text{abs}(\delta_x - CG_x)] \\ &= (0.2 \text{ ft-K}) + (1.0 \text{ kips}) [\text{abs}(0 - 6.28 \text{ ft})] + (0.875 \text{ kips}) [\text{abs}(10.0 \text{ ft} - 4.42 \text{ ft})] \\ &= 11.363 \text{ ft-K} \end{aligned}$$

$$\begin{aligned} M_z &= M_z^v + F_x^v [\text{abs}(\delta_y - CG_y)] + F_y^v [\text{abs}(\delta_x - CG_x)] \\ &= (3.1 \text{ ft-K}) + (1.0 \text{ kips}) [\text{abs}(0 - 0.09 \text{ ft})] + (1.5 \text{ kips}) [\text{abs}(10.0 \text{ ft} - 4.42 \text{ ft})] \\ &= 11.560 \text{ ft-K} \end{aligned}$$

Primary Vertical Translation:

(Attachment 10, Sections G3, G4, G5, G6)

$$a = K_{tz} - (\text{mass})(MS)^2 = (51,424 \text{ K/ft}) - (6.5 \text{ K-sec}^2/\text{ft})(31.42 \text{ rad/sec})^2 = 45,007 \text{ K/ft}$$

$$b = (C_{tz})(MS) = (1,175 \text{ K-sec/ft})(31.42 \text{ rad/sec}) = 36,919 \text{ K/ft}$$

$$\Delta_z = F_z / \sqrt{(a)^2 + (b)^2} = (0.875 \text{ kips}) / \sqrt{(45,007 \text{ K/ft})^2 + (36,919 \text{ K/ft})^2} = 1.503 \text{ E} - 5 \text{ ft}$$

by trial and error, $NF_1 = 75.5 \text{ rad/sec}$ (721 rpm)

$$FR_1 = MS / NF_1 = (31.42 \text{ rad/sec}) / (75.5 \text{ rad/sec}) = 0.42 < 0.8, \text{ OK}$$

Primary Torsional Rocking:

(Attachment 10, Section G8)

$$a = (K_{tz}) - (I_z)(MS)^2 = (5,897,315 \text{ ft-K/rad}) - (273.4 \text{ ft-K-sec}^2)(31.42 \text{ rad/sec})^2 = 5,627,410 \text{ ft-K/rad}$$

$$b = (C_{tz})(MS) = (16,074 \text{ ft-K-sec/rad})(31.42 \text{ rad/sec}) = 505,045 \text{ ft-K/rad}$$

$$\phi_z = M_z / \sqrt{(a)^2 + (b)^2} = (11.560 \text{ ft-K}) / \sqrt{(5,627,410 \text{ ft-K/rad})^2 + (505,045 \text{ ft-K/rad})^2} = 2.046 \text{ E} - 6 \text{ rad}$$

by trial and error, $NF_2 = 129.3 \text{ rad/sec}$ (1,235 rpm)

$$FR_2 = MS / NF_2 = (31.42 \text{ rad/sec}) / (129.3 \text{ rad/sec}) = 0.24 < 0.8, \text{ OK}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Primary Transverse Translation & Rocking:

(Attachment 10, Sections H3, H4, H5, H6)

$$\begin{aligned} a_1 &= [(K_{rx}) - (I_x)(MS)^2](F_y) - (K_{cx})(M_x) \\ &= [(3,575,735 \text{ ft-K/rad}) - (141.8 \text{ ft-K-sec}^2)(31.42 \text{ rad/sec})^2](1.5 \text{ kips}) - (-134,324 \text{ K/rad})(11.599 \text{ ft-K}) \\ &= 6.712 \text{ E6 ft-K}^2 \end{aligned}$$

$$\begin{aligned} a_2 &= (C_{rx})(MS)(F_y) - (C_{cx})(MS)(M_x) \\ &= (16,593 \text{ ft-K-sec/rad})(31.42 \text{ rad/sec})(1.5 \text{ kips}) - (-2,359 \text{ K-sec/rad})(31.42 \text{ rad/sec})(11.599 \text{ ft-K}) \\ &= 1.642 \text{ E6 ft-K}^2 \end{aligned}$$

$$\begin{aligned} b_1 &= [(K_{ty}) - (\text{mass})(MS)^2](M_x) - (K_{cx})(F_y) \\ &= [(44,366 \text{ K/ft}) - (6.5 \text{ K-sec}^2/\text{ft})(31.42 \text{ rad/sec})^2](11.599 \text{ ft-K}) - (-134,324 \text{ K/rad})(1.5 \text{ kips}) \\ &= 6.417 \text{ E5 kips}^2 \end{aligned}$$

$$\begin{aligned} b_2 &= (C_{ty})(MS)(M_x) - (C_{cx})(MS)(F_y) \\ &= (821 \text{ K-sec/ft})(31.42 \text{ rad/sec})(11.599 \text{ ft-K}) - (-2,359 \text{ K-sec/rad})(31.42 \text{ rad/sec})(1.5 \text{ kips}) \\ &= 4.104 \text{ E5 kips}^2 \end{aligned}$$

$$\begin{aligned} c_1 &= (\text{mass})(I_x)(MS)^4 + (K_{ty})(K_{rx}) - (K_{cx})^2 - [(\text{mass})(K_{rx}) + (I_x)(K_{ty}) + (C_{ty})(C_{rx}) - (C_{cx})^2](MS)^2 \\ &= (6.5 \text{ K-sec}^2/\text{ft})(141.8 \text{ ft-K-sec}^2)(31.42 \text{ rad/sec})^4 \\ &\quad + (44,366 \text{ K/ft})(3,575,735 \text{ ft-K/rad}) - (-134,324 \text{ K/rad})^2 \\ &\quad - [(6.5 \text{ K-sec}^2/\text{ft})(3,575,735 \text{ ft-K/rad}) + (141.8 \text{ ft-K-sec}^2)(44,366 \text{ K/ft}) \\ &\quad + (821 \text{ K-sec/ft})(16,593 \text{ ft-K-sec/rad}) - (-2,359 \text{ K-sec/rad})^2] (31.42 \text{ rad/sec})^2 \\ &= 1.044 \text{ E11 kips}^2 \end{aligned}$$

$$\begin{aligned} c_2 &= [(C_{ty})(K_{rx}) + (C_{rx})(K_{ty}) - 2(C_{cx})(K_{cx})](MS) - [(\text{mass})(C_{rx}) + (I_x)(C_{ty})](MS)^3 \\ &= [(821 \text{ K-sec/ft})(3,575,735 \text{ ft-K/rad}) + (16,593 \text{ ft-K-sec/rad})(44,366 \text{ K/ft}) \\ &\quad - 2(-2,359 \text{ K-sec/rad})(-134,324 \text{ K/rad})](31.42 \text{ rad/sec}) \\ &\quad - [(6.5 \text{ K-sec}^2/\text{ft})(16,593 \text{ ft-K-sec/rad}) + (141.8 \text{ ft-K-sec}^2)(821 \text{ K-sec/ft})] (31.42 \text{ rad/sec})^3 \\ &= 8.850 \text{ E10 kips}^2 \end{aligned}$$

$$\begin{aligned} \Delta_y &= \sqrt{[(a_1)^2 + (a_2)^2] / [(c_1)^2 + (c_2)^2]} \\ &= \sqrt{[(6.712 \text{ E6 ft-K}^2)^2 + (1.642 \text{ E6 ft-K}^2)^2] / [(1.044 \text{ E11 kips}^2)^2 + (8.850 \text{ E10 kips}^2)^2]} \\ &= 5.049 \text{ E-5 ft} \end{aligned}$$

$$\begin{aligned} \phi_x &= \sqrt{[(b_1)^2 + (b_2)^2] / [(c_1)^2 + (c_2)^2]} \\ &= \sqrt{[(6.417 \text{ E5 kips}^2)^2 + (4.104 \text{ E5 kips}^2)^2] / [(1.044 \text{ E11 kips}^2)^2 + (8.850 \text{ E10 kips}^2)^2]} \\ &= 5.565 \text{ E-6 rad} \end{aligned}$$

by trial and error, $NF_3 = 74.3 \text{ rad/sec}$ (710 rpm) and $NF_4 = 132.9 \text{ rad/sec}$ (1,269 rpm)

$$FR_3 = MS / NF_3 = (31.42 \text{ rad/sec}) / (74.3 \text{ rad/sec}) = 0.42 < 0.8, \text{ OK}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$FR_4 = MS / NF_4 = (31.42 \text{ rad/sec}) / (132.9 \text{ rad/sec}) = 0.24 < 0.8, \text{ OK}$$

Primary Longitudinal Translation & Rocking:

(Attachment 10, Section H8)

$$\begin{aligned} a_1 &= [(K_{ry}) - (I_y)(MS)^2](F_x) - (K_{cy})(M_y) \\ &= [(4,555,209 \text{ ft-K/rad}) - (227.2 \text{ ft-K-sec}^2)(31.42 \text{ rad/sec})^2](1.0 \text{ kips}) - (-134,324 \text{ K/rad})(11.363 \text{ ft-K}) \\ &= 5.857 \text{ E6 ft-K}^2 \end{aligned}$$

$$\begin{aligned} a_2 &= (C_{ry})(MS)(F_x) - (C_{cy})(MS)(M_y) \\ &= (21,717 \text{ ft-K-sec/rad})(31.42 \text{ rad/sec})(1.0 \text{ kips}) - (-2,359 \text{ K-sec/rad})(31.42 \text{ rad/sec})(11.363 \text{ ft-K}) \\ &= 1.525 \text{ E6 ft-K}^2 \end{aligned}$$

$$\begin{aligned} b_1 &= [(K_{tx}) - (\text{mass})(MS)^2](M_y) - (K_{cy})(F_x) \\ &= [(44,366 \text{ K/ft}) - (6.5 \text{ K-sec}^2/\text{ft})(31.42 \text{ rad/sec})^2](11.363 \text{ ft-K}) - (-134,324 \text{ K/rad})(1.0 \text{ kips}) \\ &= 5.655 \text{ E5 kips}^2 \end{aligned}$$

$$\begin{aligned} b_2 &= (C_{tx})(MS)(M_y) - (C_{cy})(MS)(F_x) \\ &= (821 \text{ K-sec/ft})(31.42 \text{ rad/sec})(11.363 \text{ ft-K}) - (-2,359 \text{ K-sec/rad})(31.42 \text{ rad/sec})(1.0 \text{ kips}) \\ &= 3.672 \text{ E5 kips}^2 \end{aligned}$$

$$\begin{aligned} c_1 &= (\text{mass})(I_y)(MS)^4 + (K_{tx})(K_{ry}) - (K_{cy})^2 - [(\text{mass})(K_{ry}) + (I_y)(K_{tx}) + (C_{tx})(C_{ry}) - (C_{cy})^2](MS)^2 \\ &= (6.5 \text{ K-sec}^2/\text{ft})(227.2 \text{ ft-K-sec}^2)(31.42 \text{ rad/sec})^4 \\ &\quad + (44,366 \text{ K/ft})(4,555,209 \text{ ft-K/rad}) - (-134,324 \text{ K/rad})^2 \\ &\quad - [(6.5 \text{ K-sec}^2/\text{ft})(4,555,209 \text{ ft-K/rad}) + (227.2 \text{ ft-K-sec}^2)(44,366 \text{ K/ft}) \\ &\quad + (821 \text{ K-sec/ft})(21,717 \text{ ft-K-sec/rad}) - (-2,359 \text{ K-sec/rad})^2](31.42 \text{ rad/sec})^2 \\ &= 1.342 \text{ E11 kips}^2 \end{aligned}$$

$$\begin{aligned} c_2 &= [(C_{tx})(K_{ry}) + (C_{ry})(K_{tx}) - 2(C_{cy})(K_{cy})](MS) - [(\text{mass})(C_{ry}) + (I_y)(C_{tx})](MS)^3 \\ &= [(821 \text{ K-sec/ft})(4,555,209 \text{ ft-K/rad}) + (21,717 \text{ ft-K-sec/rad})(44,366 \text{ K/ft}) \\ &\quad - 2(-2,359 \text{ K-sec/rad})(-134,324 \text{ K/rad})](31.42 \text{ rad/sec}) \\ &\quad - [(6.5 \text{ K-sec}^2/\text{ft})(21,717 \text{ ft-K-sec/rad}) + (227.2 \text{ ft-K-sec}^2)(821 \text{ K-sec/ft})](31.42 \text{ rad/sec})^3 \\ &= 1.177 \text{ E11 kips}^2 \end{aligned}$$

$$\begin{aligned} \Delta_x &= \sqrt{[(a_1)^2 + (a_2)^2] / [(c_1)^2 + (c_2)^2]} \\ &= \sqrt{[(5.857 \text{ E6 ft-K}^2)^2 + (1.525 \text{ E6 ft-K}^2)^2] / [(1.342 \text{ E11 kips}^2)^2 + (1.177 \text{ E11 kips}^2)^2]} \\ &= 3.391 \text{ E-5 ft} \end{aligned}$$

$$\begin{aligned} \phi_y &= \sqrt{[(b_1)^2 + (b_2)^2] / [(c_1)^2 + (c_2)^2]} \\ &= \sqrt{[(5.655 \text{ E5 kips}^2)^2 + (3.672 \text{ E5 kips}^2)^2] / [(1.342 \text{ E11 kips}^2)^2 + (1.177 \text{ E10 kips}^2)^2]} \\ &= 3.777 \text{ E-6 rad} \end{aligned}$$

by trial and error, $NF_5 = 75.0 \text{ rad/sec}$ (716 rpm) and $NF_6 = 119.3 \text{ rad/sec}$ (1,139 rpm)



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$FR_5 = MS / NF_5 = (31.42 \text{ rad/sec}) / (75.0 \text{ rad/sec}) = 0.42 < 0.8, \text{ OK}$$

$$FR_6 = MS / NF_6 = (31.42 \text{ rad/sec}) / (119.3 \text{ rad/sec}) = 0.26 < 0.8, \text{ OK}$$

Double Amplitudes at Point #4: (at primary machine speed)

(Attachment 10, Section I1)

$$D_x = \text{abs} [PT_x - CG_x] = \text{abs} [(13.5 \text{ ft}) - (4.42 \text{ ft})] = 9.08 \text{ ft}$$

$$D_y = \text{abs} [PT_y - CG_y] = \text{abs} [(7.0 \text{ ft}) - (0.09 \text{ ft})] = 6.91 \text{ ft}$$

$$D_z = \text{abs} [PT_z - CG_z] = \text{abs} [(-5.92 \text{ ft}) - (-6.28 \text{ ft})] = 0.36 \text{ ft}$$

$$\begin{aligned} DA_x &= \{ \Delta_x + D_y (\phi_z) + D_z (\phi_y) \} (24000) \\ &= \{ (3.391 \text{ E-5 ft}) + (6.91 \text{ ft})(2.046 \text{ E-6 rad}) + (0.36 \text{ ft})(3.777 \text{ E-6 rad}) \} (24000) \\ &= 1.19 \text{ mils} \end{aligned}$$

$$\begin{aligned} DA_y &= \{ \Delta_y + D_x (\phi_z) + D_z (\phi_x) \} (24000) \\ &= \{ (5.049 \text{ E-5 ft}) + (9.08 \text{ ft})(2.046 \text{ E-6 rad}) + (0.36 \text{ ft})(5.565 \text{ E-6 rad}) \} (24000) \\ &= 1.71 \text{ mils} \end{aligned}$$

$$\begin{aligned} DA_z &= \{ (\Delta_z + D_x (\phi_y) + D_y (\phi_x)) \} (24000) \\ &= \{ (1.503 \text{ E-5 ft}) + (9.08 \text{ ft})(3.777 \text{ E-6 rad}) + (6.91 \text{ ft})(5.565 \text{ E-6 rad}) \} (24000) \\ &= 2.11 \text{ mils} \end{aligned}$$

Dimensionless Frequencies for Base Impedance: (at secondary machine speed)

(Attachment 10, Section D4)

$$MS = (600 \text{ rpm})(2 \pi \text{ rad/rev}) / (60 \text{ sec/min}) = 62.84 \text{ rad/sec}$$

$$A_t = (MS)(R_t) \sqrt{W_s / [(G_s)(\text{gravity})]} = (62.84 \text{ rad/sec})(9.97 \text{ ft}) \sqrt{(0.11 \text{ K/ft}^3) / [(720 \text{ k/ft}^2)(32.2 \text{ ft/sec}^2)]} = 1.36 \cong 1.35$$

$$A_x = 1.36 (9.59 \text{ ft}) / (9.97 \text{ ft}) = 1.31 \cong 1.30$$

$$A_y = 1.36 (10.59 \text{ ft}) / (9.97 \text{ ft}) = 1.44 \cong 1.45$$

$$A_z = 1.36 (10.13 \text{ ft}) / (9.97 \text{ ft}) = 1.38 \cong 1.40$$

Stiffness & Damping of Underlying Soil: (at secondary machine speed)

(Attachment 10, Sections D5, D6)

Static stiffnesses are the same as computed for primary loads.

From Attachment 13 using rounded dimensionless frequencies and $\mu_s \cong 0.45$, select:

$$\begin{aligned} S_{tx}^c &= 1.000 & N_{tx}^c &= 0.615 \\ S_{ty}^c &= 1.000 & N_{ty}^c &= 0.615 \\ S_{tz}^c &= 0.715 & N_{tz}^c &= 0.888 \end{aligned}$$

$$\begin{aligned} S_{rx}^c &= 0.745 & N_{rx}^c &= 0.179 \\ S_{ry}^c &= 0.714 & N_{ry}^c &= 0.197 \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$S_{rz}^c = 0.796 \quad N_{rz}^c = 0.140$$

$$K_{tx}^c = K_{ty}^c = (S_{tx}^c)(K_{tx}^s) = (1.000)(36,812 \text{ K/ft}) = 36,812 \text{ K/ft}$$

$$K_{tz}^c = (S_{tz}^c)(K_{tz}^s) = (0.715)(51,274 \text{ K/ft}) = 36,661 \text{ K/ft}$$

$$K_{rx}^c = (S_{rx}^c)(K_{rx}^s) = (0.745)(3,023,911 \text{ ft-K/rad}) = 2,252,814 \text{ ft-K/rad}$$

$$K_{ry}^c = (S_{ry}^c)(K_{ry}^s) = (0.714)(4,071,937 \text{ ft-K/rad}) = 2,907,363 \text{ ft-K/rad}$$

$$K_{rz}^c = (S_{rz}^c)(K_{rz}^s) = (0.796)(3,991,715 \text{ ft-K/rad}) = 3,177,405 \text{ ft-K/rad}$$

$$C_{tx}^c = C_{ty}^c = (N_{tx}^c)(K_{tx}^s)(A_t) / MS = (0.615)(36,812 \text{ K/ft})(1.36) / (62.84 \text{ rad/sec}) = 490 \text{ K-sec/ft}$$

$$C_{tz}^c = (N_{tz}^c)(K_{tz}^s)(A_t) / MS = (0.888)(51,274 \text{ K/ft})(1.36) / (62.84 \text{ rad/sec}) = 985 \text{ K-sec/ft}$$

$$C_{rx}^c = (N_{rx}^c)(K_{rx}^s)(A_x) / MS = (0.179)(3,023,911 \text{ ft-K/rad})(1.31) / (62.84 \text{ rad/sec}) = 11,284 \text{ ft-K-sec/rad}$$

$$C_{ry}^c = (N_{ry}^c)(K_{ry}^s)(A_y) / MS = (0.197)(4,071,937 \text{ ft-K/rad})(1.44) / (62.84 \text{ rad/sec}) = 18,382 \text{ ft-K-sec/rad}$$

$$C_{rz}^c = (N_{rz}^c)(K_{rz}^s)(A_z) / MS = (0.140)(3,991,715 \text{ ft-K/rad})(1.38) / (62.84 \text{ rad/sec}) = 12,272 \text{ ft-K-sec/rad}$$

Dimensionless Frequencies for Embedment Impedance: (at secondary machine speed)(Attachment 10, Section E4)

$$MS = (600 \text{ rpm})(2 \pi \text{ rad/rev}) / (60 \text{ sec/min}) = 62.84 \text{ rad/sec}$$

$$A_t = (MS)(R_t) \sqrt{W_f / [(G_f)(\text{gravity})]} = (62.84 \text{ rad/sec})(9.97 \text{ ft}) \sqrt{(0.11 \text{ K/ft}^3) / [(720 \text{ k/ft}^2)(32.2 \text{ ft/sec}^2)]} = 1.36 \cong 1.35$$

$$A_x = 1.01 (9.59 \text{ ft}) / (9.97 \text{ ft}) = 0.97 \cong 1.00$$

$$A_y = 1.01 (10.59 \text{ ft}) / (9.97 \text{ ft}) = 1.07 \cong 1.05$$

$$A_z = 1.01 (10.13 \text{ ft}) / (9.97 \text{ ft}) = 1.03 \cong 1.05$$

Stiffness & Damping of Embedded Mat: (at secondary machine speed) (Attachment 10, Sections E3, E5, E6)

$$h = (2/3)(\text{depth of mat}) = (2/3)(2.0 \text{ ft}) = 1.33 \text{ ft}$$

from Attachment 14 using rounded dimensionless frequencies and $\mu_s = 0.35$, select:

$$S_{tx}^e = 4.104 \quad N_{tx}^e = 9.953$$

$$S_{ty}^e = 4.104 \quad N_{ty}^e = 9.953$$

$$S_{tz}^e = 2.836 \quad N_{tz}^e = 6.742$$

$$S_{rx}^e = 2.095 \quad N_{rx}^e = 2.488$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$\begin{aligned} S_{ry}^e &= 2.068 & N_{ry}^e &= 2.654 \\ S_{rz}^e &= 10.418 & N_{rz}^e &= 5.308 \end{aligned}$$

$$K_{tx}^e = K_{ty}^e = (S_{tx}^e)(G_f)(h) = (4.104)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) = 7,860 \text{ K/ft}$$

$$K_{tz}^e = (S_{tz}^e)(G_f)(h) = (2.836)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) = 5,432 \text{ K/ft}$$

$$\begin{aligned} K_{rx}^e &= \{(S_{rx}^e) + [(h)/(R_x)]^2 (S_{ty}^e) / 12\} (G_f)(h)(R_x)^2 \\ &= \{(2.095) + [(1.33 \text{ ft})/(9.59 \text{ ft})]^2 (4.104) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(9.59 \text{ ft})^2 \\ &= 370,166 \text{ ft-K/rad} \end{aligned}$$

$$\begin{aligned} K_{ry}^e &= \{(S_{ry}^e) + [(h)/(R_y)]^2 (S_{tx}^e) / 12\} (G_f)(h)(R_y)^2 \\ &= \{(2.068) + [(1.33 \text{ ft})/(10.59 \text{ ft})]^2 (4.104) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.59 \text{ ft})^2 \\ &= 445,336 \text{ ft-K/rad} \end{aligned}$$

$$K_{rz}^e = (S_{rz}^e)(G_f)(h)(R_z)^2 = (10.418)(1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.13 \text{ ft})^2 = 2,047,469 \text{ ft-K/rad}$$

$$C_{tx}^e = C_{ty}^e = (N_{tx}^e)(G_f)(h) / MS = (9.953)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) / (62.84 \text{ rad/sec}) = 303 \text{ K-sec/ft}$$

$$C_{tz}^e = (N_{tz}^e)(G_f)(h) / MS = (6.742)(1,440 \text{ K/ft}^2)(1.33 \text{ ft}) / (62.84 \text{ rad/sec}) = 205 \text{ K-sec/ft}$$

$$\begin{aligned} C_{rx}^e &= \{(N_{rx}^e) + [(h)/(R_x)]^2 (N_{ty}^e) / 12\} (G_f)(h)(R_x)^2 / MS \\ &= \{(2.488) + [(1.33 \text{ ft})/(9.59 \text{ ft})]^2 (9.953) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(9.59 \text{ ft})^2 / (62.84 \text{ rad/sec}) \\ &= 7,018 \text{ ft-K-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{ry}^e &= \{(N_{ry}^e) + [(h)/(R_y)]^2 (N_{tx}^e) / 12\} (G_f)(h)(R_y)^2 / MS \\ &= \{(2.654) + [(1.33 \text{ ft})/(10.59 \text{ ft})]^2 (9.953) / 12\} (1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.59 \text{ ft})^2 / (62.84 \text{ rad/sec}) \\ &= 9,116 \text{ ft-K-sec/rad} \end{aligned}$$

$$C_{rz}^e = (N_{rz}^e)(G_f)(h)(R_z)^2 / MS = (5.308)(1,440 \text{ K/ft}^2)(1.33 \text{ ft})(10.13 \text{ ft})^2 / (62.84 \text{ rad/sec}) = 16,601 \text{ ft-K-sec/rad}$$

Combined Stiffness & Damping: (at secondary machine speed)

(Attachment 10, Section F3, F4, F5)

$$\delta_c = (3.92 \text{ ft}) + (3.0 \text{ ft}) + (2.5 \text{ ft}) - (6.28 \text{ ft}) = 3.14 \text{ ft}$$

$$\delta_e = (3.14 \text{ ft}) - (1.33 \text{ ft}) / 2 = 2.48 \text{ ft}$$

$$K_{tx} = K_{ty} = K_{tx}^c + K_{tx}^e = (36,812 \text{ K/ft}) + (7,860 \text{ K/ft}) = 44,672 \text{ K/ft}$$

$$K_{tz} = K_{tz}^c + K_{tz}^e = (36,661 \text{ K/ft}) + (5,432 \text{ K/ft}) = 42,093 \text{ K/ft}$$

$$\begin{aligned} K_{rx} &= K_{rx}^c + K_{rx}^e + (K_{ty}^c)(\delta_c)^2 + (K_{ty}^e)(\delta_e)^2 \\ &= (2,252,814 \text{ ft-K/rad}) + (370,166 \text{ ft-K/rad}) + (36,812 \text{ K/ft})(3.14 \text{ ft})^2 + (7,860 \text{ K/ft})(2.48 \text{ ft})^2 \\ &= 3,034,274 \text{ ft-K/rad} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$\begin{aligned} K_{ry} &= K_{ry}^c + K_{ry}^e + (K_{tx}^c)(\delta_c)^2 + (K_{tx}^e)(\delta_e)^2 \\ &= (2,907,363 \text{ ft-K/rad}) + (445,336 \text{ ft-K/rad}) + (36,812 \text{ K/ft})(3.14 \text{ ft})^2 + (7,860 \text{ K/ft})(2.48 \text{ ft})^2 \\ &= 3,763,993 \text{ ft-K/rad} \end{aligned}$$

$$\begin{aligned} K_{rz} &= K_{rz}^c + K_{rz}^e = (3,177,405 \text{ ft-K/rad}) + (2,047,469 \text{ ft-K/rad}) = 5,224,874 \text{ ft-K/rad} \\ K_{cx} &= K_{cy} = - (K_{ty}^c)(\delta_c) - (K_{ty}^e)(\delta_e) = - (36,812 \text{ K/ft})(3.14 \text{ ft}) - (7,860 \text{ K/ft})(2.48 \text{ ft}) = -135,082 \text{ K/rad} \end{aligned}$$

$$C_{tx} = C_{ty} = C_{tx}^c + C_{tx}^e = (490 \text{ K-sec/ft}) + (303 \text{ K-sec/ft}) = 793 \text{ K-sec/ft}$$

$$C_{tz} = C_{tz}^c + C_{tz}^e = (985 \text{ K-sec/ft}) + (205 \text{ K-sec/ft}) = 1,190 \text{ K-sec/ft}$$

$$\begin{aligned} C_{rx} &= C_{rx}^c + C_{rx}^e + (C_{ty}^c)(\delta_c)^2 + (C_{ty}^e)(\delta_e)^2 \\ &= (11,284 \text{ ft-K-sec/rad}) + (7,018 \text{ ft-K-sec/rad}) + (490 \text{ K-sec/ft})(3.14 \text{ ft})^2 + (303 \text{ K-sec/ft})(2.48 \text{ ft})^2 \\ &= 24,997 \text{ ft-K-sec/rad} \end{aligned}$$

$$\begin{aligned} C_{ry} &= C_{ry}^c + C_{ry}^e + (C_{tx}^c)(\delta_c)^2 + (C_{tx}^e)(\delta_e)^2 \\ &= (18,382 \text{ ft-K-sec/rad}) + (9,116 \text{ ft-K-sec/rad}) + (490 \text{ K-sec/ft})(3.14 \text{ ft})^2 + (303 \text{ K-sec/ft})(2.48 \text{ ft})^2 \\ &= 34,193 \text{ ft-K-sec/rad} \end{aligned}$$

$$C_{rz} = C_{rz}^c + C_{rz}^e = (12,272 \text{ ft-K-sec/rad}) + (16,601 \text{ ft-K-sec/rad}) = 28,873 \text{ ft-K-sec/rad}$$

$$C_{cx} = C_{cy} = - (C_{ty}^c)(\delta_c) - (C_{ty}^e)(\delta_e) = - (490 \text{ K-sec/ft})(3.14 \text{ ft}) - (303 \text{ K-sec/ft})(2.48 \text{ ft}) = -2,290 \text{ K-sec/rad}$$

Determine Secondary Loads at Center of Gravity:

(Attachment 10, Section F6)

$$\begin{aligned} F_x &= F_x^v = 0.0 \text{ kips} \\ F_y &= F_y^v = 0.8 \text{ kips} \\ F_z &= F_z^v = 0.0 \text{ kips} \end{aligned}$$

$$\begin{aligned} M_x &= M_x^v + F_y^v [\text{abs}(\delta_z - CG_z)] + F_z^v [\text{abs}(\delta_y - CG_y)] \\ &= (0.9 \text{ ft-K}) + (0.8 \text{ kips})[\text{abs}(0 - 6.28 \text{ ft})] + (0.0 \text{ kips})[\text{abs}(0 - 0.09 \text{ ft})] \\ &= 5.924 \text{ ft-K} \end{aligned}$$

$$\begin{aligned} M_y &= M_y^v + F_x^v [\text{abs}(\delta_z - CG_z)] + F_z^v [\text{abs}(\delta_x - CG_x)] \\ &= (0.0 \text{ ft-K}) + (0.0 \text{ kips})[\text{abs}(0 - 6.28 \text{ ft})] + (0.0 \text{ kips})[\text{abs}(10.0 \text{ ft} - 4.42 \text{ ft})] \\ &= 0.0 \text{ ft-K} \end{aligned}$$

$$\begin{aligned} M_z &= M_z^v + F_x^v [\text{abs}(\delta_y - CG_y)] + F_y^v [\text{abs}(\delta_x - CG_x)] \\ &= (1.5 \text{ ft-K}) + (0.0 \text{ kips})[\text{abs}(0 - 0.09 \text{ ft})] + (0.8 \text{ kips})[\text{abs}(10.0 \text{ ft} - 4.42 \text{ ft})] \\ &= 5.964 \text{ ft-K} \end{aligned}$$

Secondary Vertical Translation:

(Attachment 10, Section G9)

$$\text{since } F_z = 0, \Delta_z = 0$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

Secondary Torsional Rocking:

(Attachment 10, Section G9)

$$a = (K_{rz}) - (I_z)(MS)^2 = (5,224,874 \text{ ft-K/rad}) - (273.4 \text{ ft-K-sec}^2)(62.84 \text{ rad/sec})^2 = 4,145,254 \text{ ft-K/rad}$$

$$b = (C_{rz})(MS) = (28,873 \text{ ft-K-sec/rad})(62.84 \text{ rad/sec}) = 1,814,379 \text{ ft-K/rad}$$

$$\phi_z = M_z / \sqrt{(a)^2 + (b)^2} = (5.964 \text{ ft-K}) / \sqrt{(4,145,254 \text{ ft-K/rad})^2 + (1,814,379 \text{ ft-K/rad})^2} = 1.318 \text{ E-6 rad}$$

$$FR_2 = MS / NF_2 = (62.84 \text{ rad/sec}) / (129.3 \text{ rad/sec}) = 0.49 < 0.8, \text{ OK}$$

Secondary Transverse Translation & Rocking:

(Attachment 10, Section H9)

$$\begin{aligned} a_1 &= [(K_{rx}) - (I_x)(MS)^2](F_y) - (K_{cx})(M_x) \\ &= [(3,034,274 \text{ ft-K/rad}) - (141.8 \text{ ft-K-sec}^2)(62.84 \text{ rad/sec})^2](0.8 \text{ kips}) - (-135,082 \text{ K/rad})(5.924 \text{ ft-K}) \\ &= 2.780 \text{ E6 ft-K}^2 \end{aligned}$$

$$\begin{aligned} a_2 &= (C_{rx})(MS)(F_y) - (C_{cx})(MS)(M_x) \\ &= (24,997 \text{ ft-K-sec/rad})(62.84 \text{ rad/sec})(0.8 \text{ kips}) - (-2,290 \text{ K-sec/rad})(62.84 \text{ rad/sec})(5.924 \text{ ft-K}) \\ &= 2.109 \text{ E6 ft-K}^2 \end{aligned}$$

$$\begin{aligned} b_1 &= [(K_{ty}) - (\text{mass})(MS)^2](M_x) - (K_{cx})(F_y) \\ &= [(44,672 \text{ K/ft}) - (6.5 \text{ K-sec}^2/\text{ft})(62.84 \text{ rad/sec})^2](5.924 \text{ ft-K}) - (-135,082 \text{ K/rad})(0.8 \text{ kips}) \\ &= 2.206 \text{ E5 kips}^2 \end{aligned}$$

$$\begin{aligned} b_2 &= (C_{ty})(MS)(M_x) - (C_{cx})(MS)(F_y) \\ &= (793 \text{ K-sec/ft})(62.84 \text{ rad/sec})(5.924 \text{ ft-K}) - (-2,290 \text{ K-sec/rad})(62.84 \text{ rad/sec})(0.8 \text{ kips}) \\ &= 4.103 \text{ E5 kips}^2 \end{aligned}$$

$$\begin{aligned} c_1 &= (\text{mass})(I_x)(MS)^4 + (K_{ty})(K_{rx}) - (K_{cx})^2 - [(\text{mass})(K_{rx}) + (I_x)(K_{ty}) + (C_{ty})(C_{rx}) - (C_{cx})^2](MS)^2 \\ &= (6.5 \text{ K-sec}^2/\text{ft})(141.8 \text{ ft-K-sec}^2)(62.84 \text{ rad/sec})^4 \\ &\quad + (44,672 \text{ K/ft})(3,034,274 \text{ ft-K/rad}) - (-135,082 \text{ K/rad})^2 \\ &\quad - [(6.5 \text{ K-sec}^2/\text{ft})(3,034,274 \text{ ft-K/rad}) + (141.8 \text{ ft-K-sec}^2)(44,672 \text{ K/ft}) \\ &\quad + (793 \text{ K-sec/ft})(24,997 \text{ ft-K-sec/rad}) - (-2,290 \text{ K-sec/rad})^2] * (62.84 \text{ rad/sec})^2 \\ &= -2.879 \text{ E10 kips}^2 \end{aligned}$$

$$\begin{aligned} c_2 &= [(C_{ty})(K_{rx}) + (C_{rx})(K_{ty}) - 2(C_{cx})(K_{cx})](MS) - [(\text{mass})(C_{rx}) + (I_x)(C_{ty})](MS)^3 \\ &= [(793 \text{ K-sec/ft})(3,034,274 \text{ ft-K/rad}) + (24,997 \text{ ft-K-sec/rad})(44,672 \text{ K/ft}) \\ &\quad - 2(-2,290 \text{ K-sec/rad})(-135,082 \text{ K/rad})](62.84 \text{ rad/sec}) \\ &\quad - [(6.5 \text{ K-sec}^2/\text{ft})(24,997 \text{ ft-K-sec/rad}) + (141.8 \text{ ft-K-sec}^2)(793 \text{ K-sec/ft})] * (62.84 \text{ rad/sec})^3 \\ &= 1.143 \text{ E11 kips}^2 \end{aligned}$$

$$\begin{aligned} \Delta_y &= \sqrt{[(a_1)^2 + (a_2)^2] / [(c_1)^2 + (c_2)^2]} \\ &= \sqrt{[(2.780 \text{ E6 ft-K}^2)^2 + (2.109 \text{ E6 ft-K}^2)^2] / [(-2.879 \text{ E10 kips}^2)^2 + (1.143 \text{ E11 kips}^2)^2]} \end{aligned}$$



VIBRATING MACHINERY FOUNDATIONS ON SOIL

Vibration Calculation

$$= 1.143 \text{ E} - 5 \text{ ft}$$

$$\begin{aligned}\phi_x &= \sqrt{[(b_1)^2 + (b_2)^2] / [(c_1)^2 + (c_2)^2]} \\ &= \sqrt{[(2.206 \text{ E} 5 \text{ kips}^2)^2 + (4.103 \text{ E} 5 \text{ kips}^2)^2] / [(-2.879 \text{ E} 10 \text{ kips}^2)^2 + (1.143 \text{ E} 11 \text{ kips}^2)^2]} \\ &= 3.952 \text{ E} - 6 \text{ rad}\end{aligned}$$

$$\text{FR}_3 = \text{MS} / \text{NF}_3 = (62.84 \text{ rad/sec}) / (74.3 \text{ rad/sec}) = 0.85 < 0.8, \text{ NG}$$

$$\text{FR}_4 = \text{MS} / \text{NF}_4 = (62.84 \text{ rad/sec}) / (132.9 \text{ rad/sec}) = 0.47 < 0.8, \text{ OK}$$

Secondary Longitudinal Translation & Rocking:

(Attachment 10, Section H9)

$$\text{since } F_x = 0 \text{ and } M_y = 0, \Delta_x = 0 \text{ and } \phi_y = 0$$

Double Amplitudes at Point #4: (at secondary machine speed)

(Attachment 10, Section I2)

$$D_x = \text{abs} [\text{PT}_x - \text{CG}_x] = \text{abs} [(13.5 \text{ ft}) - (4.42 \text{ ft})] = 9.08 \text{ ft}$$

$$D_y = \text{abs} [\text{PT}_y - \text{CG}_y] = \text{abs} [(7.0 \text{ ft}) - (0.09 \text{ ft})] = 6.91 \text{ ft}$$

$$D_z = \text{abs} [\text{PT}_z - \text{CG}_z] = \text{abs} [(-5.92 \text{ ft}) - (-6.28 \text{ ft})] = 0.36 \text{ ft}$$

$$\begin{aligned}\text{DA}_x &= \{\Delta_x + D_y (\phi_z) + D_z (\phi_y)\} (24000) \\ &= \{(0.0 \text{ ft}) + (6.91 \text{ ft})(1.318 \text{ E} - 6 \text{ rad}) + (0.36 \text{ ft})(0.0 \text{ rad})\} (24000) \\ &= 0.22 \text{ mils}\end{aligned}$$

$$\begin{aligned}\text{DA}_y &= \{\Delta_y + D_x (\phi_z) + D_z (\phi_x)\} (24000) \\ &= \{(2.960 \text{ E} - 5 \text{ ft}) + (9.08 \text{ ft})(1.318 \text{ E} - 6 \text{ rad}) + (0.36 \text{ ft})(3.952 \text{ E} - 6 \text{ rad})\} (24000) \\ &= 1.03 \text{ mils}\end{aligned}$$

$$\begin{aligned}\text{DA}_z &= \{\Delta_z + D_x (\phi_y) + D_y (\phi_x)\} (24000) \\ &= \{(0.0 \text{ ft}) + (9.08 \text{ ft})(0.0 \text{ rad}) + (6.91 \text{ ft})(3.952 \text{ E} - 6 \text{ rad})\} (24000) \\ &= 0.66 \text{ mils}\end{aligned}$$

Total Vibration Amplitudes: (check versus human tolerance, Attachment 4)

(Attachment 10, Section I2)

$$\Sigma \text{DA}_x = (1.19 \text{ mils}) + (0.22 \text{ mils}) = 1.41 \text{ mils}$$

$$\Sigma \text{DA}_y = (1.71 \text{ mils}) + (1.03 \text{ mils}) = 2.74 \text{ mils} \quad \text{clearly perceptible}$$

$$\Sigma \text{DA}_z = (2.11 \text{ mils}) + (0.66 \text{ mils}) = 2.77 \text{ mils}$$